SampleSize_n100_5Treatments/table2_nonshrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	89.66	89.66	93.10	100.00	82.76	96.55	93.1	96.55	96.55	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	89.29	100.00	89.29	96.43	100.0	96.43	92.86	100.00
20%	100.00	100.00	96.43	89.29	96.43	85.71	100.00	100.0	96.43	96.43	100.00
30%	93.10	93.10	96.55	89.66	89.66	75.86	100.00	100.0	96.55	96.55	96.55
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.0	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	89.29	96.43	100.0	96.43	92.86	100.00
20%	96.43	96.43	92.86	89.29	92.86	78.57	100.00	100.0	96.43	96.43	100.00
30%	93.10	93.10	96.55	93.10	86.21	75.86	100.00	100.0	96.55	100.00	100.00
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.0	96.55	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	92.86	100.00	89.29	96.43	100.0	96.43	92.86	100.00
20%	100.00	96.43	92.86	89.29	92.86	78.57	100.00	100.0	96.43	96.43	100.00
30%	93.10	89.66	89.66	89.66	86.21	72.41	100.00	100.0	100.00	93.10	93.10
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.0	100.00	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	89.66	89.66	93.10	93.10	72.41	100.00	100.0	96.55	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_5Treatments/table2_nonshrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	96.43	92.86	96.43	96.43	82.14	100.00	100.00	100.00	100.00	96.43
20%	96.55	96.55	96.55	93.10	100.00	89.66	100.00	100.00	100.00	96.55	100.00
30%	100.00	89.66	96.55	93.10	93.10	79.31	100.00	93.10	96.55	96.55	96.55
40%	96.43	96.43	100.00	96.43	92.86	89.29	96.43	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	93.10	96.55	96.55	93.10	100.00	86.21	100.00	100.00	100.00	96.55	100.00
30%	100.00	86.21	96.55	89.66	93.10	75.86	100.00	93.10	96.55	96.55	100.00
40%	96.43	92.86	96.43	96.43	89.29	78.57	96.43	100.00	100.00	100.00	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	92.86	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	96.55	96.55	96.55	93.10	96.55	89.66	100.00	96.55	100.00	96.55	100.00
30%	96.55	86.21	96.55	89.66	93.10	72.41	100.00	93.10	96.55	96.55	100.00
40%	96.43	92.86	96.43	96.43	92.86	82.14	96.43	100.00	100.00	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	89.66
Imputation using linear regression	NA
10%	89.29
20%	92.86
30%	93.10
40%	100.00
Imputation using predictive mean matching method	NA
10%	89.29
20%	92.86
30%	96.55
40%	96.55
Imputation using propensity−score method	NA
10%	89.29
20%	92.86
30%	86.21
40%	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	96.43
20%	96.55
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	93.10
40%	92.86
Imputation using propensity−score method	NA
10%	100.00
20%	93.10
30%	93.10
40%	92.86

SampleSize_n100_5Treatments/table2_shrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	89.66	89.66	93.10	100.00	82.76	96.55	96.55	96.55	96.55	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	89.29	100.00	89.29	96.43	100.00	96.43	96.43	100.00
20%	100.00	100.00	96.43	89.29	96.43	85.71	100.00	100.00	96.43	96.43	100.00
30%	93.10	93.10	96.55	89.66	89.66	75.86	100.00	100.00	96.55	96.55	96.55
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	92.86	100.00	89.29	96.43	100.00	96.43	92.86	100.00
20%	96.43	96.43	92.86	89.29	92.86	78.57	100.00	100.00	96.43	96.43	100.00
30%	93.10	93.10	96.55	93.10	86.21	75.86	100.00	100.00	96.55	100.00	93.10
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.00	96.55	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	100.00	96.43	92.86	100.00	89.29	96.43	100.00	96.43	96.43	100.00
20%	100.00	96.43	92.86	89.29	92.86	78.57	100.00	100.00	96.43	96.43	100.00
30%	93.10	89.66	89.66	89.66	86.21	72.41	100.00	100.00	100.00	93.10	93.10
40%	100.00	96.55	93.10	100.00	100.00	89.66	100.00	100.00	100.00	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	89.66	89.66	93.10	93.10	72.41	100.00	100.00	96.55	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n100_5Treatments/table2_shrink_5_treatments_MAR_n100

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	96.43	92.86	96.43	96.43	82.14	100.00	100.00	100.00	100.00	96.43
20%	96.55	96.55	96.55	93.10	100.00	89.66	100.00	100.00	100.00	96.55	100.00
30%	100.00	89.66	96.55	93.10	93.10	79.31	100.00	93.10	96.55	96.55	96.55
40%	96.43	96.43	100.00	96.43	92.86	89.29	96.43	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	93.10	96.55	96.55	93.10	100.00	86.21	100.00	100.00	100.00	96.55	100.00
30%	100.00	86.21	96.55	89.66	93.10	75.86	100.00	93.10	96.55	96.55	100.00
40%	96.43	92.86	96.43	96.43	89.29	78.57	96.43	100.00	100.00	100.00	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	92.86	96.43	82.14	100.00	100.00	100.00	100.00	96.43
20%	96.55	96.55	96.55	93.10	96.55	89.66	100.00	96.55	100.00	96.55	100.00
30%	96.55	86.21	96.55	89.66	93.10	72.41	100.00	93.10	96.55	93.10	100.00
40%	96.43	92.86	96.43	96.43	92.86	82.14	96.43	100.00	100.00	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	89.66
Imputation using linear regression	NA
10%	92.86
20%	92.86
30%	93.10
40%	100.00
Imputation using predictive mean matching method	NA
10%	89.29
20%	92.86
30%	93.10
40%	96.55
Imputation using propensity−score method	NA
10%	92.86
20%	92.86
30%	86.21
40%	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	96.43
20%	96.55
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	93.10
40%	92.86
Imputation using propensity−score method	NA
10%	96.43
20%	93.10
30%	89.66
40%	92.86

SampleSize_n100_5Treatments/table3_nonshrink_5_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.753	1.766	1.793	1.781	1.799	2.235	2.252	2.286	2.271	2.295
Imputation using linear regression										
10%	1.838 (4.87)	1.842 (4.30)	1.897 (5.80)	1.875 (5.31)	1.901 (5.63)	2.348 (5.03)	2.352 (4.45)	2.423 (5.96)	2.395 (5.46)	2.427 (5.79)
20%	1.928 (10.02)	1.944 (10.10)	1.978 (10.35)	1.981 (11.23)	1.988 (10.50)	2.465 (10.31)	2.486 (10.38)	2.529 (10.63)	2.532 (11.51)	2.543 (10.81)
30%	2.074 (18.33)	2.120 (20.04)	2.147 (19.77)	2.112 (18.63)	2.206 (22.63)	2.645 (18.35)	2.705 (20.09)	2.738 (19.77)	2.695 (18.68)	2.815 (22.67)
40%	2.281 (30.18)	2.262 (28.10)	2.327 (29.83)	2.312 (29.87)	2.372 (31.86)	2.916 (30.44)	2.892 (28.39)	2.975 (30.12)	2.957 (30.21)	3.033 (32.16)
Imputation using predictive mean matching method										
10%	1.824 (4.10)	1.835 (3.92)	1.882 (4.96)	1.863 (4.62)	1.897 (5.44)	2.328 (4.17)	2.342 (3.98)	2.401 (5.03)	2.377 (4.69)	2.421 (5.50)
20%	1.875 (7.01)	1.898 (7.48)	1.929 (7.60)	1.920 (7.80)	1.966 (9.26)	2.400 (7.36)	2.428 (7.82)	2.468 (7.94)	2.456 (8.15)	2.515 (9.60)
30%	1.958 (11.71)	2.039 (15.44)	2.073 (15.65)	2.017 (13.27)	2.088 (16.07)	2.499 (11.80)	2.601 (15.51)	2.646 (15.71)	2.575 (13.37)	2.666 (16.18)
40%	2.152 (22.77)	2.165 (22.62)	2.231 (24.43)	2.192 (23.09)	2.261 (25.66)	2.742 (22.67)	2.758 (22.46)	2.842 (24.28)	2.793 (23.00)	2.881 (25.55)
Imputation using propensity−score method										
10%	1.830 (4.42)	1.836 (3.99)	1.888 (5.32)	1.857 (4.31)	1.895 (5.32)	2.337 (4.55)	2.345 (4.11)	2.411 (5.44)	2.372 (4.43)	2.419 (5.44)
20%	1.892 (7.97)	1.920 (8.72)	1.952 (8.91)	1.934 (8.63)	1.983 (10.23)	2.420 (8.27)	2.455 (9.01)	2.496 (9.19)	2.474 (8.93)	2.536 (10.53)
30%	1.987 (13.36)	2.052 (16.19)	2.048 (14.22)	2.049 (15.09)	2.105 (17.00)	2.536 (13.46)	2.619 (16.29)	2.614 (14.33)	2.616 (15.18)	2.687 (17.12)
40%	2.178 (24.28)	2.196 (24.33)	2.235 (24.67)	2.216 (24.48)	2.292 (27.38)	2.782 (24.45)	2.803 (24.47)	2.854 (24.83)	2.831 (24.67)	2.927 (27.55)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.760	1.801	1.793	1.827	1.835	2.237	2.290	2.279	2.324	2.334
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.880 (6.86)	1.854 (2.94)	1.889 (5.39)	1.903 (4.12)	1.911 (4.09)	2.397 (7.15)	2.364 (3.21)	2.409 (5.67)	2.426 (4.40)	2.436 (4.36)
20%	1.959 (11.35)	1.973 (9.56)	1.985 (10.72)	2.029 (11.04)	2.021 (10.13)	2.499 (11.68)	2.517 (9.89)	2.531 (11.06)	2.588 (11.37)	2.578 (10.47)
30%	2.119 (20.40)	2.085 (15.78)	2.138 (19.25)	2.163 (18.37)	2.153 (17.30)	2.704 (20.87)	2.662 (16.23)	2.729 (19.71)	2.760 (18.80)	2.748 (17.73)
40%	2.292 (30.27)	2.311 (28.34)	2.267 (26.44)	2.318 (26.86)	2.301 (25.35)	2.932 (31.06)	2.957 (29.10)	2.901 (27.29)	2.966 (27.63)	2.945 (26.19)
Imputation using predictive mean matching method										
10%	1.847 (4.96)	1.830 (1.62)	1.877 (4.69)	1.883 (3.07)	1.888 (2.86)	2.355 (5.26)	2.334 (1.91)	2.393 (4.98)	2.402 (3.36)	2.408 (3.15)
20%	1.905 (8.26)	1.912 (6.17)	1.936 (8.00)	1.971 (7.86)	1.977 (7.73)	2.431 (8.67)	2.441 (6.58)	2.471 (8.39)	2.515 (8.26)	2.524 (8.13)
30%	2.018 (14.68)	1.979 (9.88)	2.061 (14.98)	2.078 (13.72)	2.043 (11.32)	2.574 (15.05)	2.525 (10.24)	2.628 (15.31)	2.650 (14.04)	2.606 (11.67)
40%	2.172 (23.44)	2.174 (20.71)	2.192 (22.29)	2.205 (20.69)	2.215 (20.69)	2.770 (23.80)	2.773 (21.07)	2.796 (22.67)	2.813 (21.07)	2.826 (21.10)
Imputation using propensity−score method										
10%	1.862 (5.81)	1.841 (2.22)	1.880 (4.86)	1.893 (3.57)	1.901 (3.56)	2.374 (6.10)	2.347 (2.49)	2.396 (5.14)	2.413 (3.85)	2.424 (3.83)
20%	1.936 (10.05)	1.950 (8.28)	1.954 (8.98)	1.994 (9.14)	1.999 (8.92)	2.470 (10.39)	2.487 (8.61)	2.492 (9.31)	2.544 (9.47)	2.550 (9.25)
30%	2.061 (17.12)	2.054 (14.07)	2.089 (16.55)	2.124 (16.23)	2.092 (13.96)	2.630 (17.55)	2.621 (14.44)	2.666 (16.95)	2.709 (16.60)	2.669 (14.34)
40%	2.190 (24.48)	2.213 (22.88)	2.221 (23.92)	2.239 (22.55)	2.254 (22.80)	2.802 (25.24)	2.831 (23.60)	2.842 (24.67)	2.865 (23.29)	2.884 (23.56)

SampleSize_n100_5Treatments/table3_shrink_5_treatments_MAR_n100

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.753	1.766	1.793	1.781	1.799	2.256	2.257	2.264	2.264	2.265
Imputation using linear regression										
10%	1.838 (4.87)	1.842 (4.30)	1.897 (5.80)	1.875 (5.31)	1.901 (5.63)	2.356 (4.44)	2.361 (4.62)	2.388 (5.47)	2.375 (4.91)	2.388 (5.41)
20%	1.928 (10.02)	1.944 (10.10)	1.978 (10.35)	1.981 (11.23)	1.988 (10.50)	2.482 (10.02)	2.489 (10.30)	2.523 (11.43)	2.513 (10.99)	2.514 (10.98)
30%	2.074 (18.33)	2.120 (20.04)	2.147 (19.77)	2.112 (18.63)	2.206 (22.63)	2.662 (18.02)	2.694 (19.38)	2.729 (20.54)	2.689 (18.76)	2.771 (22.32)
40%	2.281 (30.18)	2.262 (28.10)	2.327 (29.83)	2.312 (29.87)	2.372 (31.86)	2.922 (29.51)	2.898 (28.44)	2.964 (30.91)	2.942 (29.93)	3.003 (32.57)
Imputation using predictive mean matching method										
10%	1.824 (4.10)	1.835 (3.92)	1.882 (4.96)	1.863 (4.62)	1.897 (5.44)	2.335 (3.51)	2.349 (4.11)	2.362 (4.32)	2.355 (4.03)	2.378 (4.96)
20%	1.875 (7.01)	1.898 (7.48)	1.929 (7.60)	1.920 (7.80)	1.966 (9.26)	2.422 (7.34)	2.433 (7.83)	2.461 (8.71)	2.437 (7.64)	2.483 (9.59)
30%	1.958 (11.71)	2.039 (15.44)	2.073 (15.65)	2.017 (13.27)	2.088 (16.07)	2.518 (11.63)	2.587 (14.64)	2.636 (16.42)	2.565 (13.29)	2.620 (15.67)
40%	2.152 (22.77)	2.165 (22.62)	2.231 (24.43)	2.192 (23.09)	2.261 (25.66)	2.746 (21.71)	2.766 (22.56)	2.831 (25.07)	2.776 (22.62)	2.848 (25.73)
Imputation using propensity−score method										
10%	1.830 (4.42)	1.836 (3.99)	1.888 (5.32)	1.857 (4.31)	1.895 (5.32)	2.345 (3.93)	2.353 (4.28)	2.374 (4.85)	2.351 (3.82)	2.378 (4.98)
20%	1.892 (7.97)	1.920 (8.72)	1.952 (8.91)	1.934 (8.63)	1.983 (10.23)	2.439 (8.09)	2.459 (8.98)	2.490 (10.00)	2.456 (8.48)	2.505 (10.58)
30%	1.987 (13.36)	2.052 (16.19)	2.048 (14.22)	2.049 (15.09)	2.105 (17.00)	2.559 (13.43)	2.608 (15.55)	2.606 (15.09)	2.607 (15.15)	2.645 (16.77)
40%	2.178 (24.28)	2.196 (24.33)	2.235 (24.67)	2.216 (24.48)	2.292 (27.38)	2.787 (23.56)	2.812 (24.63)	2.844 (25.63)	2.817 (24.43)	2.898 (27.93)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.760	1.801	1.793	1.827	1.835	2.244	2.260	2.259	2.280	2.285
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.880 (6.86)	1.854 (2.94)	1.889 (5.39)	1.903 (4.12)	1.911 (4.09)	2.375 (5.83)	2.361 (4.45)	2.379 (5.30)	2.381 (4.43)	2.387 (4.45)
20%	1.959 (11.35)	1.973 (9.56)	1.985 (10.72)	2.029 (11.04)	2.021 (10.13)	2.502 (11.47)	2.513 (11.18)	2.520 (11.53)	2.549 (11.83)	2.542 (11.25)
30%	2.119 (20.40)	2.085 (15.78)	2.138 (19.25)	2.163 (18.37)	2.153 (17.30)	2.690 (19.87)	2.664 (17.84)	2.709 (19.87)	2.739 (20.16)	2.729 (19.44)
40%	2.292 (30.27)	2.311 (28.34)	2.267 (26.44)	2.318 (26.86)	2.301 (25.35)	2.933 (30.71)	2.954 (30.68)	2.895 (28.11)	2.959 (29.78)	2.926 (28.05)
Imputation using predictive mean matching method										
10%	1.847 (4.96)	1.830 (1.62)	1.877 (4.69)	1.883 (3.07)	1.888 (2.86)	2.335 (4.04)	2.332 (3.18)	2.359 (4.41)	2.357 (3.37)	2.356 (3.13)
20%	1.905 (8.26)	1.912 (6.17)	1.936 (8.00)	1.971 (7.86)	1.977 (7.73)	2.435 (8.48)	2.441 (7.98)	2.462 (8.96)	2.475 (8.58)	2.487 (8.85)
30%	2.018 (14.68)	1.979 (9.88)	2.061 (14.98)	2.078 (13.72)	2.043 (11.32)	2.563 (14.20)	2.525 (11.71)	2.610 (15.50)	2.630 (15.39)	2.586 (13.16)
40%	2.172 (23.44)	2.174 (20.71)	2.192 (22.29)	2.205 (20.69)	2.215 (20.69)	2.773 (23.56)	2.771 (22.61)	2.786 (23.28)	2.805 (23.04)	2.802 (22.65)
Imputation using propensity−score method										
10%	1.862 (5.81)	1.841 (2.22)	1.880 (4.86)	1.893 (3.57)	1.901 (3.56)	2.351 (4.77)	2.345 (3.75)	2.365 (4.66)	2.368 (3.86)	2.373 (3.86)
20%	1.936 (10.05)	1.950 (8.28)	1.954 (8.98)	1.994 (9.14)	1.999 (8.92)	2.474 (10.24)	2.486 (9.97)	2.484 (9.93)	2.505 (9.89)	2.514 (10.01)
30%	2.061 (17.12)	2.054 (14.07)	2.089 (16.55)	2.124 (16.23)	2.092 (13.96)	2.614 (16.48)	2.621 (15.97)	2.647 (17.15)	2.689 (17.95)	2.649 (15.93)
40%	2.190 (24.48)	2.213 (22.88)	2.221 (23.92)	2.239 (22.55)	2.254 (22.80)	2.803 (24.92)	2.826 (25.03)	2.832 (25.32)	2.856 (25.29)	2.864 (25.34)

SampleSize_n150_5Treatments/table2_nonshrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	100.00	96.55	96.55	82.76	96.55	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	86.21	86.21	96.55	82.76	93.10	68.97	93.10	96.55	96.55	96.55	100.00
40%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	92.86	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	82.76	89.66	96.55	82.76	100.00	72.41	93.10	96.55	96.55	93.10	100.00
40%	100.00	96.43	96.43	89.29	92.86	75.00	100.00	96.43	96.43	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	96.43	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	96.55	100.00	100.00
30%	82.76	93.10	96.55	79.31	93.10	72.41	93.10	96.55	96.55	89.66	100.00
40%	96.43	96.43	92.86	92.86	96.43	85.71	100.00	96.43	96.43	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	92.86	100.00	92.86	92.86	75.00	100.00	96.43	100.00	92.86	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_5Treatments/table2_nonshrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	89.66	89.66	96.55	93.10	100.00	72.41	100.00	96.55	100	96.55	100.00
30%	100.00	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100	96.55	100.00
40%	96.43	92.86	100.00	96.43	89.29	85.71	100.00	100.00	100	100.00	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	93.10	89.66	96.55	93.10	100.00	79.31	100.00	100.00	100	93.10	100.00
30%	100.00	100.00	93.10	96.55	100.00	89.66	100.00	100.00	100	96.55	100.00
40%	92.86	89.29	96.43	96.43	89.29	78.57	100.00	100.00	100	96.43	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	96.43	92.86	100.00	78.57	96.43	96.43	100	96.43	100.00
20%	93.10	86.21	96.55	93.10	100.00	75.86	100.00	100.00	100	93.10	100.00
30%	100.00	100.00	96.55	96.55	100.00	93.10	100.00	100.00	100	96.55	100.00
40%	96.43	96.43	100.00	92.86	89.29	82.14	100.00	100.00	100	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.55
Imputation using linear regression	NA
10%	92.86
20%	100.00
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	100.00
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	92.86
20%	96.55
30%	86.21
40%	92.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	92.86
Imputation using linear regression	NA

MR	Joint_many
10%	92.86
20%	93.10
30%	96.55
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	93.10
30%	96.55
40%	89.29
Imputation using propensity−score method	NA
10%	92.86
20%	93.10
30%	96.55
40%	96.43

SampleSize_n150_5Treatments/table2_shrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	93.10	96.55	100.00	96.55	96.55	82.76	96.55	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	96.43	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	86.21	86.21	96.55	82.76	93.10	68.97	93.10	96.55	96.55	96.55	100.00
40%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.43	92.86	92.86	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
30%	82.76	89.66	96.55	82.76	100.00	72.41	93.10	96.55	96.55	93.10	100.00
40%	100.00	96.43	96.43	89.29	92.86	75.00	100.00	96.43	96.43	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	96.43	96.43	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	100.00	100.00	93.10	100.00	100.00	96.55	100.00	100.00
30%	82.76	93.10	96.55	79.31	93.10	72.41	93.10	96.55	96.55	89.66	100.00
40%	96.43	96.43	92.86	92.86	96.43	85.71	100.00	96.43	96.43	100.00	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	92.86	100.00	92.86	92.86	75.00	100.00	96.43	100.00	92.86	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n150_5Treatments/table2_shrink_5_treatments_MAR_n150

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	89.66	89.66	96.55	93.10	100.00	72.41	100.00	96.55	100	93.10	100.00
30%	100.00	100.00	100.00	96.55	100.00	96.55	100.00	100.00	100	96.55	100.00
40%	96.43	92.86	100.00	96.43	89.29	85.71	100.00	100.00	100	100.00	96.43
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	100.00	92.86	100.00	82.14	100.00	96.43	100	96.43	100.00
20%	93.10	89.66	96.55	93.10	100.00	79.31	100.00	96.55	100	93.10	100.00
30%	100.00	100.00	93.10	96.55	100.00	89.66	100.00	100.00	100	96.55	100.00
40%	92.86	89.29	96.43	96.43	89.29	78.57	100.00	100.00	100	96.43	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	92.86	96.43	92.86	100.00	78.57	96.43	96.43	100	96.43	100.00
20%	93.10	86.21	96.55	93.10	100.00	75.86	100.00	100.00	100	93.10	100.00
30%	100.00	100.00	96.55	96.55	100.00	93.10	100.00	100.00	100	96.55	100.00
40%	96.43	96.43	100.00	92.86	89.29	82.14	100.00	100.00	100	100.00	96.43

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.55
Imputation using linear regression	NA
10%	92.86
20%	100.00
30%	89.66
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	100.00
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	92.86
20%	96.55
30%	86.21
40%	92.86
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	92.86
Imputation using linear regression	NA

MR	Joint_many
10%	92.86
20%	89.66
30%	96.55
40%	96.43
Imputation using predictive mean matching method	NA
10%	92.86
20%	89.66
30%	96.55
40%	89.29
Imputation using propensity−score method	NA
10%	92.86
20%	93.10
30%	96.55
40%	96.43

SampleSize_n150_5Treatments/table3_nonshrink_5_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.434	1.448	1.460	1.480	1.500	1.827	1.845	1.860	1.885	1.912
Imputation using linear regression										
10%	1.508 (5.18)	1.519 (4.94)	1.520 (4.15)	1.527 (3.18)	1.558 (3.88)	1.925 (5.38)	1.940 (5.15)	1.941 (4.35)	1.949 (3.39)	1.990 (4.09)
20%	1.577 (9.96)	1.600 (10.51)	1.589 (8.82)	1.620 (9.51)	1.644 (9.59)	2.015 (10.30)	2.045 (10.84)	2.031 (9.17)	2.071 (9.86)	2.101 (9.93)
30%	1.695 (18.22)	1.725 (19.14)	1.739 (19.12)	1.759 (18.90)	1.808 (20.53)	2.165 (18.48)	2.202 (19.38)	2.221 (19.40)	2.247 (19.16)	2.308 (20.76)
40%	1.820 (26.94)	1.851 (27.88)	1.851 (26.82)	1.871 (26.43)	1.918 (27.85)	2.328 (27.40)	2.367 (28.30)	2.368 (27.29)	2.393 (26.89)	2.452 (28.29)
Imputation using predictive mean matching method										
10%	1.499 (4.53)	1.506 (4.05)	1.511 (3.52)	1.520 (2.72)	1.546 (3.08)	1.913 (4.70)	1.923 (4.23)	1.929 (3.69)	1.940 (2.90)	1.974 (3.26)
20%	1.549 (8.03)	1.575 (8.77)	1.573 (7.75)	1.612 (8.95)	1.619 (7.92)	1.979 (8.32)	2.012 (9.06)	2.010 (8.04)	2.059 (9.23)	2.068 (8.20)
30%	1.623 (13.18)	1.633 (12.82)	1.672 (14.55)	1.688 (14.04)	1.728 (15.16)	2.074 (13.49)	2.087 (13.15)	2.137 (14.88)	2.156 (14.36)	2.207 (15.47)
40%	1.754 (22.32)	1.744 (20.47)	1.772 (21.41)	1.813 (22.52)	1.824 (21.62)	2.241 (22.64)	2.228 (20.79)	2.265 (21.76)	2.316 (22.85)	2.331 (21.94)
Imputation using propensity−score method										
10%	1.501 (4.70)	1.508 (4.18)	1.512 (3.58)	1.523 (2.91)	1.553 (3.53)	1.917 (4.90)	1.925 (4.38)	1.930 (3.78)	1.944 (3.11)	1.983 (3.73)
20%	1.571 (9.59)	1.598 (10.40)	1.592 (9.06)	1.614 (9.10)	1.650 (9.96)	2.008 (9.88)	2.042 (10.68)	2.034 (9.34)	2.062 (9.39)	2.107 (10.24)
30%	1.659 (15.73)	1.647 (13.74)	1.710 (17.15)	1.724 (16.52)	1.741 (16.09)	2.121 (16.09)	2.106 (14.15)	2.186 (17.52)	2.204 (16.88)	2.226 (16.46)
40%	1.794 (25.12)	1.818 (25.61)	1.819 (24.63)	1.843 (24.54)	1.883 (25.55)	2.296 (25.68)	2.327 (26.12)	2.329 (25.21)	2.359 (25.12)	2.410 (26.10)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.433	1.450	1.455	1.471	1.490	1.824	1.846	1.853	1.874	1.897
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.477 (3.07)	1.503 (3.68)	1.526 (4.82)	1.546 (5.09)	1.554 (4.35)	1.885 (3.34)	1.919 (3.95)	1.948 (5.10)	1.975 (5.37)	1.985 (4.63)
20%	1.586 (10.69)	1.602 (10.49)	1.625 (11.62)	1.646 (11.83)	1.629 (9.35)	2.024 (10.93)	2.044 (10.71)	2.073 (11.87)	2.100 (12.08)	2.079 (9.59)
30%	1.683 (17.47)	1.728 (19.22)	1.723 (18.37)	1.714 (16.47)	1.750 (17.49)	2.147 (17.68)	2.205 (19.43)	2.198 (18.60)	2.187 (16.72)	2.233 (17.72)
40%	1.828 (27.57)	1.823 (25.77)	1.866 (28.21)	1.851 (25.78)	1.923 (29.08)	2.332 (27.84)	2.328 (26.08)	2.381 (28.44)	2.362 (26.07)	2.453 (29.33)
Imputation using predictive mean matching method										
10%	1.466 (2.33)	1.488 (2.62)	1.513 (3.93)	1.535 (4.33)	1.534 (2.95)	1.872 (2.62)	1.900 (2.91)	1.932 (4.24)	1.961 (4.63)	1.959 (3.25)
20%	1.555 (8.58)	1.571 (8.39)	1.590 (9.22)	1.612 (9.56)	1.608 (7.99)	1.987 (8.93)	2.007 (8.72)	2.031 (9.56)	2.060 (9.91)	2.055 (8.32)
30%	1.611 (12.42)	1.672 (15.31)	1.664 (14.33)	1.668 (13.35)	1.685 (13.16)	2.056 (12.71)	2.134 (15.59)	2.124 (14.61)	2.129 (13.62)	2.152 (13.44)
40%	1.720 (20.03)	1.740 (20.04)	1.765 (21.28)	1.790 (21.64)	1.818 (22.07)	2.195 (20.32)	2.221 (20.29)	2.252 (21.50)	2.284 (21.87)	2.321 (22.35)
Imputation using propensity−score method										
10%	1.472 (2.78)	1.496 (3.17)	1.520 (4.41)	1.543 (4.84)	1.546 (3.78)	1.880 (3.08)	1.910 (3.46)	1.941 (4.70)	1.970 (5.14)	1.974 (4.07)
20%	1.576 (10.02)	1.575 (8.66)	1.594 (9.53)	1.633 (11.00)	1.629 (9.37)	2.013 (10.32)	2.011 (8.94)	2.036 (9.82)	2.085 (11.29)	2.080 (9.64)
30%	1.645 (14.83)	1.685 (16.21)	1.698 (16.65)	1.693 (15.07)	1.711 (14.87)	2.099 (15.06)	2.150 (16.44)	2.166 (16.87)	2.160 (15.30)	2.184 (15.11)
40%	1.822 (27.17)	1.798 (24.00)	1.829 (25.68)	1.844 (25.33)	1.884 (26.45)	2.323 (27.35)	2.294 (24.24)	2.333 (25.89)	2.353 (25.56)	2.403 (26.68)

SampleSize_n150_5Treatments/table3_shrink_5_treatments_MAR_n150

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.434	1.448	1.460	1.480	1.500	1.837	1.845	1.852	1.861	1.869
Imputation using linear regression										
10%	1.508 (5.18)	1.519 (4.94)	1.520 (4.15)	1.527 (3.18)	1.558 (3.88)	1.922 (4.62)	1.936 (4.94)	1.928 (4.10)	1.927 (3.54)	1.947 (4.19)
20%	1.577 (9.96)	1.600 (10.51)	1.589 (8.82)	1.620 (9.51)	1.644 (9.59)	2.026 (10.26)	2.039 (10.54)	2.029 (9.58)	2.056 (10.51)	2.073 (10.92)
30%	1.695 (18.22)	1.725 (19.14)	1.739 (19.12)	1.759 (18.90)	1.808 (20.53)	2.170 (18.13)	2.202 (19.38)	2.207 (19.18)	2.232 (19.93)	2.284 (22.23)
40%	1.820 (26.94)	1.851 (27.88)	1.851 (26.82)	1.871 (26.43)	1.918 (27.85)	2.340 (27.37)	2.375 (28.77)	2.365 (27.68)	2.385 (28.17)	2.439 (30.52)
Imputation using predictive mean matching method										
10%	1.499 (4.53)	1.506 (4.05)	1.511 (3.52)	1.520 (2.72)	1.546 (3.08)	1.911 (4.00)	1.918 (3.96)	1.916 (3.45)	1.917 (3.03)	1.930 (3.29)
20%	1.549 (8.03)	1.575 (8.77)	1.573 (7.75)	1.612 (8.95)	1.619 (7.92)	1.990 (8.33)	2.006 (8.76)	2.008 (8.44)	2.040 (9.64)	2.038 (9.05)
30%	1.623 (13.18)	1.633 (12.82)	1.672 (14.55)	1.688 (14.04)	1.728 (15.16)	2.081 (13.30)	2.089 (13.25)	2.124 (14.68)	2.144 (15.19)	2.183 (16.84)
40%	1.754 (22.32)	1.744 (20.47)	1.772 (21.41)	1.813 (22.52)	1.824 (21.62)	2.251 (22.56)	2.239 (21.37)	2.262 (22.13)	2.309 (24.06)	2.319 (24.12)
Imputation using propensity−score method										
10%	1.501 (4.70)	1.508 (4.18)	1.512 (3.58)	1.523 (2.91)	1.553 (3.53)	1.915 (4.25)	1.921 (4.15)	1.918 (3.56)	1.923 (3.33)	1.941 (3.87)
20%	1.571 (9.59)	1.598 (10.40)	1.592 (9.06)	1.614 (9.10)	1.650 (9.96)	2.018 (9.84)	2.038 (10.51)	2.033 (9.76)	2.047 (9.98)	2.079 (11.25)
30%	1.659 (15.73)	1.647 (13.74)	1.710 (17.15)	1.724 (16.52)	1.741 (16.09)	2.127 (15.77)	2.108 (14.26)	2.174 (17.37)	2.190 (17.67)	2.203 (17.90)
40%	1.794 (25.12)	1.818 (25.61)	1.819 (24.63)	1.843 (24.54)	1.883 (25.55)	2.306 (25.54)	2.335 (26.59)	2.326 (25.58)	2.353 (26.43)	2.400 (28.45)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.433	1.450	1.455	1.471	1.490	1.828	1.836	1.841	1.848	1.858
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.477 (3.07)	1.503 (3.68)	1.526 (4.82)	1.546 (5.09)	1.554 (4.35)	1.900 (3.90)	1.919 (4.53)	1.931 (4.90)	1.943 (5.19)	1.954 (5.14)
20%	1.586 (10.69)	1.602 (10.49)	1.625 (11.62)	1.646 (11.83)	1.629 (9.35)	2.028 (10.93)	2.044 (11.31)	2.052 (11.46)	2.070 (12.05)	2.053 (10.49)
30%	1.683 (17.47)	1.728 (19.22)	1.723 (18.37)	1.714 (16.47)	1.750 (17.49)	2.152 (17.72)	2.190 (19.30)	2.183 (18.60)	2.171 (17.50)	2.209 (18.89)
40%	1.828 (27.57)	1.823 (25.77)	1.866 (28.21)	1.851 (25.78)	1.923 (29.08)	2.336 (27.75)	2.326 (26.69)	2.378 (29.16)	2.356 (27.51)	2.435 (31.04)
Imputation using predictive mean matching method										
10%	1.466 (2.33)	1.488 (2.62)	1.513 (3.93)	1.535 (4.33)	1.534 (2.95)	1.887 (3.22)	1.901 (3.52)	1.914 (4.00)	1.929 (4.41)	1.928 (3.74)
20%	1.555 (8.58)	1.571 (8.39)	1.590 (9.22)	1.612 (9.56)	1.608 (7.99)	1.992 (8.95)	2.006 (9.27)	2.011 (9.25)	2.028 (9.78)	2.029 (9.15)
30%	1.611 (12.42)	1.672 (15.31)	1.664 (14.33)	1.668 (13.35)	1.685 (13.16)	2.062 (12.80)	2.117 (15.28)	2.107 (14.44)	2.114 (14.41)	2.125 (14.35)
40%	1.720 (20.03)	1.740 (20.04)	1.765 (21.28)	1.790 (21.64)	1.818 (22.07)	2.199 (20.27)	2.219 (20.87)	2.251 (22.27)	2.276 (23.21)	2.302 (23.85)
Imputation using propensity−score method										
10%	1.472 (2.78)	1.496 (3.17)	1.520 (4.41)	1.543 (4.84)	1.546 (3.78)	1.895 (3.64)	1.911 (4.08)	1.924 (4.53)	1.939 (4.93)	1.944 (4.60)
20%	1.576 (10.02)	1.575 (8.66)	1.594 (9.53)	1.633 (11.00)	1.629 (9.37)	2.015 (10.21)	2.013 (9.61)	2.016 (9.52)	2.056 (11.27)	2.057 (10.67)
30%	1.645 (14.83)	1.685 (16.21)	1.698 (16.65)	1.693 (15.07)	1.711 (14.87)	2.105 (15.15)	2.134 (16.24)	2.151 (16.84)	2.144 (16.06)	2.158 (16.13)
40%	1.822 (27.17)	1.798 (24.00)	1.829 (25.68)	1.844 (25.33)	1.884 (26.45)	2.327 (27.28)	2.293 (24.87)	2.331 (26.64)	2.347 (27.01)	2.386 (28.41)

SampleSize_n20_5Treatments/table2_nonshrink_5_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	100.00	100.00	93.10	100.00	100.00	96.55	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	93.10	96.55	96.55	96.55	96.55
20%	92.86	100.00	100.00	96.43	96.43	85.71	96.43	100.00	100.00	100.00	96.43
30%	100.00	100.00	96.43	96.43	100.00	92.86	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	93.10	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	62.07	93.10	96.55	96.55	96.55	96.55
20%	85.71	100.00	100.00	96.43	92.86	75.00	96.43	100.00	100.00	100.00	92.86
30%	89.29	92.86	96.43	96.43	100.00	75.00	100.00	100.00	100.00	100.00	100.00
40%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	93.10	96.55	96.55	96.55	93.10
20%	89.29	92.86	100.00	96.43	89.29	71.43	92.86	100.00	100.00	100.00	96.43
30%	89.29	92.86	96.43	92.86	96.43	71.43	96.43	100.00	100.00	100.00	100.00
40%	89.66	93.10	93.10	86.21	93.10	75.86	100.00	96.55	96.55	89.66	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	96.55	89.66	100.00	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_5Treatments/table2_nonshrink_5_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	86.21	89.66	96.55	96.55	75.86	96.55	96.55	96.55	96.55	100.00
20%	92.86	92.86	96.43	92.86	85.71	75.00	96.43	100.00	96.43	92.86	96.43
30%	93.10	96.55	96.55	100.00	100.00	89.66	96.55	100.00	100.00	100.00	100.00
40%	96.55	89.66	96.55	100.00	93.10	79.31	100.00	96.55	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	93.10	93.10	68.97	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	93.10	93.10	96.55	96.55	75.86	96.55	96.55	96.55	100.00	100.00
40%	93.10	82.76	89.66	93.10	89.66	68.97	96.55	93.10	96.55	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	96.55	93.10	72.41	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	96.55	93.10	93.10	100.00	75.86	96.55	100.00	100.00	100.00	100.00
40%	93.10	82.76	89.66	89.66	82.76	62.07	96.55	89.66	100.00	100.00	93.10

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.55
Imputation using linear regression	NA
10%	93.10
20%	92.86
30%	100.00
40%	100.00
Imputation using predictive mean matching method	NA
10%	93.10
20%	89.29
30%	100.00
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	89.29
30%	96.43
40%	82.76
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	89.29
30%	96.55
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	85.71
30%	89.66
40%	89.66
Imputation using propensity−score method	NA
10%	89.66
20%	85.71
30%	96.55
40%	86.21

SampleSize_n20_5Treatments/table2_shrink_5_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	100.00	93.10	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	96.55	96.55	100.00	96.55	96.55
20%	92.86	100.00	100.00	96.43	96.43	85.71	100.00	100.00	100.00	100.00	100.00
30%	100.00	100.00	96.43	96.43	100.00	92.86	100.00	100.00	100.00	100.00	100.00
40%	100.00	100.00	100.00	93.10	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	62.07	96.55	96.55	100.00	96.55	93.10
20%	85.71	100.00	100.00	96.43	92.86	75.00	100.00	100.00	100.00	100.00	96.43
30%	89.29	92.86	96.43	96.43	100.00	75.00	100.00	100.00	100.00	100.00	100.00
40%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	79.31	58.62	93.10	96.55	96.55	96.55	93.10
20%	89.29	92.86	100.00	96.43	89.29	71.43	96.43	100.00	100.00	100.00	96.43
30%	89.29	92.86	96.43	92.86	96.43	71.43	100.00	100.00	100.00	100.00	96.43
40%	89.66	93.10	93.10	86.21	93.10	75.86	96.55	96.55	96.55	93.10	96.55
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	100.00	96.55	100.00	96.55	96.55	89.66	100.00	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n20_5Treatments/table2_shrink_5_treatments_MAR_n20

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	86.21	89.66	96.55	96.55	75.86	96.55	96.55	96.55	96.55	100.00
20%	92.86	92.86	96.43	92.86	85.71	75.00	96.43	100.00	96.43	92.86	96.43
30%	93.10	96.55	96.55	100.00	100.00	89.66	96.55	100.00	100.00	100.00	100.00
40%	96.55	89.66	96.55	100.00	93.10	79.31	100.00	96.55	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	93.10	93.10	68.97	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	93.10	93.10	96.55	96.55	75.86	96.55	96.55	100.00	100.00	100.00
40%	93.10	82.76	89.66	93.10	89.66	68.97	96.55	93.10	96.55	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	86.21	93.10	96.55	93.10	72.41	96.55	96.55	96.55	96.55	100.00
20%	92.86	89.29	96.43	89.29	85.71	71.43	92.86	100.00	96.43	92.86	96.43
30%	89.66	96.55	93.10	93.10	100.00	75.86	96.55	100.00	100.00	100.00	100.00
40%	93.10	82.76	89.66	89.66	82.76	62.07	96.55	89.66	100.00	96.55	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	100.00
Imputation using linear regression	NA
10%	96.55
20%	100.00
30%	100.00
40%	100.00
Imputation using predictive mean matching method	NA
10%	93.10
20%	96.43
30%	100.00
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	92.86
30%	96.43
40%	86.21
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	89.29
30%	96.55
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	85.71
30%	93.10
40%	93.10
Imputation using propensity−score method	NA
10%	89.66
20%	85.71
30%	96.55
40%	86.21

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.066	4.131	4.153	4.259	4.175	5.086	5.169	5.196	5.329	5.221
Imputation using linear regression										
10%	4.177 (2.73)	4.318 (4.51)	4.427 (6.58)	4.385 (2.97)	4.492 (7.58)	5.335 (4.90)	5.516 (6.73)	5.652 (8.78)	5.600 (5.08)	5.737 (9.87)
20%	4.480 (10.18)	4.707 (13.95)	4.633 (11.54)	4.703 (10.43)	4.808 (15.16)	5.716 (12.38)	6.007 (16.22)	5.909 (13.73)	6.000 (12.60)	6.135 (17.51)
30%	4.788 (17.76)	4.993 (20.85)	4.941 (18.98)	5.110 (19.98)	5.005 (19.87)	6.119 (20.31)	6.374 (23.32)	6.314 (21.52)	6.527 (22.48)	6.396 (22.50)
40%	6.032 (48.36)	5.911 (43.07)	6.048 (45.61)	6.056 (42.19)	6.129 (46.80)	7.648 (50.36)	7.499 (45.08)	7.673 (47.67)	7.681 (44.13)	7.767 (48.75)
Imputation using predictive mean matching method										
10%	4.028 (−0.93)	4.173 (1.00)	4.217 (1.53)	4.183 (−1.77)	4.316 (3.38)	5.140 (1.05)	5.325 (3.03)	5.379 (3.52)	5.337 (0.14)	5.507 (5.48)
20%	4.017 (−1.21)	4.241 (2.66)	4.179 (0.61)	4.234 (−0.58)	4.436 (6.24)	5.126 (0.78)	5.413 (4.72)	5.331 (2.61)	5.403 (1.39)	5.660 (8.40)
30%	4.088 (0.54)	3.987 (−3.49)	4.177 (0.57)	4.251 (−0.19)	4.260 (2.02)	5.222 (2.67)	5.091 (−1.50)	5.335 (2.69)	5.427 (1.84)	5.442 (4.24)
40%	4.616 (13.53)	4.474 (8.29)	4.591 (10.54)	4.605 (8.14)	4.634 (10.98)	5.872 (15.46)	5.695 (10.19)	5.843 (12.47)	5.863 (10.01)	5.898 (12.96)
Imputation using propensity−score method										
10%	4.006 (−1.49)	4.153 (0.52)	4.204 (1.22)	4.151 (−2.52)	4.297 (2.91)	5.112 (0.51)	5.301 (2.57)	5.364 (3.24)	5.298 (−0.59)	5.484 (5.03)
20%	4.056 (−0.26)	4.219 (2.13)	4.182 (0.68)	4.264 (0.12)	4.442 (6.40)	5.171 (1.67)	5.380 (4.08)	5.330 (2.59)	5.435 (1.99)	5.664 (8.48)
30%	4.097 (0.76)	3.996 (−3.28)	4.174 (0.49)	4.192 (−1.58)	4.329 (3.68)	5.232 (2.86)	5.100 (−1.33)	5.331 (2.60)	5.352 (0.43)	5.528 (5.89)
40%	4.403 (8.29)	4.320 (4.58)	4.342 (4.53)	4.447 (4.42)	4.479 (7.28)	5.620 (10.50)	5.514 (6.67)	5.540 (6.63)	5.676 (6.51)	5.718 (9.51)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	3.940	3.942	4.079	3.994	4.006	4.938	4.938	5.112	5.006	5.020
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.367 (10.84)	4.435 (12.50)	4.447 (9.02)	4.438 (11.14)	4.465 (11.47)	5.544 (12.28)	5.628 (13.98)	5.644 (10.39)	5.632 (12.50)	5.670 (12.95)
20%	4.590 (16.50)	4.731 (20.03)	4.730 (15.96)	4.912 (22.99)	4.876 (21.73)	5.824 (17.94)	6.004 (21.59)	6.009 (17.53)	6.238 (24.61)	6.192 (23.36)
30%	5.398 (36.99)	5.337 (35.39)	5.299 (29.90)	5.504 (37.81)	5.713 (42.63)	6.831 (38.32)	6.759 (36.89)	6.711 (31.27)	6.963 (39.10)	7.228 (44.00)
40%	6.002 (52.32)	6.132 (55.57)	5.978 (46.55)	5.940 (48.73)	5.830 (45.53)	7.570 (53.30)	7.734 (56.63)	7.543 (47.54)	7.497 (49.76)	7.356 (46.54)
Imputation using predictive mean matching method										
10%	4.138 (5.03)	4.231 (7.33)	4.235 (3.83)	4.235 (6.04)	4.240 (5.85)	5.258 (6.48)	5.374 (8.83)	5.379 (5.23)	5.380 (7.48)	5.388 (7.33)
20%	4.158 (5.53)	4.275 (8.45)	4.268 (4.62)	4.405 (10.31)	4.482 (11.88)	5.268 (6.68)	5.418 (9.73)	5.411 (5.85)	5.584 (11.55)	5.680 (13.16)
30%	4.429 (12.39)	4.491 (13.92)	4.463 (9.42)	4.432 (10.98)	4.673 (16.67)	5.634 (14.09)	5.714 (15.71)	5.680 (11.10)	5.636 (12.59)	5.944 (18.42)
40%	4.227 (7.27)	4.465 (13.26)	4.332 (6.19)	4.425 (10.80)	4.319 (7.82)	5.392 (9.19)	5.692 (15.28)	5.526 (8.09)	5.646 (12.80)	5.511 (9.79)
Imputation using propensity−score method										
10%	4.174 (5.94)	4.269 (8.29)	4.266 (4.57)	4.255 (6.55)	4.274 (6.69)	5.300 (7.33)	5.418 (9.73)	5.414 (5.90)	5.402 (7.91)	5.426 (8.10)
20%	4.181 (6.11)	4.301 (9.12)	4.271 (4.71)	4.464 (11.77)	4.507 (12.51)	5.297 (7.26)	5.451 (10.39)	5.415 (5.92)	5.658 (13.02)	5.712 (13.79)
30%	4.454 (13.05)	4.624 (17.31)	4.554 (11.65)	4.545 (13.81)	4.710 (17.57)	5.653 (14.47)	5.871 (18.89)	5.783 (13.12)	5.767 (15.20)	5.978 (19.09)
40%	4.194 (6.45)	4.403 (11.71)	4.354 (6.72)	4.323 (8.24)	4.272 (6.63)	5.366 (8.67)	5.633 (14.08)	5.568 (8.91)	5.532 (10.51)	5.466 (8.89)

SampleSize_n20_5Treatments/table3_shrink_5_treatments_MAR_n20

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.066	4.131	4.153	4.259	4.175	5.139	5.160	5.171	5.183	5.163
Imputation using linear regression										
10%	4.177 (2.73)	4.318 (4.51)	4.427 (6.58)	4.385 (2.97)	4.492 (7.58)	5.419 (5.45)	5.515 (6.88)	5.611 (8.52)	5.607 (8.17)	5.607 (8.58)
20%	4.480 (10.18)	4.707 (13.95)	4.633 (11.54)	4.703 (10.43)	4.808 (15.16)	5.805 (12.96)	5.959 (15.49)	5.903 (14.16)	5.956 (14.90)	5.944 (15.12)
30%	4.788 (17.76)	4.993 (20.85)	4.941 (18.98)	5.110 (19.98)	5.005 (19.87)	6.191 (20.48)	6.464 (25.28)	6.313 (22.09)	6.478 (24.97)	6.379 (23.54)
40%	6.032 (48.36)	5.911 (43.07)	6.048 (45.61)	6.056 (42.19)	6.129 (46.80)	7.674 (49.34)	7.526 (45.86)	7.701 (48.93)	7.674 (48.04)	7.728 (49.67)
Imputation using predictive mean matching method										
10%	4.028 (−0.93)	4.173 (1.00)	4.217 (1.53)	4.183 (−1.77)	4.316 (3.38)	5.215 (1.49)	5.316 (3.03)	5.348 (3.43)	5.350 (3.21)	5.370 (4.00)
20%	4.017 (−1.21)	4.241 (2.66)	4.179 (0.61)	4.234 (−0.58)	4.436 (6.24)	5.243 (2.04)	5.359 (3.87)	5.325 (2.99)	5.342 (3.05)	5.436 (5.28)
30%	4.088 (0.54)	3.987 (−3.49)	4.177 (0.57)	4.251 (−0.19)	4.260 (2.02)	5.276 (2.68)	5.226 (1.28)	5.338 (3.23)	5.384 (3.86)	5.389 (4.37)
40%	4.616 (13.53)	4.474 (8.29)	4.591 (10.54)	4.605 (8.14)	4.634 (10.98)	5.892 (14.65)	5.727 (10.98)	5.846 (13.05)	5.847 (12.79)	5.850 (13.29)
Imputation using propensity−score method										
10%	4.006 (−1.49)	4.153 (0.52)	4.204 (1.22)	4.151 (−2.52)	4.297 (2.91)	5.195 (1.09)	5.287 (2.47)	5.331 (3.09)	5.313 (2.51)	5.342 (3.47)
20%	4.056 (−0.26)	4.219 (2.13)	4.182 (0.68)	4.264 (0.12)	4.442 (6.40)	5.275 (2.66)	5.326 (3.22)	5.328 (3.03)	5.380 (3.80)	5.432 (5.19)
30%	4.097 (0.76)	3.996 (−3.28)	4.174 (0.49)	4.192 (−1.58)	4.329 (3.68)	5.278 (2.70)	5.246 (1.67)	5.328 (3.04)	5.317 (2.58)	5.477 (6.07)
40%	4.403 (8.29)	4.320 (4.58)	4.342 (4.53)	4.447 (4.42)	4.479 (7.28)	5.639 (9.73)	5.577 (8.07)	5.571 (7.75)	5.689 (9.74)	5.687 (10.14)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	3.940	3.942	4.079	3.994	4.006	4.983	4.989	5.026	5.008	4.987
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.367 (10.84)	4.435 (12.50)	4.447 (9.02)	4.438 (11.14)	4.465 (11.47)	5.536 (11.08)	5.578 (11.82)	5.569 (10.82)	5.567 (11.17)	5.560 (11.50)
20%	4.590 (16.50)	4.731 (20.03)	4.730 (15.96)	4.912 (22.99)	4.876 (21.73)	5.890 (18.19)	5.979 (19.85)	5.977 (18.93)	6.134 (22.50)	6.046 (21.24)
30%	5.398 (36.99)	5.337 (35.39)	5.299 (29.90)	5.504 (37.81)	5.713 (42.63)	6.862 (37.69)	6.755 (35.41)	6.674 (32.80)	6.959 (38.98)	7.110 (42.59)
40%	6.002 (52.32)	6.132 (55.57)	5.978 (46.55)	5.940 (48.73)	5.830 (45.53)	7.579 (52.09)	7.706 (54.46)	7.537 (49.98)	7.466 (49.11)	7.382 (48.03)
Imputation using predictive mean matching method										
10%	4.138 (5.03)	4.231 (7.33)	4.235 (3.83)	4.235 (6.04)	4.240 (5.85)	5.252 (5.39)	5.314 (6.53)	5.311 (5.68)	5.290 (5.64)	5.287 (6.02)
20%	4.158 (5.53)	4.275 (8.45)	4.268 (4.62)	4.405 (10.31)	4.482 (11.88)	5.321 (6.78)	5.377 (7.79)	5.359 (6.64)	5.463 (9.09)	5.493 (10.16)
30%	4.429 (12.39)	4.491 (13.92)	4.463 (9.42)	4.432 (10.98)	4.673 (16.67)	5.679 (13.95)	5.705 (14.36)	5.636 (12.14)	5.664 (13.12)	5.849 (17.29)
40%	4.227 (7.27)	4.465 (13.26)	4.332 (6.19)	4.425 (10.80)	4.319 (7.82)	5.480 (9.97)	5.677 (13.80)	5.559 (10.61)	5.599 (11.81)	5.528 (10.85)
Imputation using propensity−score method										
10%	4.174 (5.94)	4.269 (8.29)	4.266 (4.57)	4.255 (6.55)	4.274 (6.69)	5.288 (6.11)	5.351 (7.27)	5.352 (6.50)	5.315 (6.13)	5.327 (6.83)
20%	4.181 (6.11)	4.301 (9.12)	4.271 (4.71)	4.464 (11.77)	4.507 (12.51)	5.358 (7.52)	5.423 (8.71)	5.374 (6.93)	5.544 (10.71)	5.531 (10.92)
30%	4.454 (13.05)	4.624 (17.31)	4.554 (11.65)	4.545 (13.81)	4.710 (17.57)	5.712 (14.61)	5.857 (17.40)	5.740 (14.21)	5.757 (14.96)	5.883 (17.98)
40%	4.194 (6.45)	4.403 (11.71)	4.354 (6.72)	4.323 (8.24)	4.272 (6.63)	5.468 (9.72)	5.646 (13.18)	5.619 (11.82)	5.537 (10.57)	5.515 (10.60)

SampleSize_n200_5Treatments/table2_nonshrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	85.71	100.00	92.86	96.43	75.00	96.43	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	96.55	100.00	86.21	75.86	100.00	100.00	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	92.86	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	89.66	96.55	100.00	100.00	79.31	68.97	96.55	96.55	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	85.71	100.00	100.00	96.43	96.43	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	100.00	96.55	96.55	89.66	100.00	86.21	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	100.00	96.55	86.21	79.31	96.55	96.55	100.00	100.00	93.10
40%	96.43	96.43	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	85.71	92.86	92.86	89.29	64.29	92.86	92.86	100.00	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_5Treatments/table2_nonshrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	100.00	93.10	100.00	93.10	86.21	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	86.21	79.31	100.00	96.55	96.55	100	100.00
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	96.43
40%	92.86	96.43	96.43	100.00	100.00	92.86	92.86	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	100.00	93.10	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	89.66	96.55	89.66	100.00	82.76	68.97	100.00	96.55	93.10	100	96.55
30%	96.43	100.00	100.00	100.00	89.29	85.71	100.00	100.00	100.00	100	100.00
40%	92.86	100.00	92.86	100.00	92.86	85.71	100.00	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	89.66	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	82.76	75.86	100.00	100.00	93.10	100	96.55
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	100.00
40%	92.86	92.86	96.43	100.00	96.43	85.71	96.43	100.00	96.43	100	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.43
Imputation using linear regression	NA
10%	96.43
20%	93.10
30%	93.10
40%	96.43
Imputation using predictive mean matching method	NA
10%	96.43
20%	93.10
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	100.00
20%	93.10
30%	89.66
40%	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	82.14
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	96.55
30%	96.43
40%	92.86
Imputation using predictive mean matching method	NA
10%	89.66
20%	89.66
30%	100.00
40%	100.00
Imputation using propensity−score method	NA
10%	89.66
20%	89.66
30%	100.00
40%	96.43

SampleSize_n200_5Treatments/table2_shrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	85.71	100.00	92.86	96.43	75.00	100.00	100.00	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	96.55	100.00	86.21	75.86	100.00	100.00	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	92.86	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	96.43
20%	100.00	96.55	96.55	93.10	100.00	89.66	100.00	96.55	96.55	100.00	100.00
30%	89.66	96.55	100.00	100.00	79.31	68.97	96.55	96.55	100.00	100.00	93.10
40%	92.86	92.86	96.43	92.86	100.00	85.71	100.00	100.00	96.43	96.43	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	100.00	96.43	96.43	96.43	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	100.00	96.55	96.55	89.66	100.00	86.21	100.00	96.55	96.55	100.00	100.00
30%	93.10	96.55	100.00	96.55	86.21	79.31	96.55	96.55	100.00	100.00	93.10
40%	96.43	96.43	96.43	92.86	100.00	89.29	100.00	100.00	100.00	96.43	100.00
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	85.71	85.71	92.86	92.86	89.29	64.29	96.43	92.86	100.00	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n200_5Treatments/table2_shrink_5_treatments_MAR_n200

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.55	100.00	93.10	100.00	93.10	86.21	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	86.21	79.31	100.00	96.55	96.55	100	96.55
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	100.00
40%	92.86	96.43	96.43	100.00	100.00	92.86	92.86	100.00	100.00	100	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	100.00	93.10	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	89.66	96.55	89.66	100.00	82.76	68.97	100.00	96.55	93.10	100	96.55
30%	96.43	100.00	100.00	100.00	89.29	85.71	100.00	100.00	100.00	100	100.00
40%	92.86	100.00	92.86	100.00	92.86	85.71	100.00	100.00	100.00	100	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	89.66	96.55	93.10	82.76	100.00	100.00	93.10	100	96.55
20%	96.55	96.55	89.66	100.00	82.76	75.86	100.00	100.00	93.10	100	96.55
30%	96.43	96.43	100.00	96.43	92.86	89.29	100.00	100.00	100.00	100	100.00
40%	92.86	92.86	96.43	100.00	96.43	85.71	96.43	100.00	96.43	100	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	100.00
Imputation using linear regression	NA
10%	96.43
20%	93.10
30%	93.10
40%	96.43
Imputation using predictive mean matching method	NA
10%	96.43
20%	93.10
30%	89.66
40%	92.86
Imputation using propensity−score method	NA
10%	100.00
20%	93.10
30%	89.66
40%	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	85.71
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	93.10
30%	100.00
40%	92.86
Imputation using predictive mean matching method	NA
10%	89.66
20%	89.66
30%	100.00
40%	100.00
Imputation using propensity−score method	NA
10%	89.66
20%	89.66
30%	100.00
40%	96.43

SampleSize_n200_5Treatments/table3_nonshrink_5_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.233	1.249	1.257	1.271	1.282	1.573	1.593	1.604	1.622	1.636
Imputation using linear regression										
10%	1.305 (5.83)	1.314 (5.26)	1.327 (5.57)	1.340 (5.45)	1.352 (5.48)	1.664 (5.81)	1.677 (5.24)	1.693 (5.56)	1.710 (5.43)	1.725 (5.47)
20%	1.363 (10.56)	1.401 (12.17)	1.390 (10.58)	1.423 (11.92)	1.408 (9.81)	1.741 (10.72)	1.789 (12.31)	1.776 (10.73)	1.817 (12.05)	1.799 (9.97)
30%	1.455 (17.99)	1.477 (18.28)	1.482 (17.90)	1.515 (19.17)	1.520 (18.59)	1.861 (18.33)	1.890 (18.63)	1.896 (18.23)	1.938 (19.50)	1.945 (18.92)
40%	1.567 (27.11)	1.611 (28.98)	1.588 (26.37)	1.639 (28.95)	1.651 (28.81)	2.004 (27.41)	2.060 (29.29)	2.032 (26.69)	2.096 (29.22)	2.112 (29.14)
Imputation using predictive mean matching method										
10%	1.296 (5.13)	1.308 (4.77)	1.319 (4.98)	1.336 (5.07)	1.346 (4.95)	1.653 (5.11)	1.669 (4.75)	1.683 (4.96)	1.704 (5.05)	1.716 (4.94)
20%	1.342 (8.83)	1.367 (9.47)	1.368 (8.84)	1.404 (10.45)	1.396 (8.90)	1.713 (8.89)	1.745 (9.54)	1.746 (8.89)	1.792 (10.50)	1.782 (8.95)
30%	1.413 (14.60)	1.431 (14.61)	1.438 (14.38)	1.454 (14.39)	1.459 (13.83)	1.808 (14.92)	1.831 (14.95)	1.839 (14.70)	1.861 (14.73)	1.868 (14.19)
40%	1.518 (23.11)	1.571 (25.84)	1.526 (21.38)	1.546 (21.64)	1.579 (23.19)	1.939 (23.31)	2.008 (26.06)	1.951 (21.65)	1.977 (21.89)	2.020 (23.48)
Imputation using propensity−score method										
10%	1.295 (5.04)	1.309 (4.80)	1.322 (5.20)	1.337 (5.17)	1.344 (4.83)	1.652 (5.03)	1.669 (4.79)	1.687 (5.18)	1.705 (5.15)	1.714 (4.82)
20%	1.358 (10.13)	1.390 (11.29)	1.397 (11.15)	1.416 (11.41)	1.407 (9.71)	1.733 (10.18)	1.774 (11.35)	1.783 (11.19)	1.807 (11.46)	1.795 (9.76)
30%	1.429 (15.91)	1.442 (15.44)	1.463 (16.38)	1.480 (16.41)	1.492 (16.41)	1.829 (16.28)	1.845 (15.83)	1.872 (16.73)	1.894 (16.79)	1.910 (16.78)
40%	1.551 (25.77)	1.599 (28.02)	1.577 (25.45)	1.587 (24.81)	1.616 (26.08)	1.983 (26.06)	2.044 (28.32)	2.017 (25.77)	2.029 (25.13)	2.068 (26.43)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.227	1.242	1.257	1.266	1.286	1.566	1.585	1.604	1.616	1.641
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.303 (6.22)	1.304 (4.98)	1.329 (5.72)	1.335 (5.43)	1.332 (3.59)	1.665 (6.30)	1.665 (5.06)	1.697 (5.79)	1.705 (5.51)	1.701 (3.66)
20%	1.381 (12.51)	1.386 (11.59)	1.399 (11.34)	1.400 (10.53)	1.424 (10.75)	1.762 (12.55)	1.769 (11.63)	1.786 (11.38)	1.787 (10.58)	1.818 (10.79)
30%	1.454 (18.53)	1.480 (19.12)	1.485 (18.14)	1.510 (19.23)	1.511 (17.52)	1.856 (18.53)	1.888 (19.11)	1.895 (18.13)	1.927 (19.24)	1.928 (17.52)
40%	1.588 (29.40)	1.599 (28.72)	1.641 (30.58)	1.620 (27.97)	1.660 (29.15)	2.029 (29.56)	2.043 (28.91)	2.097 (30.74)	2.072 (28.20)	2.121 (29.29)
Imputation using predictive mean matching method										
10%	1.290 (5.16)	1.290 (3.90)	1.322 (5.16)	1.327 (4.76)	1.321 (2.76)	1.648 (5.24)	1.648 (3.98)	1.688 (5.23)	1.694 (4.84)	1.687 (2.83)
20%	1.360 (10.88)	1.355 (9.13)	1.385 (10.19)	1.381 (9.09)	1.402 (9.06)	1.736 (10.87)	1.730 (9.12)	1.767 (10.17)	1.763 (9.09)	1.789 (9.05)
30%	1.419 (15.64)	1.446 (16.42)	1.450 (15.40)	1.461 (15.36)	1.478 (15.00)	1.812 (15.73)	1.846 (16.48)	1.852 (15.46)	1.866 (15.47)	1.888 (15.08)
40%	1.508 (22.86)	1.550 (24.79)	1.580 (25.75)	1.572 (24.14)	1.576 (22.59)	1.925 (22.92)	1.979 (24.82)	2.017 (25.77)	2.007 (24.20)	2.012 (22.65)
Imputation using propensity−score method										
10%	1.289 (5.08)	1.291 (3.96)	1.317 (4.78)	1.326 (4.70)	1.324 (3.00)	1.647 (5.19)	1.650 (4.07)	1.682 (4.88)	1.694 (4.81)	1.692 (3.11)
20%	1.366 (11.33)	1.364 (9.84)	1.377 (9.54)	1.381 (9.06)	1.412 (9.81)	1.744 (11.39)	1.742 (9.89)	1.758 (9.61)	1.764 (9.13)	1.802 (9.86)
30%	1.439 (17.24)	1.458 (17.39)	1.464 (16.49)	1.483 (17.13)	1.495 (16.33)	1.837 (17.33)	1.862 (17.47)	1.870 (16.57)	1.895 (17.23)	1.910 (16.42)
40%	1.543 (25.76)	1.588 (27.82)	1.594 (26.84)	1.599 (26.25)	1.643 (27.83)	1.971 (25.85)	2.027 (27.88)	2.036 (26.95)	2.042 (26.35)	2.097 (27.84)

SampleSize_n200_5Treatments/table3_shrink_5_treatments_MAR_n200

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.233	1.249	1.257	1.271	1.282	1.583	1.589	1.591	1.598	1.604
Imputation using linear regression										
10%	1.305 (5.83)	1.314 (5.26)	1.327 (5.57)	1.340 (5.45)	1.352 (5.48)	1.666 (5.28)	1.670 (5.13)	1.675 (5.25)	1.683 (5.33)	1.690 (5.38)
20%	1.363 (10.56)	1.401 (12.17)	1.390 (10.58)	1.423 (11.92)	1.408 (9.81)	1.746 (10.29)	1.774 (11.68)	1.767 (11.02)	1.794 (12.25)	1.776 (10.71)
30%	1.455 (17.99)	1.477 (18.28)	1.482 (17.90)	1.515 (19.17)	1.520 (18.59)	1.867 (17.98)	1.883 (18.52)	1.893 (18.94)	1.921 (20.23)	1.925 (20.02)
40%	1.567 (27.11)	1.611 (28.98)	1.588 (26.37)	1.639 (28.95)	1.651 (28.81)	2.012 (27.10)	2.052 (29.12)	2.028 (27.46)	2.089 (30.71)	2.090 (30.33)
Imputation using predictive mean matching method										
10%	1.296 (5.13)	1.308 (4.77)	1.319 (4.98)	1.336 (5.07)	1.346 (4.95)	1.656 (4.62)	1.663 (4.65)	1.666 (4.72)	1.676 (4.87)	1.682 (4.87)
20%	1.342 (8.83)	1.367 (9.47)	1.368 (8.84)	1.404 (10.45)	1.396 (8.90)	1.718 (8.52)	1.731 (8.93)	1.737 (9.19)	1.767 (10.58)	1.759 (9.68)
30%	1.413 (14.60)	1.431 (14.61)	1.438 (14.38)	1.454 (14.39)	1.459 (13.83)	1.814 (14.58)	1.827 (15.00)	1.840 (15.62)	1.845 (15.47)	1.850 (15.33)
40%	1.518 (23.11)	1.571 (25.84)	1.526 (21.38)	1.546 (21.64)	1.579 (23.19)	1.948 (23.05)	1.999 (25.81)	1.946 (22.32)	1.970 (23.31)	1.996 (24.45)
Imputation using propensity−score method										
10%	1.295 (5.04)	1.309 (4.80)	1.322 (5.20)	1.337 (5.17)	1.344 (4.83)	1.654 (4.51)	1.662 (4.62)	1.669 (4.92)	1.679 (5.10)	1.679 (4.70)
20%	1.358 (10.13)	1.390 (11.29)	1.397 (11.15)	1.416 (11.41)	1.407 (9.71)	1.737 (9.77)	1.760 (10.77)	1.773 (11.45)	1.783 (11.62)	1.772 (10.50)
30%	1.429 (15.91)	1.442 (15.44)	1.463 (16.38)	1.480 (16.41)	1.492 (16.41)	1.835 (15.95)	1.841 (15.89)	1.871 (17.56)	1.877 (17.48)	1.891 (17.93)
40%	1.551 (25.77)	1.599 (28.02)	1.577 (25.45)	1.587 (24.81)	1.616 (26.08)	1.989 (25.69)	2.036 (28.15)	2.012 (26.47)	2.022 (26.54)	2.047 (27.64)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.227	1.242	1.257	1.266	1.286	1.582	1.584	1.590	1.596	1.604
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.303 (6.22)	1.304 (4.98)	1.329 (5.72)	1.335 (5.43)	1.332 (3.59)	1.664 (5.22)	1.664 (5.07)	1.679 (5.57)	1.683 (5.42)	1.679 (4.66)
20%	1.381 (12.51)	1.386 (11.59)	1.399 (11.34)	1.400 (10.53)	1.424 (10.75)	1.759 (11.21)	1.768 (11.64)	1.772 (11.42)	1.769 (10.86)	1.790 (11.56)
30%	1.454 (18.53)	1.480 (19.12)	1.485 (18.14)	1.510 (19.23)	1.511 (17.52)	1.859 (17.53)	1.883 (18.89)	1.884 (18.46)	1.905 (19.38)	1.906 (18.83)
40%	1.588 (29.40)	1.599 (28.72)	1.641 (30.58)	1.620 (27.97)	1.660 (29.15)	2.037 (28.78)	2.041 (28.86)	2.085 (31.11)	2.062 (29.21)	2.110 (31.55)
Imputation using predictive mean matching method										
10%	1.290 (5.16)	1.290 (3.90)	1.322 (5.16)	1.327 (4.76)	1.321 (2.76)	1.648 (4.17)	1.647 (4.03)	1.669 (4.93)	1.671 (4.71)	1.664 (3.71)
20%	1.360 (10.88)	1.355 (9.13)	1.385 (10.19)	1.381 (9.09)	1.402 (9.06)	1.732 (9.51)	1.730 (9.24)	1.753 (10.24)	1.746 (9.38)	1.760 (9.73)
30%	1.419 (15.64)	1.446 (16.42)	1.450 (15.40)	1.461 (15.36)	1.478 (15.00)	1.815 (14.74)	1.841 (16.29)	1.842 (15.80)	1.842 (15.43)	1.864 (16.22)
40%	1.508 (22.86)	1.550 (24.79)	1.580 (25.75)	1.572 (24.14)	1.576 (22.59)	1.937 (22.50)	1.977 (24.83)	2.006 (26.12)	1.997 (25.12)	2.001 (24.77)
Imputation using propensity−score method										
10%	1.289 (5.08)	1.291 (3.96)	1.317 (4.78)	1.326 (4.70)	1.324 (3.00)	1.647 (4.14)	1.648 (4.04)	1.664 (4.65)	1.671 (4.69)	1.669 (4.02)
20%	1.366 (11.33)	1.364 (9.84)	1.377 (9.54)	1.381 (9.06)	1.412 (9.81)	1.740 (10.03)	1.741 (9.92)	1.745 (9.69)	1.747 (9.44)	1.775 (10.65)
30%	1.439 (17.24)	1.458 (17.39)	1.464 (16.49)	1.483 (17.13)	1.495 (16.33)	1.840 (16.36)	1.859 (17.39)	1.859 (16.90)	1.873 (17.37)	1.889 (17.75)
40%	1.543 (25.76)	1.588 (27.82)	1.594 (26.84)	1.599 (26.25)	1.643 (27.83)	1.979 (25.15)	2.026 (27.92)	2.025 (27.30)	2.032 (27.32)	2.087 (30.07)

SampleSize_n250_5Treatments/table2_nonshrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	86.21	96.55	82.76	100.00	100.00	100.00	93.10	100.00
20%	89.66	89.66	96.55	96.55	86.21	79.31	96.55	96.55	96.55	96.55	93.10
30%	92.86	96.43	92.86	89.29	89.29	75.00	100.00	100.00	100.00	100.00	96.43
40%	89.29	89.29	96.43	96.43	89.29	71.43	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	86.21	100.00	79.31	100.00	100.00	100.00	93.10	100.00
20%	93.10	89.66	96.55	93.10	93.10	86.21	96.55	96.55	96.55	96.55	93.10
30%	92.86	92.86	92.86	78.57	85.71	64.29	96.43	100.00	96.43	96.43	96.43
40%	78.57	92.86	96.43	92.86	92.86	64.29	100.00	100.00	100.00	96.43	96.43
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	86.21	96.55	79.31	100.00	100.00	100.00	93.10	100.00
20%	96.55	89.66	96.55	96.55	89.66	86.21	96.55	96.55	96.55	96.55	93.10
30%	85.71	96.43	89.29	85.71	92.86	60.71	100.00	100.00	96.43	100.00	96.43
40%	92.86	100.00	96.43	89.29	96.43	78.57	100.00	100.00	96.43	96.43	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	100.00	96.43	100.00	85.71	82.14	92.86	100.00	100.00	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_5Treatments/table2_nonshrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	100.00	93.10	86.21	86.21	89.66	68.97	100.00	100.00	96.55	93.10	100.00
20%	89.66	100.00	93.10	96.55	100.00	86.21	96.55	100.00	93.10	100.00	100.00
30%	96.43	89.29	89.29	96.43	96.43	75.00	100.00	96.43	96.43	100.00	100.00
40%	96.55	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	86.21	82.76	86.21	72.41	100.00	100.00	96.55	96.55	100.00
20%	93.10	100.00	93.10	100.00	96.55	86.21	96.55	100.00	96.55	100.00	100.00
30%	92.86	82.14	89.29	96.43	96.43	67.86	100.00	89.29	96.43	100.00	100.00
40%	100.00	93.10	96.55	93.10	100.00	86.21	100.00	96.55	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	86.21	82.76	89.66	68.97	100.00	96.55	93.10	93.10	96.55
20%	89.66	100.00	93.10	100.00	96.55	82.76	96.55	100.00	96.55	100.00	100.00
30%	96.43	85.71	89.29	96.43	96.43	71.43	100.00	96.43	96.43	100.00	100.00
40%	96.55	89.66	93.10	96.55	100.00	79.31	100.00	96.55	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	96.43
Imputation using linear regression	NA
10%	93.10
20%	89.66
30%	96.43
40%	96.43
Imputation using predictive mean matching method	NA
10%	93.10
20%	89.66
30%	89.29
40%	92.86
Imputation using propensity−score method	NA
10%	93.10
20%	89.66
30%	92.86
40%	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	89.29
Imputation using linear regression	NA

MR	Joint_many
10%	93.10
20%	93.10
30%	92.86
40%	100.00
Imputation using predictive mean matching method	NA
10%	96.55
20%	93.10
30%	85.71
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	93.10
30%	92.86
40%	96.55

SampleSize_n250_5Treatments/table2_shrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.43	96.43	96.43	96.43	100.00	89.29	100.00	100.00	96.43	96.43	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	100.00	86.21	96.55	82.76	100.00	100.00	100.00	93.10	100.00
20%	89.66	89.66	96.55	96.55	86.21	79.31	96.55	96.55	96.55	96.55	93.10
30%	92.86	96.43	92.86	89.29	89.29	75.00	100.00	100.00	100.00	100.00	96.43
40%	89.29	89.29	96.43	96.43	89.29	71.43	100.00	100.00	96.43	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	96.55	96.55	86.21	100.00	79.31	100.00	100.00	100.00	93.10	100.00
20%	93.10	89.66	96.55	93.10	93.10	86.21	96.55	96.55	96.55	96.55	93.10
30%	92.86	92.86	92.86	78.57	85.71	64.29	96.43	100.00	96.43	96.43	96.43
40%	78.57	92.86	96.43	92.86	92.86	64.29	100.00	100.00	100.00	96.43	92.86
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	96.55	86.21	96.55	79.31	100.00	100.00	100.00	93.10	100.00
20%	96.55	89.66	96.55	96.55	89.66	86.21	96.55	96.55	96.55	96.55	93.10
30%	85.71	96.43	89.29	85.71	92.86	60.71	100.00	100.00	96.43	100.00	96.43
40%	92.86	100.00	96.43	89.29	96.43	78.57	100.00	100.00	96.43	96.43	96.43
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	92.86	100.00	96.43	100.00	85.71	82.14	96.43	100.00	96.43	100.00	96.43
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n250_5Treatments/table2_shrink_5_treatments_MAR_n250

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	100.00	93.10	86.21	86.21	89.66	68.97	100.00	100.00	96.55	93.10	100.00
20%	89.66	100.00	93.10	96.55	100.00	86.21	96.55	100.00	93.10	100.00	100.00
30%	96.43	89.29	89.29	96.43	96.43	75.00	100.00	96.43	96.43	100.00	100.00
40%	96.55	96.55	96.55	100.00	100.00	93.10	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	96.55	86.21	82.76	86.21	72.41	100.00	100.00	96.55	96.55	100.00
20%	93.10	100.00	93.10	100.00	96.55	86.21	96.55	100.00	96.55	100.00	100.00
30%	92.86	82.14	89.29	96.43	96.43	67.86	100.00	89.29	96.43	100.00	100.00
40%	100.00	93.10	96.55	93.10	100.00	86.21	100.00	96.55	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.10	93.10	86.21	82.76	89.66	68.97	100.00	96.55	93.10	93.10	96.55
20%	89.66	100.00	93.10	100.00	96.55	82.76	96.55	100.00	96.55	100.00	100.00
30%	96.43	85.71	89.29	96.43	96.43	71.43	100.00	96.43	96.43	100.00	100.00
40%	96.55	89.66	93.10	96.55	100.00	79.31	100.00	96.55	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	92.86
Imputation using linear regression	NA
10%	93.10
20%	89.66
30%	96.43
40%	96.43
Imputation using predictive mean matching method	NA
10%	93.10
20%	89.66
30%	89.29
40%	89.29
Imputation using propensity−score method	NA
10%	93.10
20%	89.66
30%	92.86
40%	89.29
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	89.29
Imputation using linear regression	NA

MR	Joint_many
10%	93.10
20%	93.10
30%	92.86
40%	100.00
Imputation using predictive mean matching method	NA
10%	96.55
20%	93.10
30%	85.71
40%	96.55
Imputation using propensity−score method	NA
10%	89.66
20%	93.10
30%	92.86
40%	96.55

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.111	1.125	1.136	1.137	1.147	1.417	1.435	1.448	1.450	1.463
Imputation using linear regression										
10%	1.156 (3.99)	1.165 (3.47)	1.185 (4.35)	1.183 (4.00)	1.211 (5.57)	1.477 (4.18)	1.488 (3.66)	1.514 (4.54)	1.511 (4.19)	1.547 (5.76)
20%	1.230 (10.66)	1.220 (8.40)	1.242 (9.35)	1.248 (9.72)	1.259 (9.76)	1.572 (10.88)	1.559 (8.63)	1.587 (9.55)	1.595 (9.96)	1.608 (9.98)
30%	1.317 (18.46)	1.336 (18.74)	1.363 (19.96)	1.372 (20.61)	1.386 (20.83)	1.681 (18.59)	1.706 (18.85)	1.739 (20.09)	1.751 (20.73)	1.769 (20.94)
40%	1.439 (29.48)	1.433 (27.28)	1.465 (28.99)	1.454 (27.84)	1.528 (33.26)	1.838 (29.65)	1.830 (27.48)	1.871 (29.18)	1.857 (28.05)	1.950 (33.35)
Imputation using predictive mean matching method										
10%	1.144 (2.97)	1.157 (2.80)	1.177 (3.66)	1.175 (3.32)	1.203 (4.89)	1.462 (3.18)	1.478 (3.01)	1.504 (3.87)	1.501 (3.53)	1.537 (5.10)
20%	1.217 (9.49)	1.218 (8.21)	1.221 (7.54)	1.240 (9.05)	1.240 (8.14)	1.554 (9.67)	1.555 (8.37)	1.560 (7.70)	1.584 (9.23)	1.584 (8.32)
30%	1.283 (15.40)	1.302 (15.68)	1.315 (15.74)	1.315 (15.66)	1.325 (15.57)	1.640 (15.68)	1.664 (15.95)	1.681 (16.05)	1.682 (15.99)	1.695 (15.88)
40%	1.398 (25.76)	1.407 (25.01)	1.418 (24.86)	1.423 (25.10)	1.468 (27.97)	1.779 (25.55)	1.791 (24.79)	1.806 (24.66)	1.811 (24.91)	1.868 (27.71)
Imputation using propensity−score method										
10%	1.149 (3.40)	1.162 (3.20)	1.183 (4.18)	1.182 (3.93)	1.207 (5.22)	1.468 (3.59)	1.484 (3.40)	1.512 (4.38)	1.510 (4.13)	1.542 (5.42)
20%	1.218 (9.59)	1.223 (8.69)	1.223 (7.64)	1.245 (9.50)	1.242 (8.30)	1.557 (9.84)	1.563 (8.93)	1.562 (7.87)	1.592 (9.75)	1.588 (8.55)
30%	1.287 (15.82)	1.304 (15.84)	1.343 (18.24)	1.333 (17.26)	1.360 (18.62)	1.644 (16.02)	1.665 (16.02)	1.715 (18.43)	1.703 (17.47)	1.737 (18.80)
40%	1.422 (27.95)	1.458 (29.51)	1.474 (29.80)	1.457 (28.11)	1.495 (30.34)	1.815 (28.04)	1.859 (29.55)	1.880 (29.82)	1.859 (28.18)	1.908 (30.43)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.098	1.111	1.126	1.136	1.145	1.401	1.419	1.437	1.450	1.462
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.163 (6.00)	1.168 (5.08)	1.176 (4.50)	1.191 (4.90)	1.205 (5.24)	1.485 (5.99)	1.491 (5.07)	1.502 (4.49)	1.521 (4.90)	1.538 (5.22)
20%	1.214 (10.63)	1.238 (11.38)	1.248 (10.90)	1.262 (11.11)	1.280 (11.81)	1.551 (10.67)	1.581 (11.42)	1.594 (10.95)	1.611 (11.15)	1.635 (11.84)
30%	1.302 (18.58)	1.303 (17.27)	1.317 (16.96)	1.339 (17.89)	1.363 (19.07)	1.664 (18.77)	1.667 (17.47)	1.684 (17.15)	1.712 (18.07)	1.743 (19.24)
40%	1.428 (30.10)	1.438 (29.35)	1.429 (26.90)	1.467 (29.14)	1.454 (26.97)	1.826 (30.33)	1.838 (29.57)	1.827 (27.15)	1.875 (29.35)	1.860 (27.25)
Imputation using predictive mean matching method										
10%	1.159 (5.61)	1.167 (5.03)	1.173 (4.19)	1.185 (4.35)	1.197 (4.52)	1.480 (5.62)	1.490 (5.03)	1.497 (4.20)	1.513 (4.36)	1.528 (4.52)
20%	1.209 (10.13)	1.224 (10.12)	1.242 (10.33)	1.255 (10.48)	1.263 (10.31)	1.543 (10.14)	1.563 (10.15)	1.586 (10.35)	1.602 (10.49)	1.613 (10.33)
30%	1.269 (15.65)	1.274 (14.65)	1.279 (13.63)	1.314 (15.67)	1.338 (16.82)	1.623 (15.80)	1.629 (14.83)	1.636 (13.82)	1.679 (15.82)	1.710 (16.97)
40%	1.391 (26.69)	1.390 (25.05)	1.369 (21.59)	1.404 (23.60)	1.436 (25.38)	1.778 (26.87)	1.777 (25.26)	1.751 (21.84)	1.795 (23.83)	1.836 (25.58)
Imputation using propensity−score method										
10%	1.158 (5.53)	1.159 (4.33)	1.168 (3.80)	1.187 (4.54)	1.196 (4.48)	1.479 (5.55)	1.480 (4.35)	1.492 (3.81)	1.516 (4.55)	1.528 (4.50)
20%	1.216 (10.79)	1.233 (10.96)	1.248 (10.85)	1.257 (10.70)	1.277 (11.56)	1.552 (10.77)	1.574 (10.95)	1.593 (10.84)	1.605 (10.68)	1.631 (11.54)
30%	1.280 (16.63)	1.291 (16.19)	1.315 (16.80)	1.325 (16.66)	1.355 (18.30)	1.637 (16.85)	1.651 (16.40)	1.681 (16.99)	1.694 (16.86)	1.732 (18.49)
40%	1.418 (29.20)	1.399 (25.89)	1.404 (24.76)	1.453 (27.90)	1.460 (27.51)	1.813 (29.40)	1.790 (26.17)	1.797 (25.02)	1.857 (28.11)	1.867 (27.75)

SampleSize_n250_5Treatments/table3_shrink_5_treatments_MAR_n250

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.111	1.125	1.136	1.137	1.147	1.421	1.425	1.433	1.432	1.440
Imputation using linear regression										
10%	1.156 (3.99)	1.165 (3.47)	1.185 (4.35)	1.183 (4.00)	1.211 (5.57)	1.483 (4.43)	1.490 (4.56)	1.500 (4.69)	1.500 (4.74)	1.524 (5.85)
20%	1.230 (10.66)	1.220 (8.40)	1.242 (9.35)	1.248 (9.72)	1.259 (9.76)	1.569 (10.46)	1.557 (9.29)	1.583 (10.45)	1.574 (9.95)	1.589 (10.39)
30%	1.317 (18.46)	1.336 (18.74)	1.363 (19.96)	1.372 (20.61)	1.386 (20.83)	1.686 (18.70)	1.707 (19.76)	1.729 (20.63)	1.736 (21.26)	1.752 (21.68)
40%	1.439 (29.48)	1.433 (27.28)	1.465 (28.99)	1.454 (27.84)	1.528 (33.26)	1.840 (29.49)	1.828 (28.28)	1.864 (30.05)	1.850 (29.17)	1.937 (34.52)
Imputation using predictive mean matching method										
10%	1.144 (2.97)	1.157 (2.80)	1.177 (3.66)	1.175 (3.32)	1.203 (4.89)	1.469 (3.43)	1.481 (3.91)	1.491 (4.05)	1.491 (4.11)	1.515 (5.25)
20%	1.217 (9.49)	1.218 (8.21)	1.221 (7.54)	1.240 (9.05)	1.240 (8.14)	1.551 (9.20)	1.553 (8.95)	1.556 (8.61)	1.563 (9.15)	1.567 (8.83)
30%	1.283 (15.40)	1.302 (15.68)	1.315 (15.74)	1.315 (15.66)	1.325 (15.57)	1.646 (15.89)	1.664 (16.80)	1.670 (16.54)	1.667 (16.45)	1.677 (16.47)
40%	1.398 (25.76)	1.407 (25.01)	1.418 (24.86)	1.423 (25.10)	1.468 (27.97)	1.779 (25.26)	1.789 (25.53)	1.800 (25.62)	1.803 (25.89)	1.852 (28.62)
Imputation using propensity−score method										
10%	1.149 (3.40)	1.162 (3.20)	1.183 (4.18)	1.182 (3.93)	1.207 (5.22)	1.475 (3.84)	1.486 (4.31)	1.499 (4.59)	1.499 (4.68)	1.520 (5.56)
20%	1.218 (9.59)	1.223 (8.69)	1.223 (7.64)	1.245 (9.50)	1.242 (8.30)	1.555 (9.46)	1.560 (9.47)	1.559 (8.82)	1.570 (9.67)	1.569 (9.02)
30%	1.287 (15.82)	1.304 (15.84)	1.343 (18.24)	1.333 (17.26)	1.360 (18.62)	1.651 (16.19)	1.666 (16.91)	1.705 (18.99)	1.691 (18.07)	1.720 (19.50)
40%	1.422 (27.95)	1.458 (29.51)	1.474 (29.80)	1.457 (28.11)	1.495 (30.34)	1.816 (27.85)	1.857 (30.30)	1.875 (30.85)	1.852 (29.33)	1.893 (31.51)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.098	1.111	1.126	1.136	1.145	1.414	1.418	1.424	1.429	1.429
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.163 (6.00)	1.168 (5.08)	1.176 (4.50)	1.191 (4.90)	1.205 (5.24)	1.486 (5.09)	1.484 (4.66)	1.492 (4.77)	1.504 (5.25)	1.514 (5.94)
20%	1.214 (10.63)	1.238 (11.38)	1.248 (10.90)	1.262 (11.11)	1.280 (11.81)	1.560 (10.31)	1.578 (11.28)	1.585 (11.28)	1.600 (11.95)	1.614 (12.93)
30%	1.302 (18.58)	1.303 (17.27)	1.317 (16.96)	1.339 (17.89)	1.363 (19.07)	1.666 (17.85)	1.665 (17.39)	1.678 (17.85)	1.701 (19.02)	1.723 (20.55)
40%	1.428 (30.10)	1.438 (29.35)	1.429 (26.90)	1.467 (29.14)	1.454 (26.97)	1.829 (29.36)	1.837 (29.49)	1.825 (28.18)	1.868 (30.66)	1.850 (29.43)
Imputation using predictive mean matching method										
10%	1.159 (5.61)	1.167 (5.03)	1.173 (4.19)	1.185 (4.35)	1.197 (4.52)	1.481 (4.72)	1.483 (4.58)	1.488 (4.46)	1.497 (4.72)	1.504 (5.26)
20%	1.209 (10.13)	1.224 (10.12)	1.242 (10.33)	1.255 (10.48)	1.263 (10.31)	1.552 (9.76)	1.561 (10.03)	1.576 (10.69)	1.590 (11.27)	1.591 (11.34)
30%	1.269 (15.65)	1.274 (14.65)	1.279 (13.63)	1.314 (15.67)	1.338 (16.82)	1.626 (15.03)	1.629 (14.82)	1.630 (14.45)	1.668 (16.71)	1.688 (18.14)
40%	1.391 (26.69)	1.390 (25.05)	1.369 (21.59)	1.404 (23.60)	1.436 (25.38)	1.783 (26.11)	1.777 (25.27)	1.752 (23.05)	1.788 (25.13)	1.826 (27.75)
Imputation using propensity−score method										
10%	1.158 (5.53)	1.159 (4.33)	1.168 (3.80)	1.187 (4.54)	1.196 (4.48)	1.480 (4.68)	1.474 (3.94)	1.483 (4.14)	1.499 (4.89)	1.504 (5.23)
20%	1.216 (10.79)	1.233 (10.96)	1.248 (10.85)	1.257 (10.70)	1.277 (11.56)	1.560 (10.36)	1.572 (10.82)	1.584 (11.23)	1.593 (11.48)	1.609 (12.62)
30%	1.280 (16.63)	1.291 (16.19)	1.315 (16.80)	1.325 (16.66)	1.355 (18.30)	1.641 (16.03)	1.651 (16.37)	1.676 (17.69)	1.684 (17.85)	1.712 (19.77)
40%	1.418 (29.20)	1.399 (25.89)	1.404 (24.76)	1.453 (27.90)	1.460 (27.51)	1.817 (28.52)	1.790 (26.19)	1.797 (26.19)	1.850 (29.41)	1.858 (29.98)

SampleSize_n300_5Treatments/table2_nonshrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	96.43	100.00	92.86	92.86	82.14	100.00	100.00	100.00	92.86	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	100.00	100.00	96.55	96.55	93.10	100.00	100.00	100.00	100.00	100
30%	89.29	96.43	96.43	96.43	96.43	78.57	96.43	100.00	100.00	100.00	100
40%	89.66	96.55	96.55	96.55	96.55	75.86	96.55	100.00	100.00	100.00	100
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	96.55	100
30%	89.29	92.86	96.43	92.86	92.86	71.43	92.86	100.00	96.43	100.00	100
40%	86.21	86.21	93.10	96.55	100.00	68.97	96.55	96.55	100.00	100.00	100
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	100.00	100
30%	92.86	96.43	92.86	92.86	92.86	78.57	96.43	100.00	96.43	100.00	100
40%	89.66	96.55	96.55	96.55	100.00	79.31	96.55	96.55	100.00	100.00	100
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	96.55	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_5Treatments/table2_nonshrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	86.21	93.10	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	85.71	89.29	85.71	89.29	92.86	67.86	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	82.14	89.29	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	96.55	89.66	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	79.31	93.10	96.55	96.55	96.55	93.10
20%	82.14	92.86	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	96.43	96.43	96.43	92.86	82.14	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	92.86
Imputation using linear regression	NA
10%	100.00
20%	100.00
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	92.86
40%	93.10
Imputation using propensity−score method	NA
10%	100.00
20%	100.00
30%	96.43
40%	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using propensity−score method	NA
10%	86.21
20%	82.14
30%	96.43
40%	100.00

SampleSize_n300_5Treatments/table2_shrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.29	96.43	100.00	92.86	92.86	82.14	100.00	100.00	100.00	92.86	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	100.00	100.00	96.55	96.55	93.10	100.00	100.00	100.00	100.00	100
30%	89.29	96.43	96.43	96.43	96.43	78.57	96.43	100.00	100.00	100.00	100
40%	89.66	96.55	96.55	96.55	96.55	75.86	96.55	100.00	100.00	100.00	100
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	96.55	100
30%	89.29	92.86	96.43	92.86	92.86	71.43	92.86	100.00	96.43	100.00	100
40%	86.21	86.21	93.10	96.55	100.00	68.97	96.55	96.55	100.00	100.00	100
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	96.55	100.00	100.00	100.00	96.55	96.55	100.00	100.00	100.00	100.00	100
20%	96.55	96.55	96.55	96.55	100.00	86.21	100.00	100.00	100.00	100.00	100
30%	92.86	96.43	92.86	92.86	92.86	78.57	96.43	100.00	96.43	100.00	100
40%	89.66	96.55	96.55	96.55	100.00	79.31	96.55	96.55	96.55	100.00	100
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	100.00	96.55	93.10	96.55	82.76	100.00	100.00	100.00	96.55	100
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n300_5Treatments/table2_shrink_5_treatments_MAR_n300

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	86.21	93.10	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	85.71	89.29	85.71	89.29	92.86	67.86	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	96.55	89.66	86.21	79.31	93.10	96.55	96.55	96.55	96.55
20%	82.14	89.29	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	100.00	96.43	96.43	96.43	85.71	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	96.55	89.66	82.76	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	93.10	96.55	89.66	82.76	79.31	93.10	96.55	96.55	93.10	93.10
20%	82.14	92.86	85.71	89.29	92.86	60.71	92.86	92.86	92.86	96.43	96.43
30%	96.43	96.43	96.43	96.43	92.86	82.14	100.00	100.00	96.43	100.00	100.00
40%	100.00	100.00	96.55	100.00	89.66	86.21	100.00	100.00	100.00	100.00	100.00

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	92.86
Imputation using linear regression	NA
10%	100.00
20%	100.00
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	100.00
20%	96.55
30%	92.86
40%	93.10
Imputation using propensity−score method	NA
10%	100.00
20%	100.00
30%	96.43
40%	89.66
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using predictive mean matching method	NA
10%	89.66
20%	82.14
30%	96.43
40%	96.55
Imputation using propensity−score method	NA
10%	86.21
20%	82.14
30%	96.43
40%	100.00

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.025	1.034	1.036	1.053	1.056	1.307	1.319	1.321	1.343	1.347
Imputation using linear regression										
10%	1.051 (2.53)	1.062 (2.72)	1.079 (4.17)	1.090 (3.53)	1.098 (3.94)	1.342 (2.66)	1.357 (2.87)	1.378 (4.32)	1.392 (3.67)	1.402 (4.08)
20%	1.105 (7.85)	1.127 (8.93)	1.129 (8.98)	1.134 (7.66)	1.164 (10.14)	1.413 (8.12)	1.440 (9.20)	1.443 (9.26)	1.449 (7.92)	1.487 (10.41)
30%	1.183 (15.40)	1.212 (17.18)	1.213 (17.14)	1.238 (17.58)	1.247 (18.00)	1.513 (15.75)	1.550 (17.49)	1.552 (17.48)	1.583 (17.89)	1.594 (18.34)
40%	1.289 (25.79)	1.305 (26.20)	1.314 (26.83)	1.337 (26.97)	1.335 (26.36)	1.650 (26.23)	1.670 (26.63)	1.681 (27.27)	1.711 (27.37)	1.708 (26.81)
Imputation using predictive mean matching method										
10%	1.045 (1.97)	1.063 (2.75)	1.073 (3.63)	1.087 (3.22)	1.094 (3.55)	1.334 (2.08)	1.357 (2.87)	1.370 (3.75)	1.388 (3.33)	1.397 (3.66)
20%	1.091 (6.44)	1.114 (7.66)	1.118 (7.92)	1.125 (6.80)	1.148 (8.71)	1.395 (6.70)	1.423 (7.91)	1.429 (8.18)	1.438 (7.05)	1.468 (8.97)
30%	1.155 (12.67)	1.160 (12.13)	1.175 (13.46)	1.196 (13.53)	1.211 (14.59)	1.477 (13.03)	1.484 (12.49)	1.503 (13.82)	1.529 (13.86)	1.548 (14.93)
40%	1.245 (21.45)	1.246 (20.42)	1.267 (22.30)	1.289 (22.43)	1.296 (22.69)	1.594 (21.97)	1.595 (20.96)	1.623 (22.83)	1.651 (22.93)	1.660 (23.21)
Imputation using propensity−score method										
10%	1.044 (1.86)	1.063 (2.77)	1.078 (4.06)	1.090 (3.46)	1.098 (3.89)	1.333 (2.00)	1.357 (2.91)	1.376 (4.19)	1.391 (3.60)	1.401 (4.02)
20%	1.100 (7.31)	1.120 (8.26)	1.120 (8.12)	1.134 (7.71)	1.156 (9.40)	1.406 (7.53)	1.431 (8.47)	1.431 (8.35)	1.449 (7.92)	1.477 (9.63)
30%	1.164 (13.58)	1.183 (14.41)	1.195 (15.36)	1.203 (14.25)	1.222 (15.65)	1.489 (13.93)	1.513 (14.73)	1.528 (15.70)	1.539 (14.58)	1.563 (15.99)
40%	1.295 (26.36)	1.290 (24.70)	1.303 (25.84)	1.325 (25.81)	1.327 (25.57)	1.655 (26.64)	1.649 (25.00)	1.666 (26.16)	1.693 (26.09)	1.696 (25.88)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.014	1.017	1.029	1.037	1.048	1.294	1.298	1.314	1.323	1.337
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.055 (4.02)	1.068 (5.02)	1.079 (4.88)	1.092 (5.38)	1.094 (4.43)	1.345 (3.93)	1.362 (4.93)	1.377 (4.79)	1.393 (5.29)	1.395 (4.34)
20%	1.112 (9.68)	1.125 (10.66)	1.135 (10.28)	1.140 (9.97)	1.149 (9.70)	1.421 (9.79)	1.438 (10.77)	1.450 (10.40)	1.457 (10.08)	1.469 (9.81)
30%	1.194 (17.71)	1.209 (18.94)	1.225 (18.98)	1.225 (18.18)	1.224 (16.86)	1.526 (17.86)	1.546 (19.09)	1.565 (19.14)	1.566 (18.35)	1.565 (17.05)
40%	1.286 (26.79)	1.296 (27.53)	1.303 (26.55)	1.308 (26.18)	1.324 (26.34)	1.644 (27.03)	1.658 (27.79)	1.666 (26.81)	1.673 (26.46)	1.693 (26.59)
Imputation using predictive mean matching method										
10%	1.050 (3.54)	1.061 (4.38)	1.076 (4.56)	1.086 (4.76)	1.089 (3.91)	1.340 (3.49)	1.354 (4.34)	1.373 (4.51)	1.386 (4.71)	1.389 (3.87)
20%	1.105 (8.94)	1.113 (9.46)	1.123 (9.10)	1.125 (8.52)	1.142 (9.01)	1.411 (8.98)	1.421 (9.50)	1.434 (9.15)	1.437 (8.57)	1.458 (9.05)
30%	1.181 (16.44)	1.189 (16.90)	1.199 (16.46)	1.215 (17.24)	1.218 (16.29)	1.507 (16.38)	1.517 (16.87)	1.530 (16.43)	1.550 (17.18)	1.555 (16.24)
40%	1.255 (23.80)	1.267 (24.61)	1.281 (24.44)	1.285 (23.99)	1.271 (21.33)	1.605 (23.98)	1.620 (24.82)	1.638 (24.65)	1.644 (24.21)	1.626 (21.60)
Imputation using propensity−score method										
10%	1.050 (3.59)	1.061 (4.38)	1.074 (4.35)	1.085 (4.67)	1.086 (3.71)	1.340 (3.55)	1.354 (4.34)	1.370 (4.31)	1.384 (4.63)	1.386 (3.66)
20%	1.110 (9.46)	1.122 (10.32)	1.139 (10.70)	1.142 (10.22)	1.156 (10.33)	1.417 (9.48)	1.432 (10.35)	1.455 (10.72)	1.459 (10.23)	1.476 (10.35)
30%	1.179 (16.28)	1.197 (17.74)	1.207 (17.24)	1.230 (18.64)	1.234 (17.76)	1.506 (16.37)	1.529 (17.85)	1.542 (17.35)	1.571 (18.71)	1.576 (17.84)
40%	1.272 (25.45)	1.288 (26.68)	1.313 (27.54)	1.292 (24.67)	1.290 (23.16)	1.625 (25.56)	1.646 (26.80)	1.677 (27.64)	1.652 (24.84)	1.650 (23.34)

SampleSize_n300_5Treatments/table3_shrink_5_treatments_MAR_n300

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	1.025	1.034	1.036	1.053	1.056	1.307	1.312	1.313	1.318	1.320
Imputation using linear regression										
10%	1.051 (2.53)	1.062 (2.72)	1.079 (4.17)	1.090 (3.53)	1.098 (3.94)	1.351 (3.33)	1.354 (3.22)	1.369 (4.21)	1.376 (4.38)	1.384 (4.82)
20%	1.105 (7.85)	1.127 (8.93)	1.129 (8.98)	1.134 (7.66)	1.164 (10.14)	1.419 (8.51)	1.435 (9.43)	1.436 (9.36)	1.439 (9.23)	1.466 (11.05)
30%	1.183 (15.40)	1.212 (17.18)	1.213 (17.14)	1.238 (17.58)	1.247 (18.00)	1.518 (16.14)	1.549 (18.08)	1.547 (17.80)	1.576 (19.56)	1.582 (19.80)
40%	1.289 (25.79)	1.305 (26.20)	1.314 (26.83)	1.337 (26.97)	1.335 (26.36)	1.653 (26.45)	1.670 (27.34)	1.676 (27.63)	1.704 (29.27)	1.697 (28.52)
Imputation using predictive mean matching method										
10%	1.045 (1.97)	1.063 (2.75)	1.073 (3.63)	1.087 (3.22)	1.094 (3.55)	1.343 (2.72)	1.355 (3.27)	1.363 (3.77)	1.371 (4.05)	1.378 (4.39)
20%	1.091 (6.44)	1.114 (7.66)	1.118 (7.92)	1.125 (6.80)	1.148 (8.71)	1.400 (7.04)	1.418 (8.10)	1.421 (8.18)	1.427 (8.26)	1.447 (9.61)
30%	1.155 (12.67)	1.160 (12.13)	1.175 (13.46)	1.196 (13.53)	1.211 (14.59)	1.484 (13.48)	1.485 (13.20)	1.500 (14.21)	1.522 (15.50)	1.536 (16.36)
40%	1.245 (21.45)	1.246 (20.42)	1.267 (22.30)	1.289 (22.43)	1.296 (22.69)	1.599 (22.27)	1.596 (21.72)	1.617 (23.15)	1.645 (24.79)	1.647 (24.77)
Imputation using propensity−score method										
10%	1.044 (1.86)	1.063 (2.77)	1.078 (4.06)	1.090 (3.46)	1.098 (3.89)	1.343 (2.68)	1.355 (3.31)	1.368 (4.20)	1.375 (4.36)	1.383 (4.79)
20%	1.100 (7.31)	1.120 (8.26)	1.120 (8.12)	1.134 (7.71)	1.156 (9.40)	1.410 (7.87)	1.425 (8.65)	1.424 (8.41)	1.439 (9.21)	1.456 (10.32)
30%	1.164 (13.58)	1.183 (14.41)	1.195 (15.36)	1.203 (14.25)	1.222 (15.65)	1.496 (14.39)	1.513 (15.35)	1.524 (16.02)	1.532 (16.28)	1.550 (17.44)
40%	1.295 (26.36)	1.290 (24.70)	1.303 (25.84)	1.325 (25.81)	1.327 (25.57)	1.659 (26.86)	1.649 (25.75)	1.661 (26.48)	1.687 (28.05)	1.685 (27.64)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	1.014	1.017	1.029	1.037	1.048	1.297	1.300	1.302	1.305	1.312
Imputation using linear regression										

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	1.055 (4.02)	1.068 (5.02)	1.079 (4.88)	1.092 (5.38)	1.094 (4.43)	1.346 (3.79)	1.357 (4.43)	1.366 (4.93)	1.375 (5.34)	1.378 (5.03)
20%	1.112 (9.68)	1.125 (10.66)	1.135 (10.28)	1.140 (9.97)	1.149 (9.70)	1.424 (9.81)	1.433 (10.26)	1.442 (10.73)	1.446 (10.78)	1.449 (10.43)
30%	1.194 (17.71)	1.209 (18.94)	1.225 (18.98)	1.225 (18.18)	1.224 (16.86)	1.529 (17.93)	1.542 (18.67)	1.557 (19.59)	1.555 (19.18)	1.555 (18.47)
40%	1.286 (26.79)	1.296 (27.53)	1.303 (26.55)	1.308 (26.18)	1.324 (26.34)	1.648 (27.09)	1.656 (27.39)	1.660 (27.50)	1.664 (27.52)	1.682 (28.20)
Imputation using predictive mean matching method										
10%	1.050 (3.54)	1.061 (4.38)	1.076 (4.56)	1.086 (4.76)	1.089 (3.91)	1.340 (3.35)	1.350 (3.86)	1.362 (4.63)	1.367 (4.76)	1.372 (4.54)
20%	1.105 (8.94)	1.113 (9.46)	1.123 (9.10)	1.125 (8.52)	1.142 (9.01)	1.413 (8.97)	1.417 (8.99)	1.425 (9.44)	1.426 (9.24)	1.439 (9.67)
30%	1.181 (16.44)	1.189 (16.90)	1.199 (16.46)	1.215 (17.24)	1.218 (16.29)	1.511 (16.53)	1.513 (16.39)	1.522 (16.88)	1.540 (18.00)	1.546 (17.78)
40%	1.255 (23.80)	1.267 (24.61)	1.281 (24.44)	1.285 (23.99)	1.271 (21.33)	1.609 (24.08)	1.619 (24.54)	1.631 (25.30)	1.635 (25.26)	1.617 (23.25)
Imputation using propensity−score method										
10%	1.050 (3.59)	1.061 (4.38)	1.074 (4.35)	1.085 (4.67)	1.086 (3.71)	1.341 (3.41)	1.350 (3.91)	1.360 (4.47)	1.366 (4.67)	1.369 (4.32)
20%	1.110 (9.46)	1.122 (10.32)	1.139 (10.70)	1.142 (10.22)	1.156 (10.33)	1.419 (9.45)	1.427 (9.78)	1.445 (10.99)	1.448 (10.96)	1.457 (11.03)
30%	1.179 (16.28)	1.197 (17.74)	1.207 (17.24)	1.230 (18.64)	1.234 (17.76)	1.510 (16.44)	1.526 (17.42)	1.534 (17.84)	1.560 (19.55)	1.566 (19.37)
40%	1.272 (25.45)	1.288 (26.68)	1.313 (27.54)	1.292 (24.67)	1.290 (23.16)	1.629 (25.62)	1.643 (26.41)	1.670 (28.30)	1.643 (25.92)	1.639 (24.91)

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	93.10	93.10	100.00	100.00	82.76	96.55	96.55	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	82.76	89.66	93.10	96.55	93.10	68.97	93.10	96.55	100.00	96.55	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	93.10	89.66	93.10	93.10	72.41	100.00	96.55	100.00	100.00	96.55
40%	96.55	100.00	96.55	100.00	93.10	86.21	100.00	100.00	100.00	100.00	96.55
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	93.10	96.55	93.10	72.41	93.10	96.55	96.55	100.00	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	96.55	86.21	93.10	82.76	68.97	100.00	96.55	93.10	100.00	93.10
40%	93.10	96.55	89.66	93.10	89.66	68.97	100.00	100.00	100.00	100.00	100.00
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	93.10	96.55	93.10	72.41	93.10	96.55	96.55	96.55	100.00
20%	96.43	96.43	96.43	92.86	96.43	82.14	100.00	96.43	100.00	96.43	100.00
30%	96.55	89.66	86.21	93.10	86.21	58.62	100.00	96.55	96.55	100.00	93.10
40%	93.10	96.55	89.66	96.55	89.66	79.31	100.00	100.00	100.00	100.00	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	93.10	96.55	100.00	93.10	82.76	96.55	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_5Treatments/table2_nonshrink_5_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	100.00	96.43	100.00	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	89.29	96.43	96.43	85.71	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	96.55	82.76	96.55	100.00	100.00	96.55	100.00
40%	86.21	89.66	100.00	89.66	96.55	79.31	100.00	96.55	100.00	93.10	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	100.00	96.43	85.71	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	85.71	96.43	96.43	82.14	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	82.76	72.41	96.55	96.55	100.00	96.55	100.00
40%	82.76	86.21	93.10	82.76	93.10	79.31	93.10	89.66	100.00	89.66	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	96.43	96.43	89.29	96.43	96.43	85.71	96.43	100.00	92.86	100.00	100.00
30%	89.66	96.55	93.10	96.55	86.21	75.86	96.55	96.55	100.00	96.55	100.00
40%	82.76	89.66	100.00	89.66	89.66	82.76	93.10	96.55	100.00	93.10	96.55

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.10
Imputation using linear regression	NA
10%	86.21
20%	96.43
30%	93.10
40%	96.55
Imputation using predictive mean matching method	NA
10%	86.21
20%	96.43
30%	82.76
40%	100.00
Imputation using propensity−score method	NA
10%	82.76
20%	92.86
30%	86.21
40%	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	100.00
20%	92.86
30%	96.55
40%	89.66
Imputation using predictive mean matching method	NA
10%	100.00
20%	92.86
30%	93.10
40%	89.66
Imputation using propensity−score method	NA
10%	100.00
20%	89.29
30%	93.10
40%	89.66

SampleSize_n50_5Treatments/table2_shrink_5_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	96.55	93.10	93.10	100.00	100.00	82.76	96.55	96.55	100.00	100.00	100.00
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	82.76	89.66	93.10	96.55	93.10	68.97	96.55	96.55	100.00	100.00	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	93.10	89.66	93.10	93.10	72.41	100.00	96.55	93.10	100.00	96.55
40%	96.55	100.00	96.55	100.00	93.10	86.21	100.00	100.00	100.00	100.00	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	89.66	93.10	96.55	93.10	72.41	93.10	96.55	96.55	100.00	100.00
20%	100.00	96.43	96.43	92.86	96.43	85.71	100.00	96.43	100.00	100.00	100.00
30%	100.00	96.55	86.21	93.10	82.76	68.97	100.00	96.55	93.10	100.00	93.10
40%	93.10	96.55	89.66	93.10	89.66	68.97	100.00	100.00	100.00	100.00	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	86.21	82.76	93.10	96.55	93.10	72.41	93.10	96.55	96.55	100.00	100.00
20%	96.43	96.43	96.43	92.86	96.43	82.14	100.00	96.43	100.00	96.43	100.00
30%	96.55	89.66	86.21	93.10	86.21	58.62	100.00	96.55	96.55	100.00	93.10
40%	93.10	96.55	89.66	96.55	89.66	79.31	100.00	100.00	100.00	100.00	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	89.66	93.10	96.55	100.00	93.10	82.76	96.55	100.00	100.00	100.00	96.55
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

SampleSize_n50_5Treatments/table2_shrink_5_treatments_MAR_n50

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	96.43	100.00	96.43	100.00	96.43	89.29	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	89.29	96.43	96.43	85.71	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	96.55	82.76	96.55	100.00	100.00	96.55	100.00
40%	86.21	89.66	100.00	89.66	96.55	79.31	100.00	96.55	100.00	93.10	100.00
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	96.43	100.00	96.43	85.71	100.00	100.00	100.00	100.00	100.00
20%	96.43	100.00	85.71	96.43	96.43	82.14	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	82.76	72.41	96.55	96.55	100.00	96.55	100.00
40%	82.76	86.21	93.10	82.76	93.10	79.31	96.55	93.10	100.00	89.66	96.55
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.86	100.00	92.86	100.00	96.43	82.14	100.00	100.00	100.00	100.00	100.00
20%	96.43	96.43	89.29	96.43	96.43	85.71	96.43	100.00	96.43	100.00	100.00
30%	89.66	96.55	93.10	96.55	86.21	75.86	93.10	96.55	96.55	96.55	100.00
40%	82.76	89.66	100.00	89.66	89.66	82.76	96.55	96.55	100.00	93.10	96.55

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.10
Imputation using linear regression	NA
10%	93.10
20%	96.43
30%	86.21
40%	100.00
Imputation using predictive mean matching method	NA
10%	86.21
20%	96.43
30%	82.76
40%	96.55
Imputation using propensity−score method	NA
10%	86.21
20%	92.86
30%	86.21
40%	93.10
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	96.55
Imputation using linear regression	NA

MR	Joint_many
10%	100.00
20%	92.86
30%	96.55
40%	89.66
Imputation using predictive mean matching method	NA
10%	100.00
20%	92.86
30%	93.10
40%	89.66
Imputation using propensity−score method	NA
10%	100.00
20%	92.86
30%	89.66
40%	89.66

SampleSize_n50_5Treatments/table3_nonshrink_5_treatments_MAR_n50

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	2.519	2.502	2.547	2.551	2.593	3.199	3.178	3.235	3.240	3.293
Imputation using linear regression										
10%	2.628 (4.34)	2.621 (4.77)	2.654 (4.20)	2.719 (6.58)	2.710 (4.50)	3.356 (4.90)	3.347 (5.32)	3.389 (4.75)	3.471 (7.13)	3.460 (5.06)
20%	2.836 (12.58)	2.819 (12.70)	2.906 (14.09)	2.888 (13.22)	2.914 (12.37)	3.618 (13.07)	3.597 (13.20)	3.707 (14.59)	3.685 (13.73)	3.718 (12.89)
30%	2.872 (14.02)	3.000 (19.90)	2.981 (17.03)	3.089 (21.08)	3.038 (17.19)	3.672 (14.76)	3.834 (20.67)	3.811 (17.79)	3.950 (21.92)	3.885 (17.95)
40%	3.358 (33.29)	3.261 (30.35)	3.255 (27.79)	3.431 (34.48)	3.412 (31.59)	4.301 (34.42)	4.181 (31.58)	4.173 (28.98)	4.397 (35.69)	4.370 (32.70)
Imputation using predictive mean matching method										
10%	2.573 (2.16)	2.600 (3.94)	2.612 (2.56)	2.674 (4.83)	2.661 (2.63)	3.286 (2.69)	3.320 (4.47)	3.335 (3.08)	3.414 (5.35)	3.397 (3.15)
20%	2.721 (8.04)	2.699 (7.87)	2.820 (10.72)	2.792 (9.47)	2.806 (8.23)	3.471 (8.48)	3.442 (8.33)	3.596 (11.16)	3.562 (9.93)	3.580 (8.71)
30%	2.683 (6.52)	2.784 (11.28)	2.802 (10.00)	2.884 (13.08)	2.811 (8.41)	3.429 (7.19)	3.559 (12.01)	3.582 (10.72)	3.689 (13.87)	3.595 (9.16)
40%	2.982 (18.39)	2.964 (18.49)	3.017 (18.45)	3.020 (18.39)	3.043 (17.35)	3.815 (19.23)	3.794 (19.39)	3.859 (19.27)	3.866 (19.32)	3.892 (18.19)
Imputation using propensity−score method										
10%	2.586 (2.65)	2.591 (3.55)	2.618 (2.78)	2.678 (5.00)	2.664 (2.74)	3.302 (3.22)	3.309 (4.13)	3.344 (3.35)	3.420 (5.56)	3.402 (3.30)
20%	2.746 (9.02)	2.707 (8.22)	2.815 (10.52)	2.800 (9.77)	2.838 (9.44)	3.502 (9.47)	3.454 (8.69)	3.591 (10.99)	3.572 (10.23)	3.620 (9.91)
30%	2.679 (6.35)	2.811 (12.35)	2.849 (11.85)	2.951 (15.70)	2.855 (10.10)	3.428 (7.14)	3.596 (13.17)	3.644 (12.64)	3.776 (16.55)	3.653 (10.93)
40%	3.041 (20.74)	3.034 (21.29)	3.063 (20.27)	3.166 (24.12)	3.190 (23.04)	3.890 (21.59)	3.882 (22.16)	3.918 (21.11)	4.051 (25.02)	4.079 (23.86)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	2.477	2.539	2.550	2.584	2.582	3.134	3.213	3.227	3.269	3.267
Imputation using linear regression										

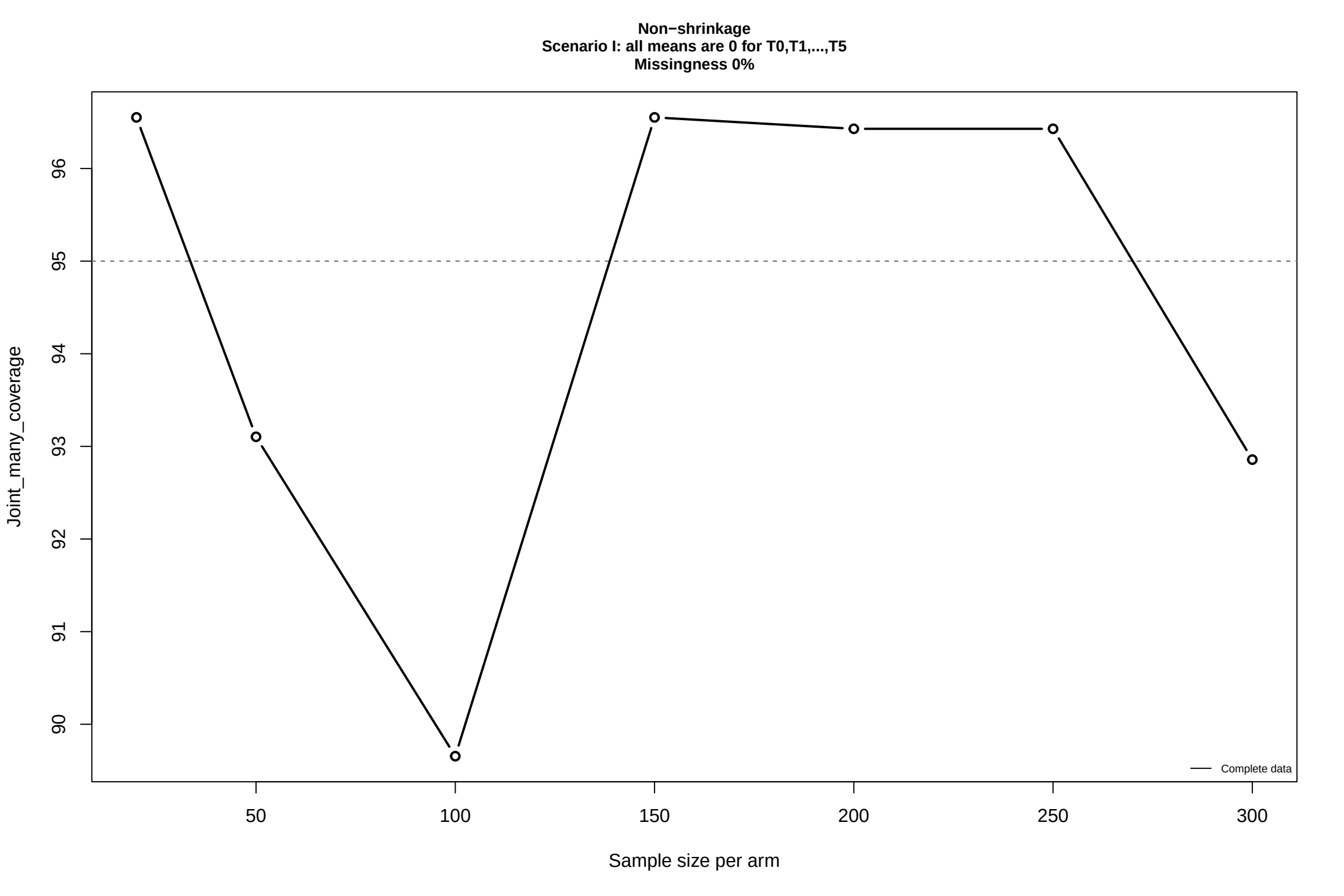
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	2.642 (6.65)	2.702 (6.43)	2.715 (6.47)	2.729 (5.59)	2.749 (6.49)	3.376 (7.72)	3.453 (7.49)	3.470 (7.52)	3.486 (6.63)	3.513 (7.55)
20%	2.785 (12.44)	2.858 (12.56)	2.759 (8.18)	2.853 (10.39)	2.885 (11.73)	3.545 (13.12)	3.638 (13.25)	3.513 (8.86)	3.632 (11.08)	3.673 (12.45)
30%	2.902 (17.15)	3.011 (18.57)	3.018 (18.32)	3.003 (16.22)	3.074 (19.07)	3.703 (18.18)	3.841 (19.57)	3.852 (19.38)	3.832 (17.20)	3.923 (20.09)
40%	3.258 (31.53)	3.331 (31.19)	3.418 (34.03)	3.494 (35.22)	3.409 (32.03)	4.158 (32.69)	4.250 (32.28)	4.362 (35.18)	4.460 (36.41)	4.350 (33.15)
Imputation using predictive mean matching method										
10%	2.587 (4.45)	2.654 (4.50)	2.652 (3.98)	2.696 (4.34)	2.676 (3.64)	3.306 (5.49)	3.391 (5.54)	3.388 (5.00)	3.445 (5.35)	3.419 (4.68)
20%	2.656 (7.21)	2.736 (7.75)	2.694 (5.61)	2.783 (7.71)	2.791 (8.09)	3.380 (7.88)	3.482 (8.39)	3.428 (6.23)	3.543 (8.35)	3.552 (8.74)
30%	2.709 (9.34)	2.769 (9.06)	2.809 (10.14)	2.759 (6.77)	2.859 (10.75)	3.454 (10.24)	3.532 (9.94)	3.584 (11.07)	3.520 (7.65)	3.647 (11.65)
40%	2.943 (18.80)	2.988 (17.68)	3.024 (18.55)	3.194 (23.61)	3.018 (16.89)	3.756 (19.87)	3.815 (18.74)	3.862 (19.67)	4.075 (24.65)	3.853 (17.95)
Imputation using propensity−score method										
10%	2.611 (5.41)	2.669 (5.10)	2.677 (4.95)	2.696 (4.33)	2.704 (4.73)	3.336 (6.46)	3.410 (6.14)	3.420 (5.98)	3.444 (5.35)	3.455 (5.77)
20%	2.718 (9.73)	2.760 (8.71)	2.701 (5.90)	2.820 (9.12)	2.794 (8.22)	3.460 (10.43)	3.516 (9.43)	3.439 (6.58)	3.591 (9.82)	3.559 (8.96)
30%	2.747 (10.88)	2.802 (10.34)	2.880 (12.91)	2.823 (9.26)	2.942 (13.96)	3.504 (11.83)	3.575 (11.27)	3.674 (13.85)	3.602 (10.17)	3.752 (14.87)
40%	2.981 (20.34)	3.019 (18.89)	3.069 (20.32)	3.177 (22.95)	3.033 (17.46)	3.813 (21.68)	3.862 (20.22)	3.926 (21.67)	4.065 (24.32)	3.881 (18.80)

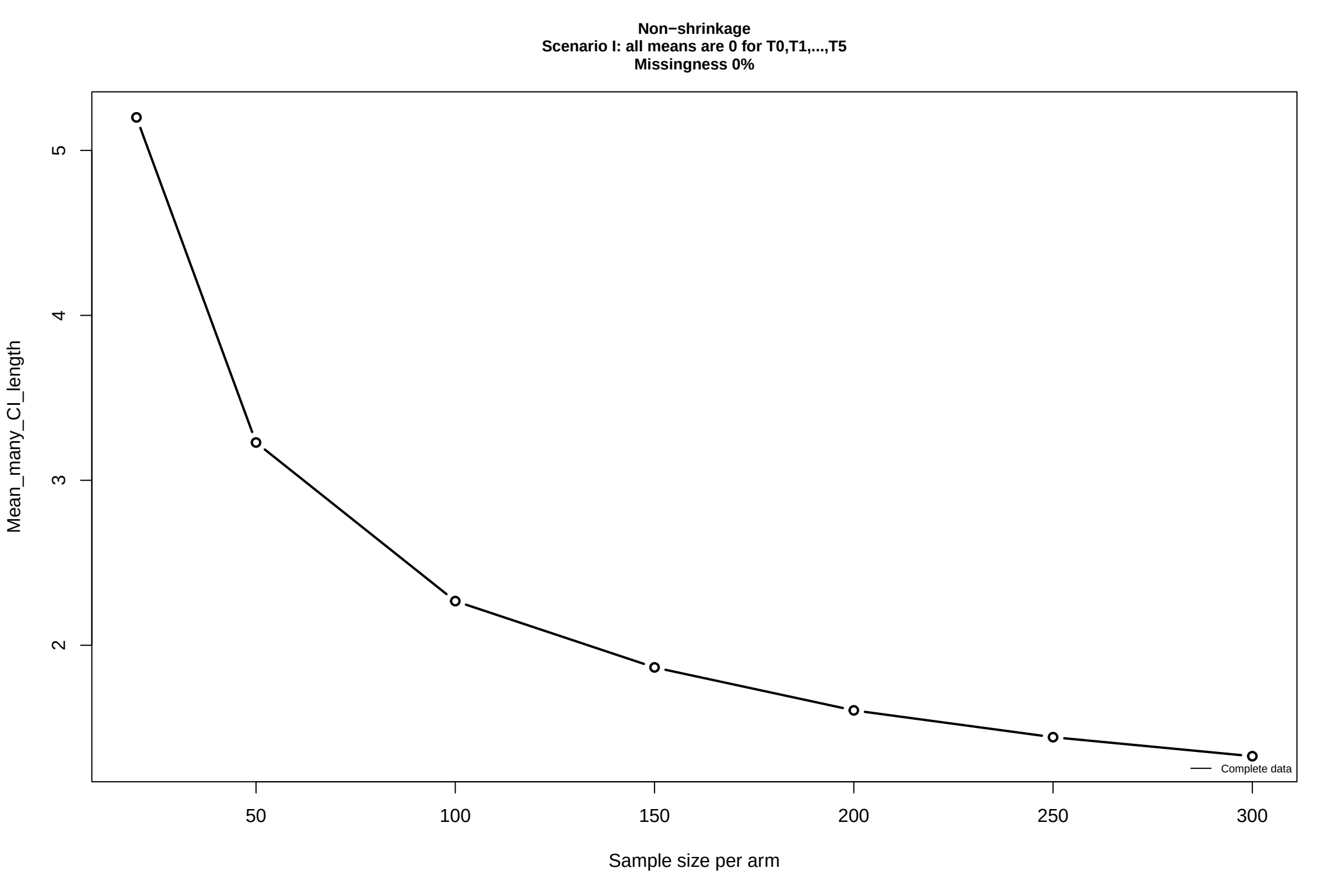
SampleSize_n50_5Treatments/table3_shrink_5_treatments_MAR_n50

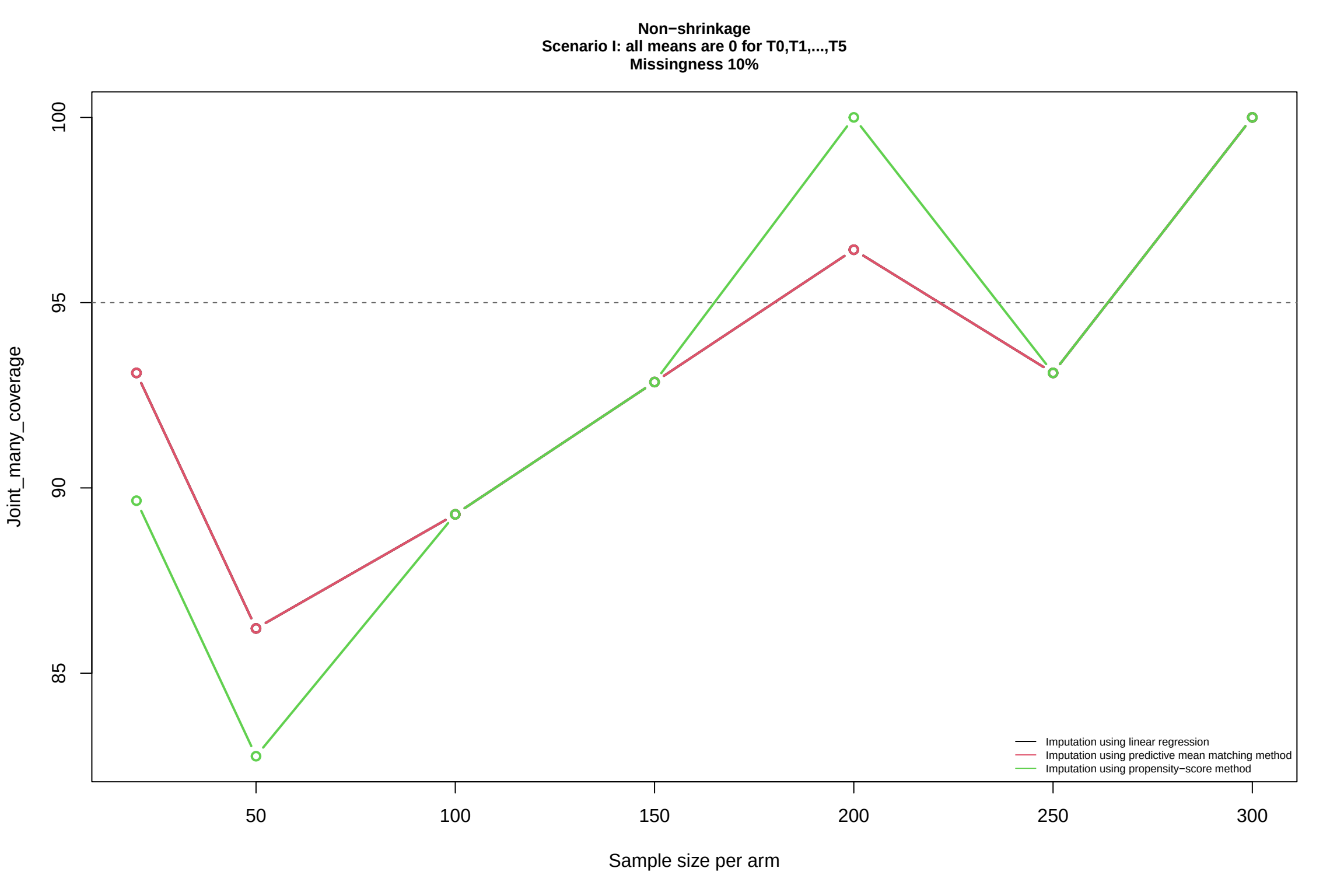
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	2.519	2.502	2.547	2.551	2.593	3.226	3.213	3.226	3.226	3.245
Imputation using linear regression										
10%	2.628 (4.34)	2.621 (4.77)	2.654 (4.20)	2.719 (6.58)	2.710 (4.50)	3.357 (4.06)	3.367 (4.80)	3.348 (3.79)	3.419 (5.99)	3.388 (4.40)
20%	2.836 (12.58)	2.819 (12.70)	2.906 (14.09)	2.888 (13.22)	2.914 (12.37)	3.624 (12.33)	3.594 (11.86)	3.644 (12.96)	3.627 (12.40)	3.659 (12.75)
30%	2.872 (14.02)	3.000 (19.90)	2.981 (17.03)	3.089 (21.08)	3.038 (17.19)	3.720 (15.30)	3.829 (19.17)	3.808 (18.05)	3.920 (21.51)	3.867 (19.18)
40%	3.358 (33.29)	3.261 (30.35)	3.255 (27.79)	3.431 (34.48)	3.412 (31.59)	4.320 (33.91)	4.189 (30.38)	4.178 (29.52)	4.365 (35.31)	4.356 (34.25)
Imputation using predictive mean matching method										
10%	2.573 (2.16)	2.600 (3.94)	2.612 (2.56)	2.674 (4.83)	2.661 (2.63)	3.289 (1.95)	3.330 (3.64)	3.293 (2.07)	3.352 (3.88)	3.320 (2.32)
20%	2.721 (8.04)	2.699 (7.87)	2.820 (10.72)	2.792 (9.47)	2.806 (8.23)	3.477 (7.78)	3.439 (7.04)	3.530 (9.43)	3.493 (8.27)	3.524 (8.59)
30%	2.683 (6.52)	2.784 (11.28)	2.802 (10.00)	2.884 (13.08)	2.811 (8.41)	3.482 (7.92)	3.556 (10.68)	3.577 (10.87)	3.653 (13.23)	3.578 (10.26)
40%	2.982 (18.39)	2.964 (18.49)	3.017 (18.45)	3.020 (18.39)	3.043 (17.35)	3.843 (19.12)	3.805 (18.42)	3.862 (19.73)	3.841 (19.06)	3.873 (19.35)
Imputation using propensity−score method										
10%	2.586 (2.65)	2.591 (3.55)	2.618 (2.78)	2.678 (5.00)	2.664 (2.74)	3.307 (2.50)	3.325 (3.47)	3.304 (2.41)	3.362 (4.20)	3.328 (2.57)
20%	2.746 (9.02)	2.707 (8.22)	2.815 (10.52)	2.800 (9.77)	2.838 (9.44)	3.505 (8.62)	3.448 (7.32)	3.523 (9.21)	3.507 (8.71)	3.563 (9.79)
30%	2.679 (6.35)	2.811 (12.35)	2.849 (11.85)	2.951 (15.70)	2.855 (10.10)	3.482 (7.91)	3.595 (11.89)	3.643 (12.93)	3.745 (16.07)	3.637 (12.09)
40%	3.041 (20.74)	3.034 (21.29)	3.063 (20.27)	3.166 (24.12)	3.190 (23.04)	3.913 (21.29)	3.897 (21.29)	3.921 (21.55)	4.028 (24.84)	4.063 (25.21)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	2.477	2.539	2.550	2.584	2.582	3.165	3.180	3.188	3.197	3.206
Imputation using linear regression										

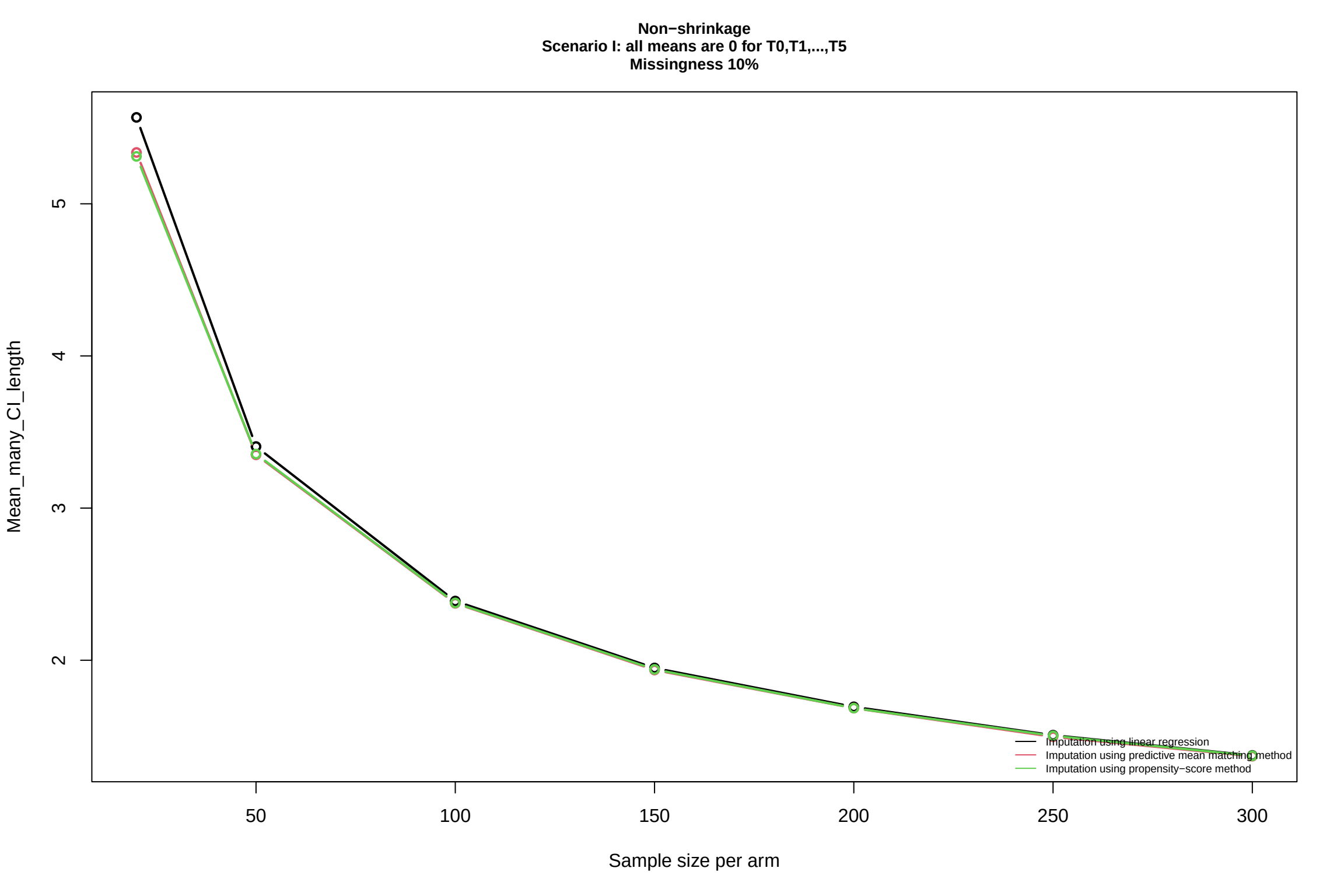
SampleSize_n50_5Treatments/table3_shrink_5_treatments_MAR_n50

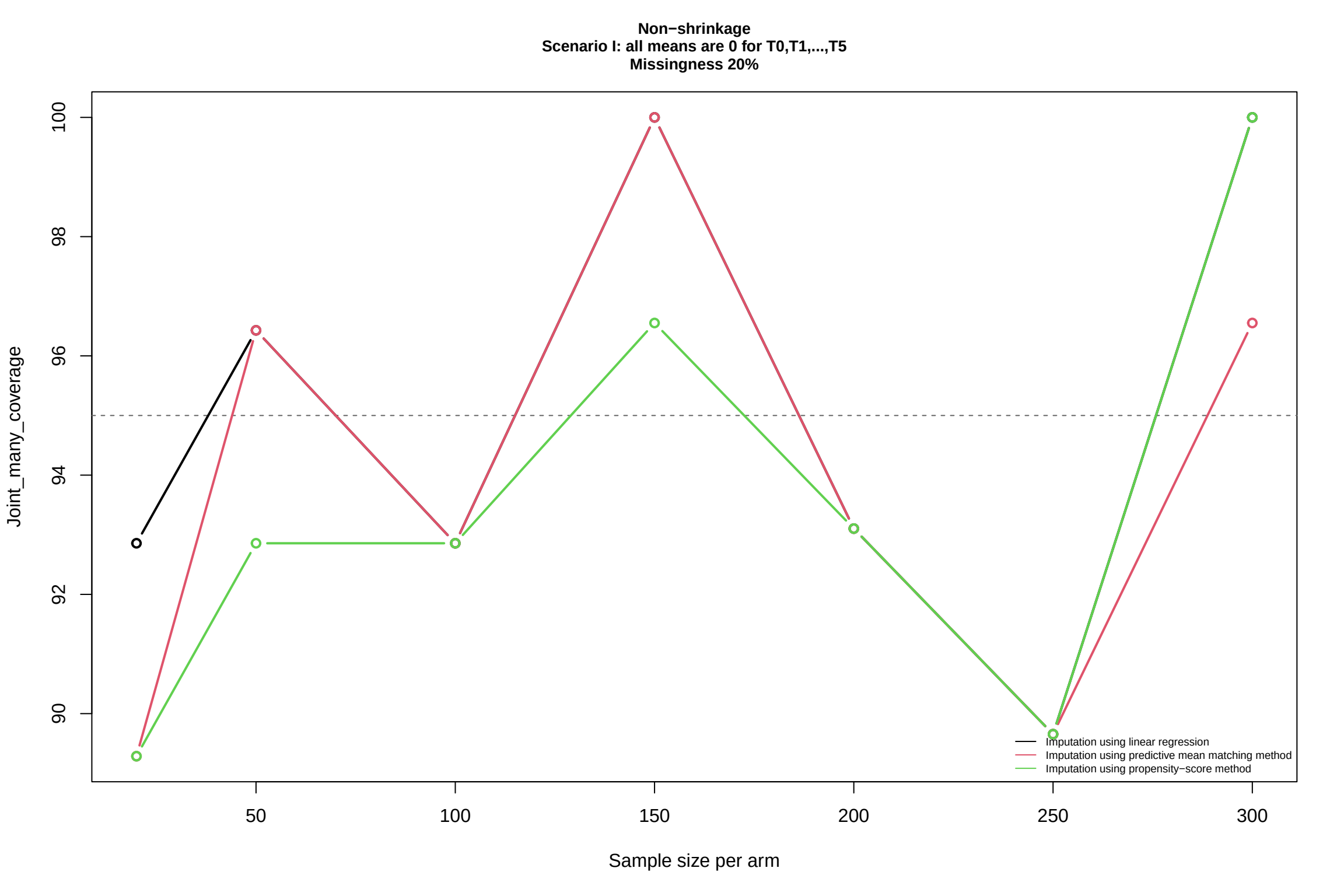
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	2.642 (6.65)	2.702 (6.43)	2.715 (6.47)	2.729 (5.59)	2.749 (6.49)	3.410 (7.73)	3.424 (7.69)	3.445 (8.05)	3.451 (7.95)	3.455 (7.77)
20%	2.785 (12.44)	2.858 (12.56)	2.759 (8.18)	2.853 (10.39)	2.885 (11.73)	3.566 (12.66)	3.604 (13.35)	3.519 (10.38)	3.585 (12.17)	3.606 (12.47)
30%	2.902 (17.15)	3.011 (18.57)	3.018 (18.32)	3.003 (16.22)	3.074 (19.07)	3.731 (17.89)	3.847 (20.99)	3.817 (19.73)	3.821 (19.55)	3.871 (20.73)
40%	3.258 (31.53)	3.331 (31.19)	3.418 (34.03)	3.494 (35.22)	3.409 (32.03)	4.176 (31.94)	4.256 (33.85)	4.342 (36.21)	4.412 (38.01)	4.327 (34.97)
Imputation using predictive mean matching method										
10%	2.587 (4.45)	2.654 (4.50)	2.652 (3.98)	2.696 (4.34)	2.676 (3.64)	3.342 (5.60)	3.358 (5.60)	3.364 (5.51)	3.399 (6.33)	3.365 (4.96)
20%	2.656 (7.21)	2.736 (7.75)	2.694 (5.61)	2.783 (7.71)	2.791 (8.09)	3.410 (7.75)	3.452 (8.57)	3.432 (7.63)	3.485 (9.02)	3.482 (8.60)
30%	2.709 (9.34)	2.769 (9.06)	2.809 (10.14)	2.759 (6.77)	2.859 (10.75)	3.480 (9.94)	3.541 (11.38)	3.539 (11.01)	3.511 (9.84)	3.591 (12.00)
40%	2.943 (18.80)	2.988 (17.68)	3.024 (18.55)	3.194 (23.61)	3.018 (16.89)	3.782 (19.48)	3.817 (20.06)	3.846 (20.64)	4.026 (25.94)	3.834 (19.59)
Imputation using propensity−score method										
10%	2.611 (5.41)	2.669 (5.10)	2.677 (4.95)	2.696 (4.33)	2.704 (4.73)	3.374 (6.60)	3.378 (6.25)	3.395 (6.50)	3.402 (6.42)	3.403 (6.14)
20%	2.718 (9.73)	2.760 (8.71)	2.701 (5.90)	2.820 (9.12)	2.794 (8.22)	3.489 (10.25)	3.484 (9.58)	3.443 (7.98)	3.536 (10.61)	3.491 (8.87)
30%	2.747 (10.88)	2.802 (10.34)	2.880 (12.91)	2.823 (9.26)	2.942 (13.96)	3.527 (11.44)	3.581 (12.62)	3.630 (13.86)	3.598 (12.56)	3.699 (15.38)
40%	2.981 (20.34)	3.019 (18.89)	3.069 (20.32)	3.177 (22.95)	3.033 (17.46)	3.840 (21.31)	3.873 (21.80)	3.917 (22.87)	4.026 (25.94)	3.872 (20.76)

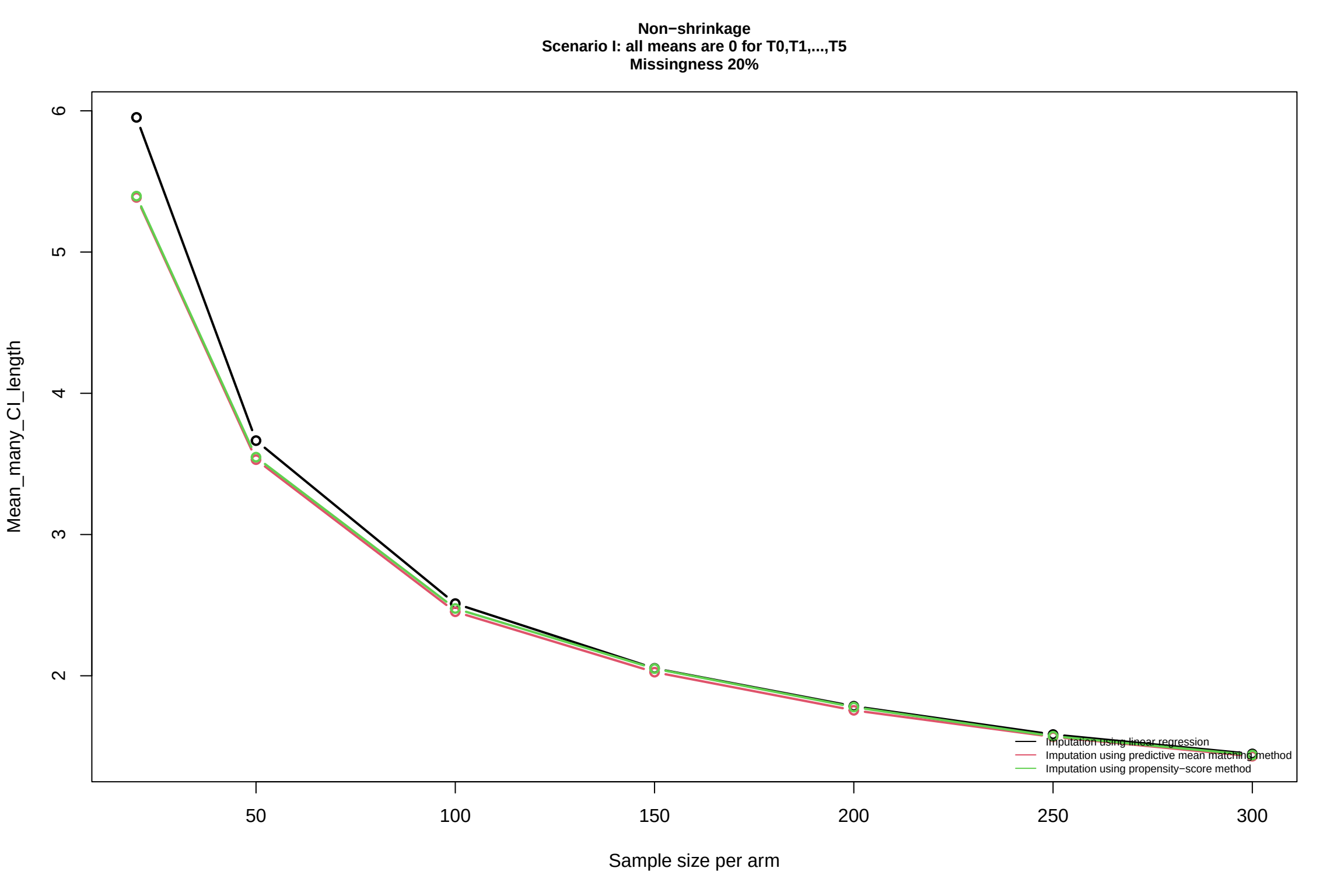

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Joint_many_coverage
90
91
92
93
94
95
96
50
100
150
200
250
300
Sample size per arm
Complete data

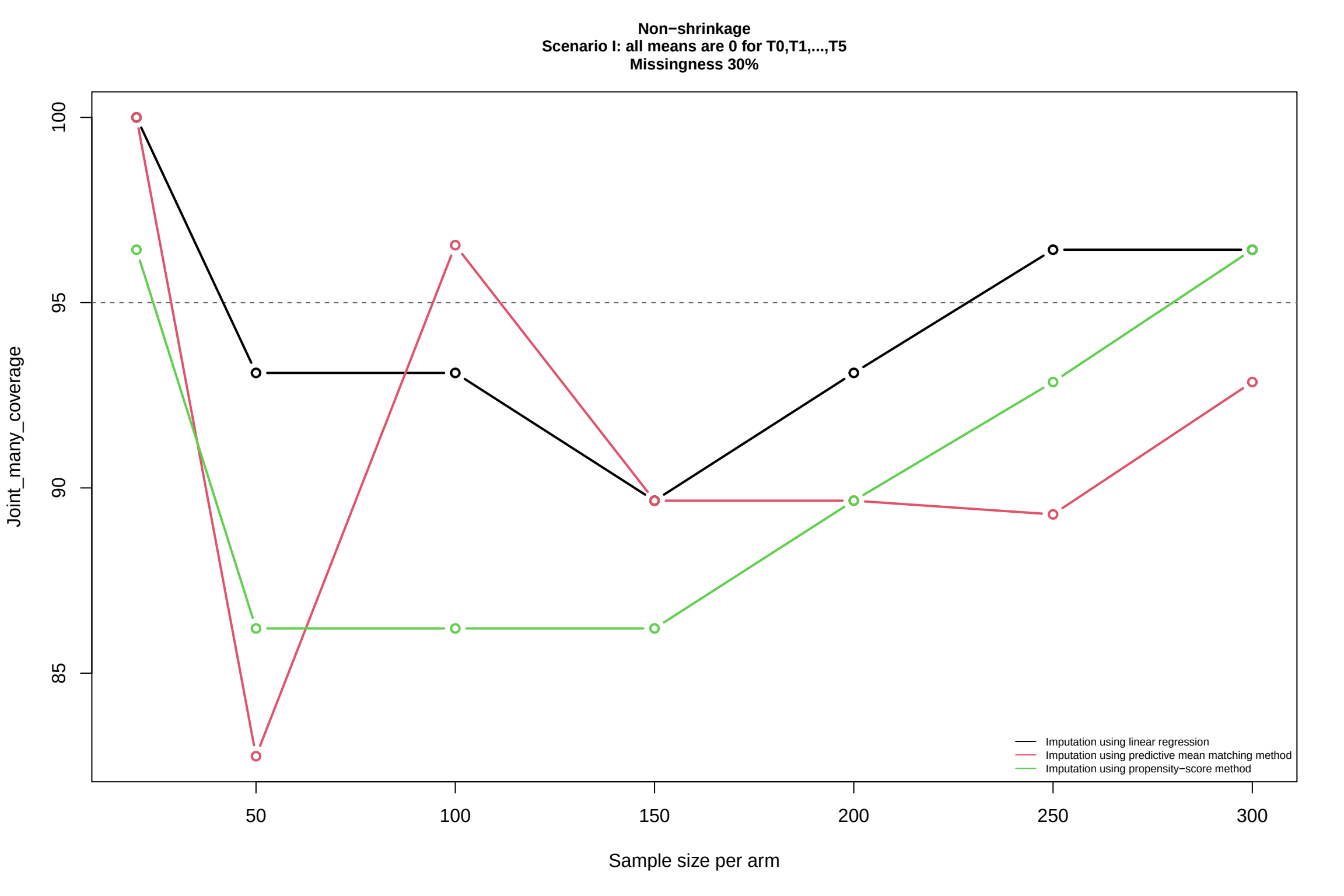

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

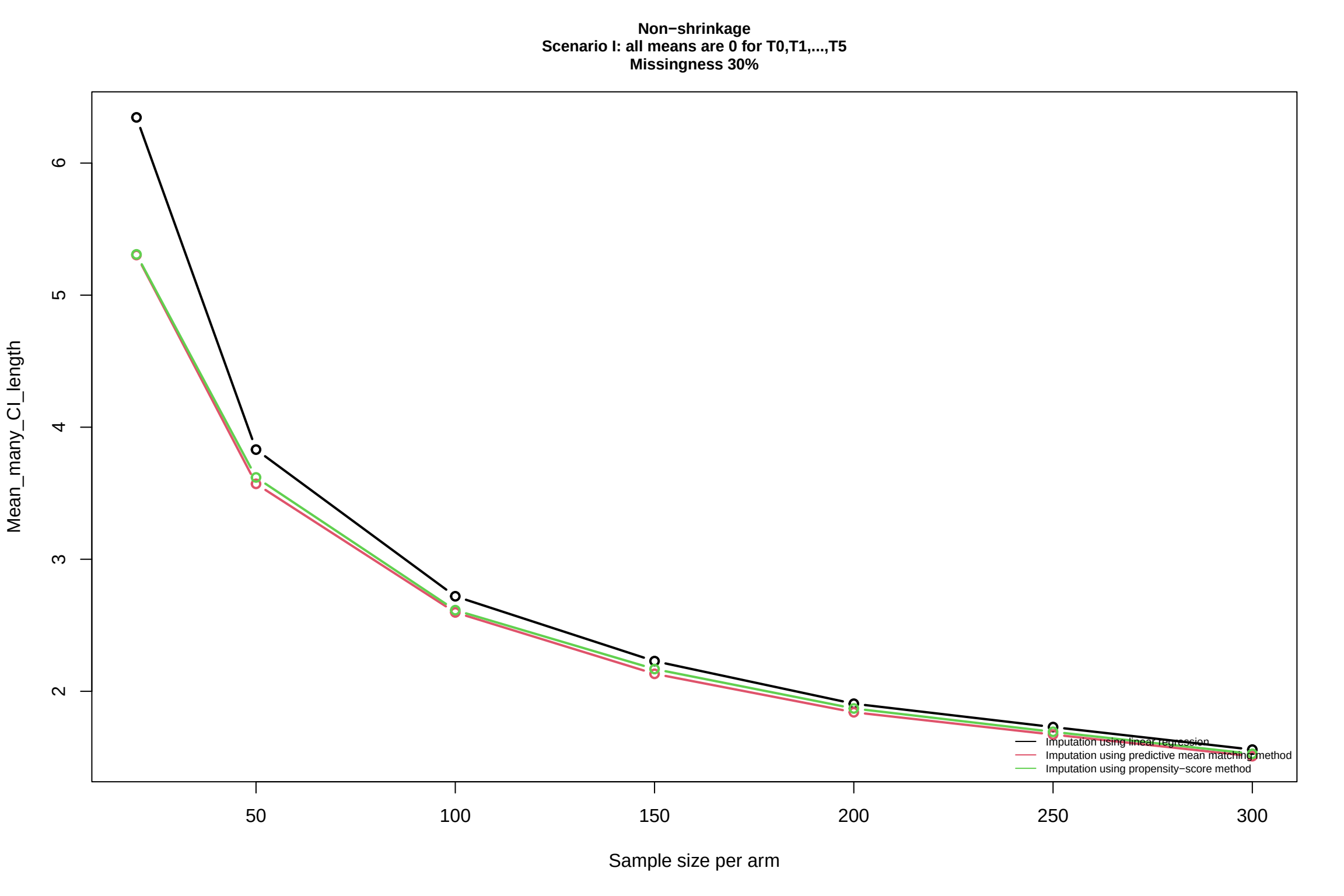

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

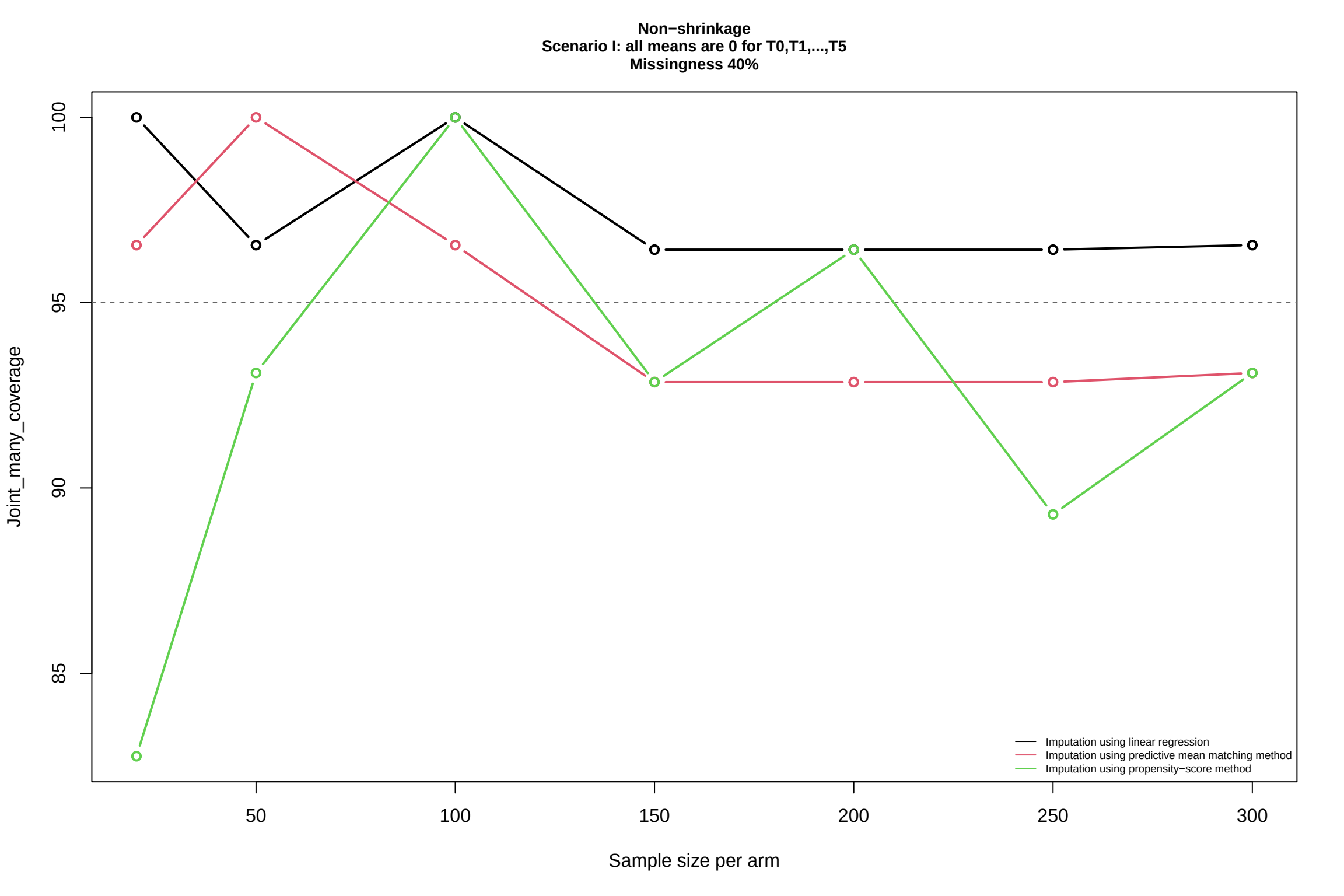

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

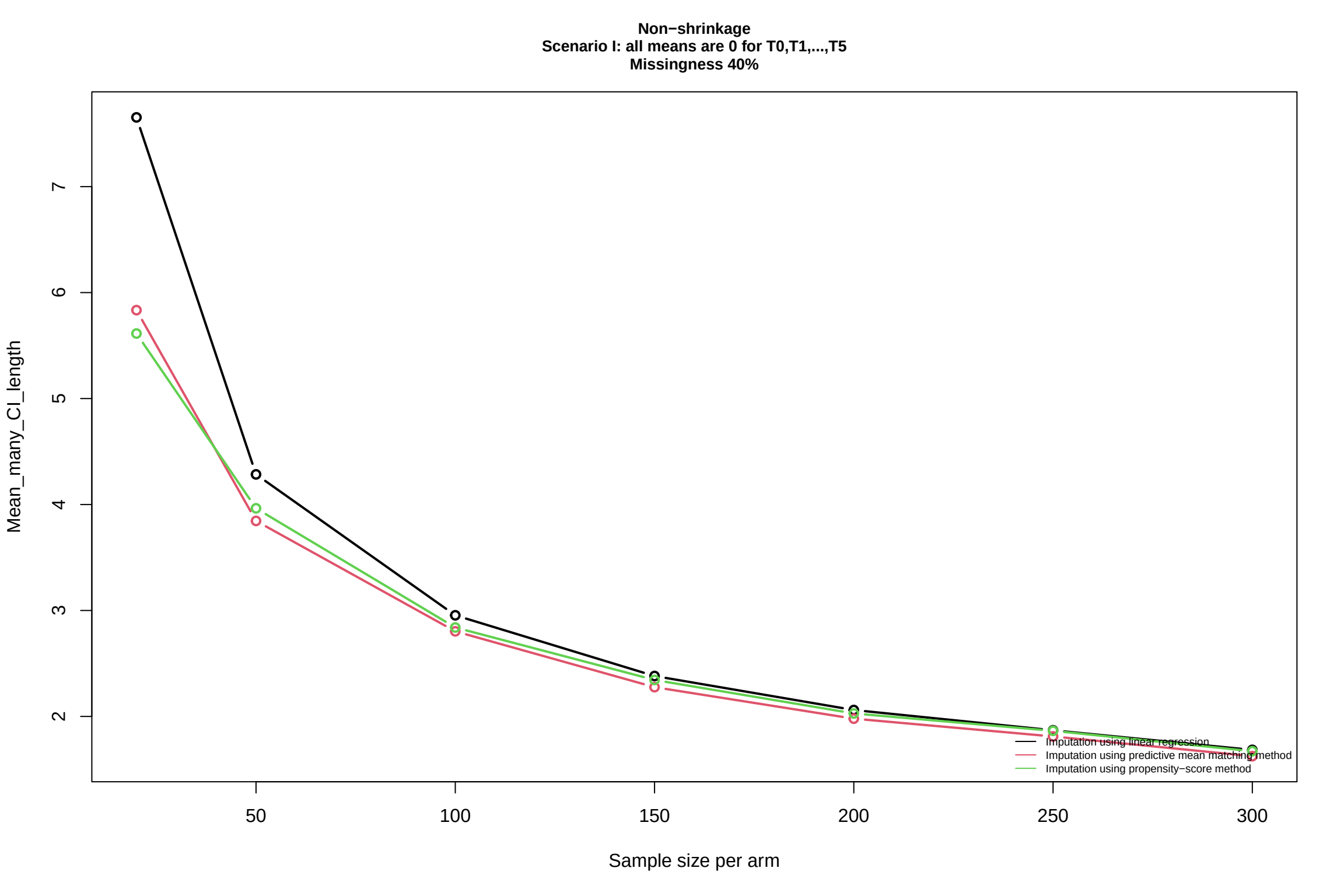

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

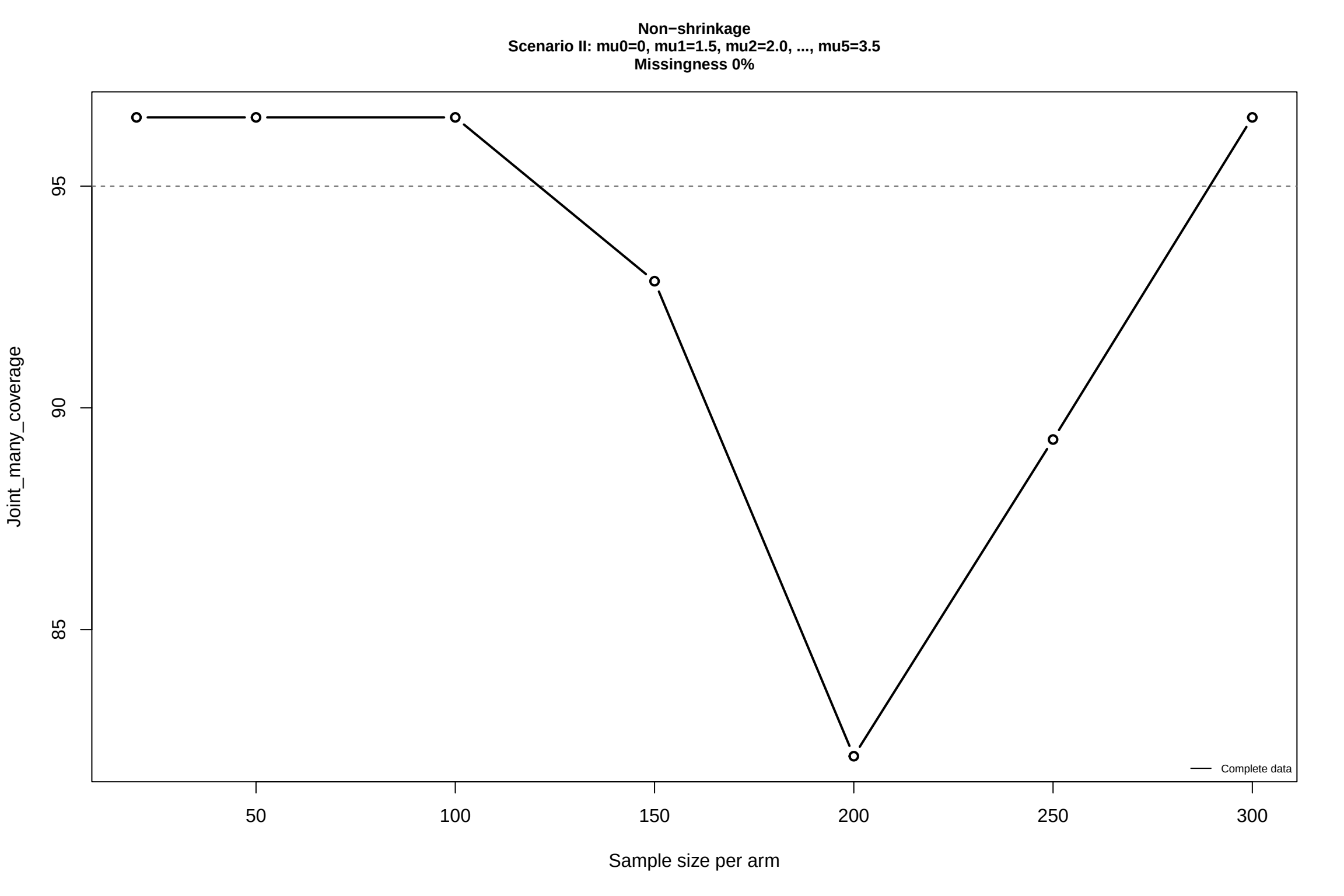

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

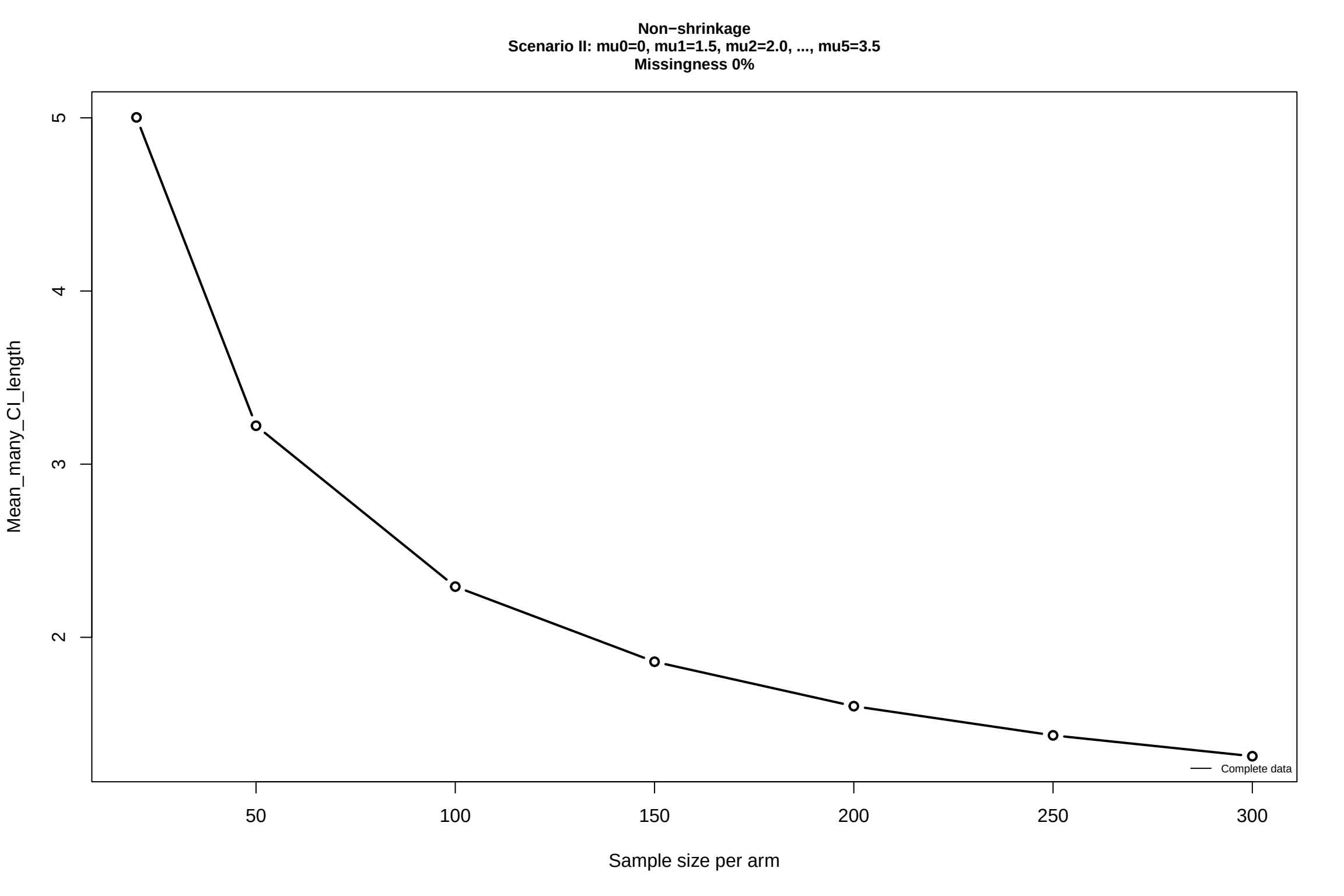

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

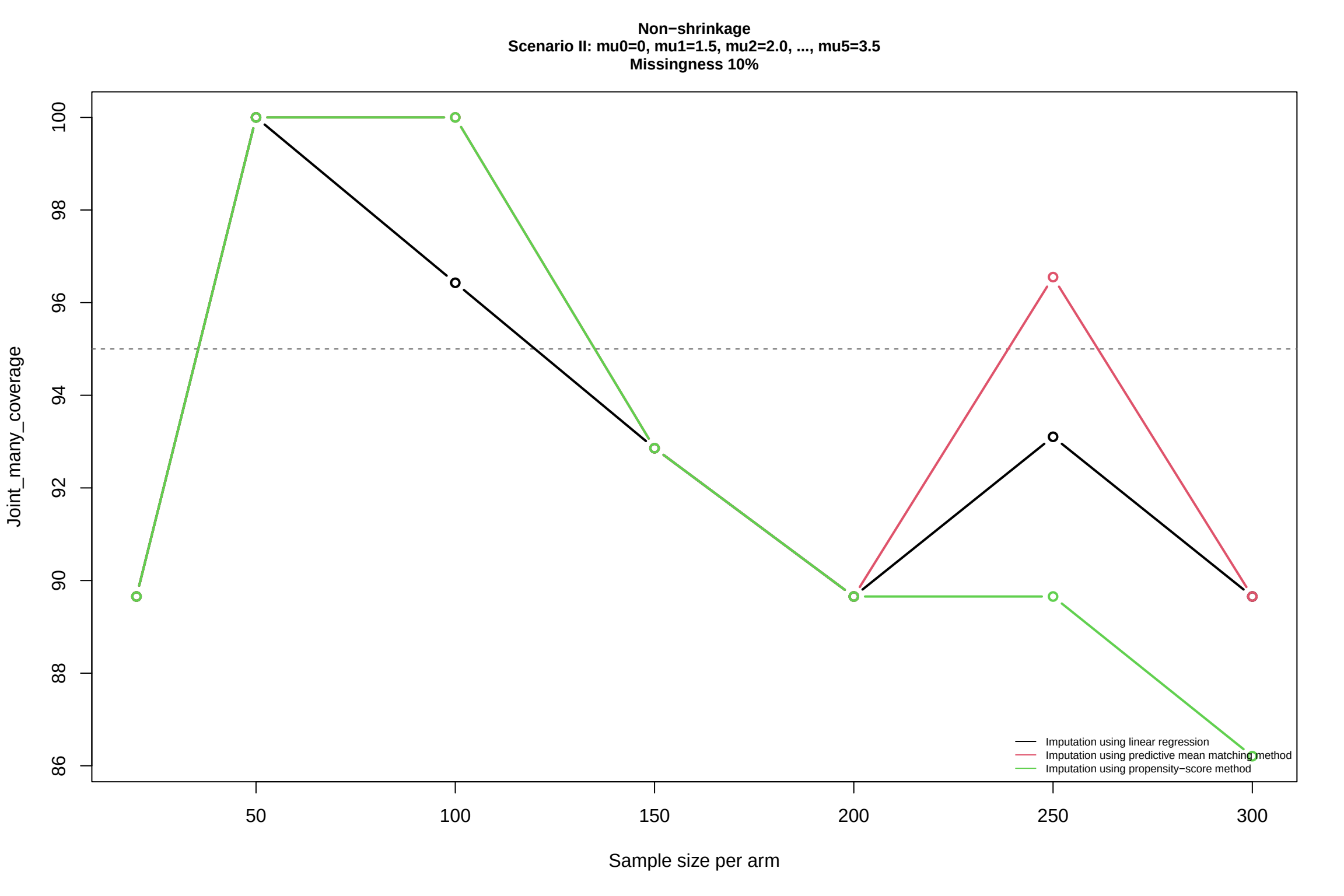

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

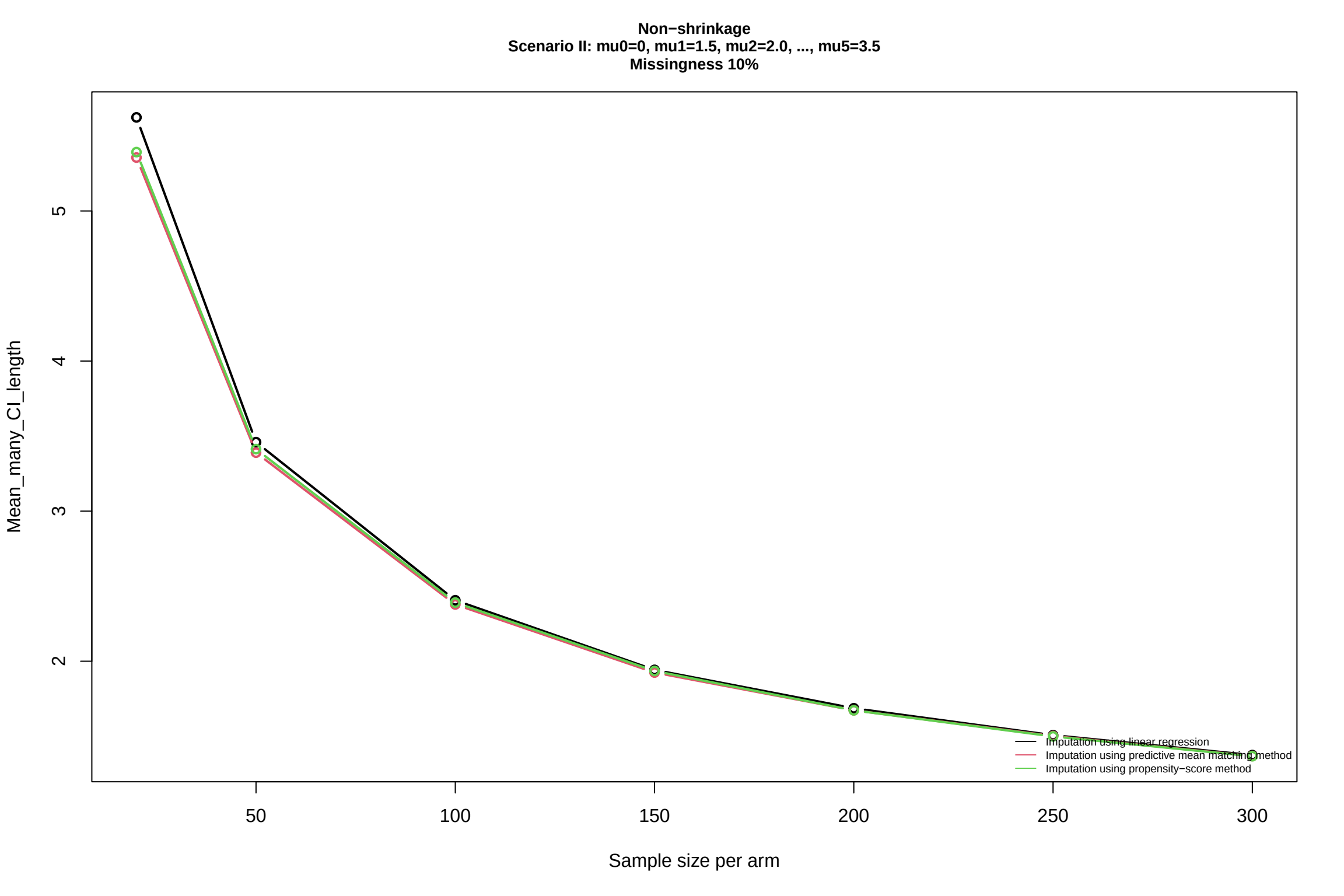

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

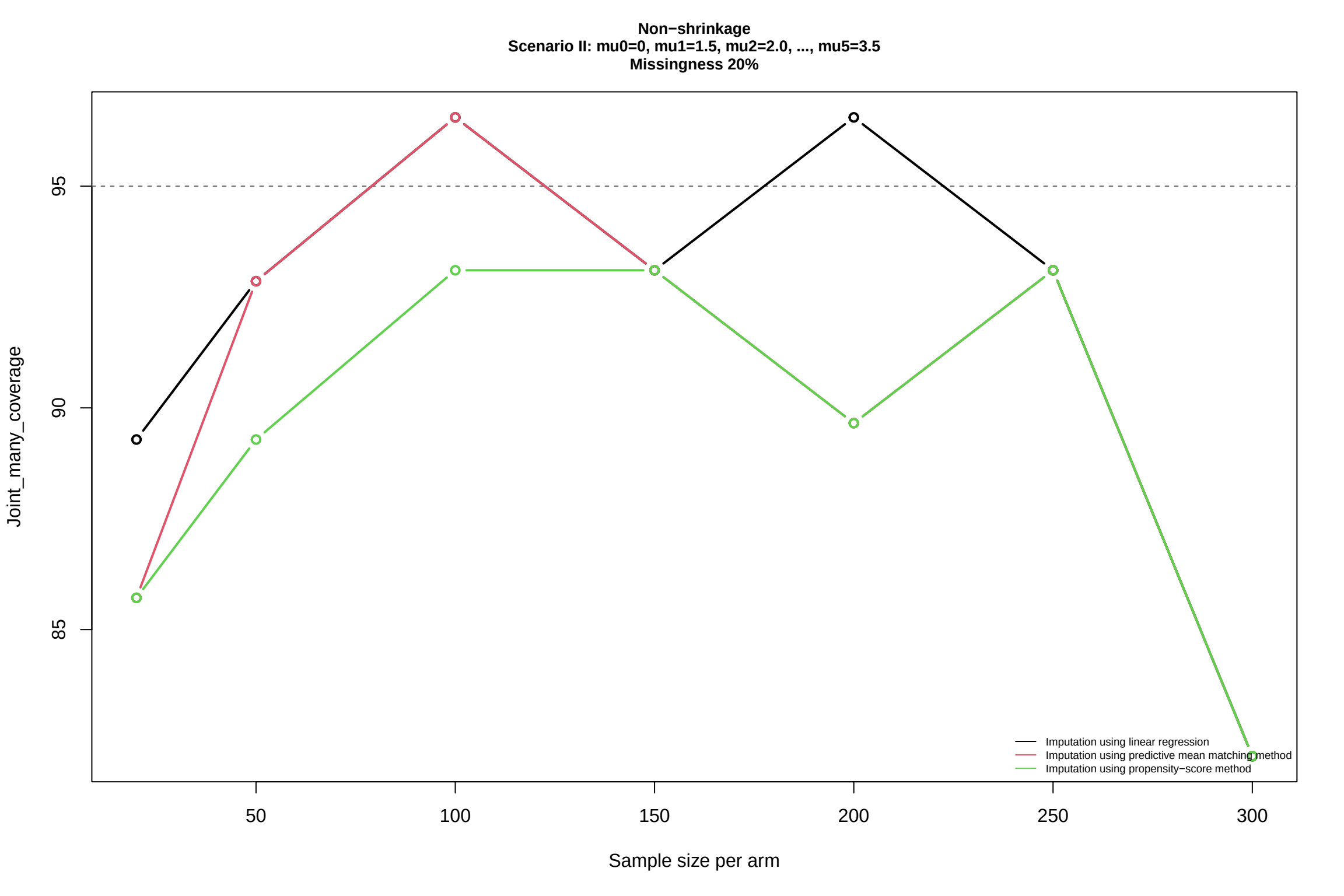

Non−shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

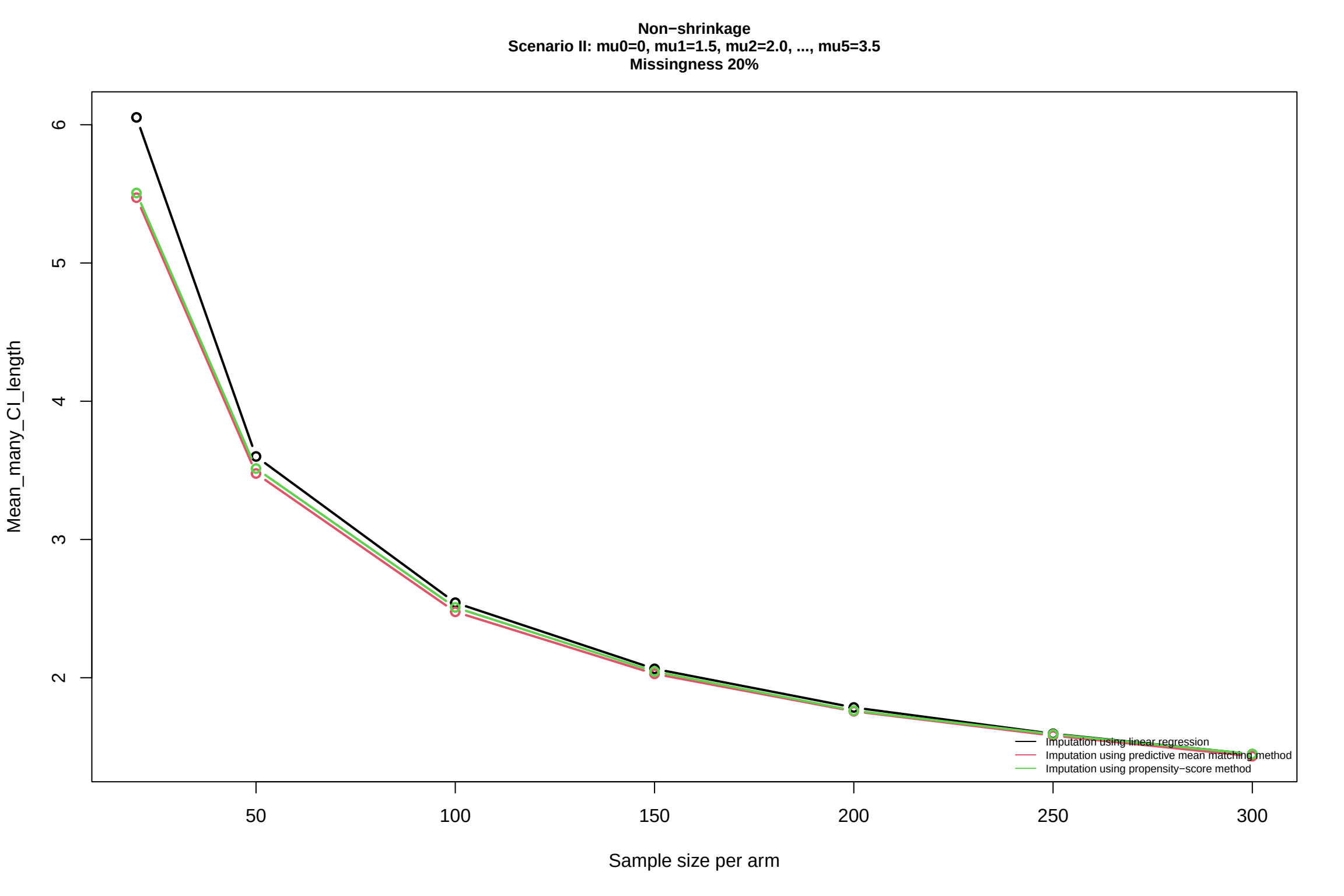

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Joint_many_coverage
95
90
85
50
100
150
200
250
300
Sample size per arm
Complete data

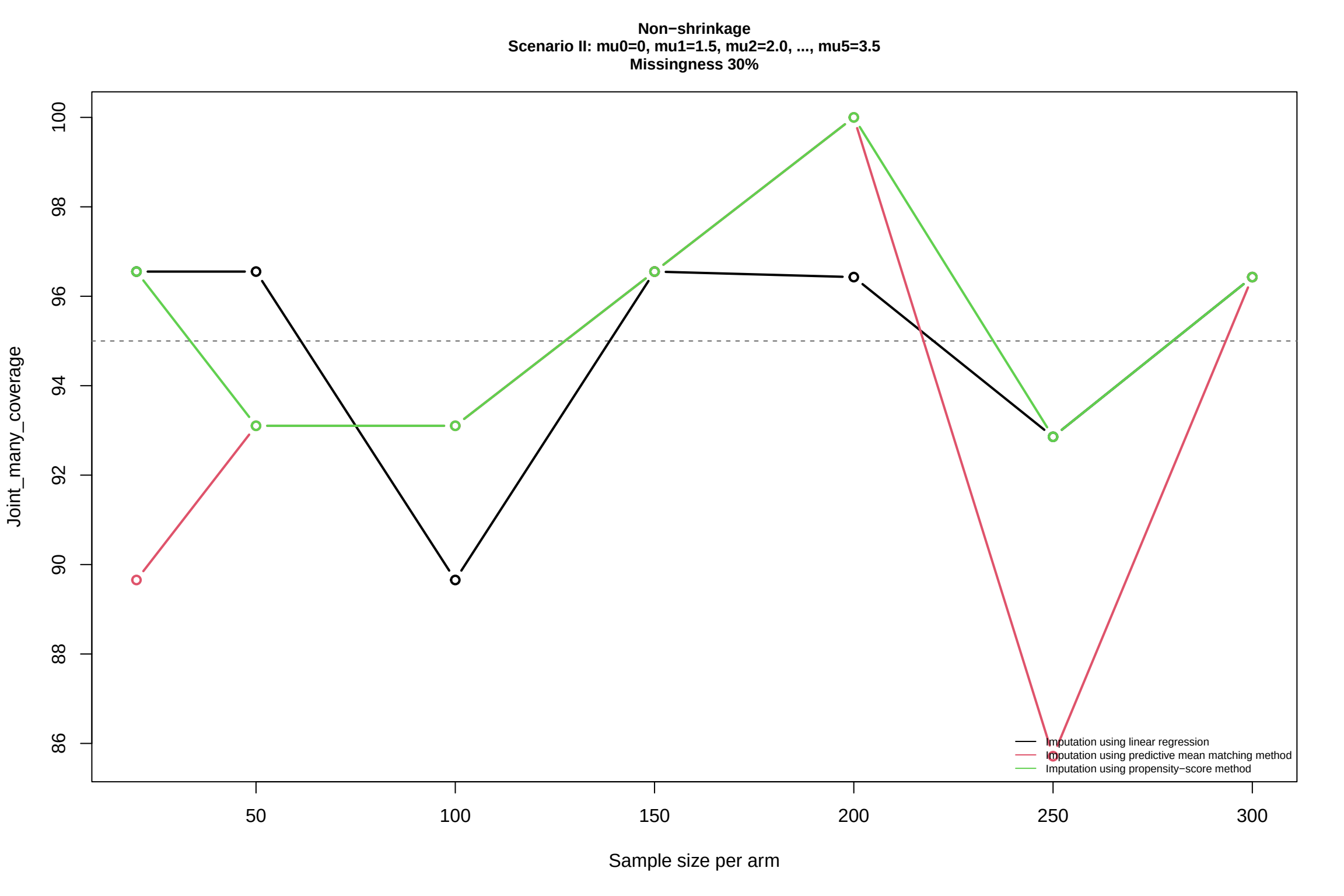

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

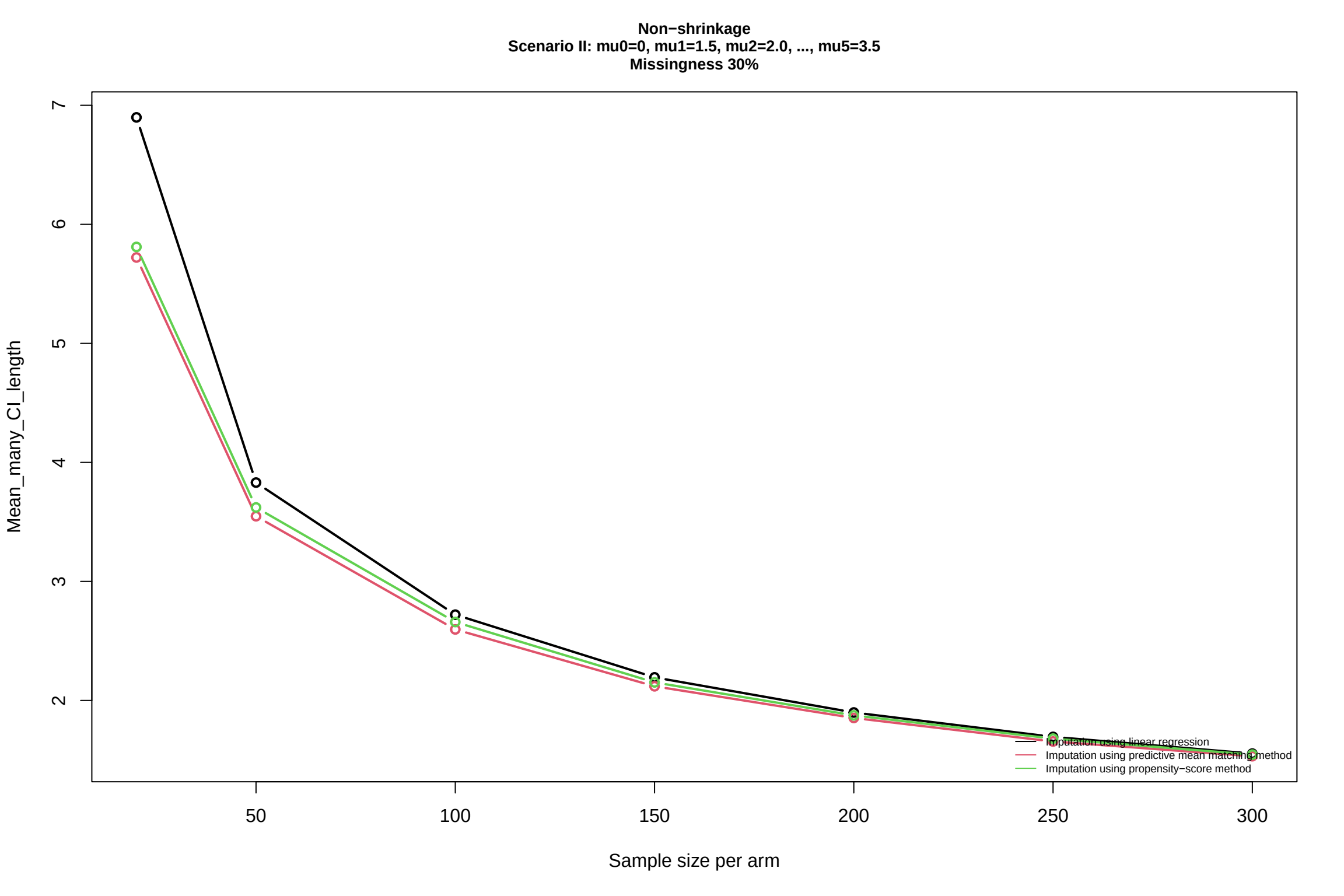

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

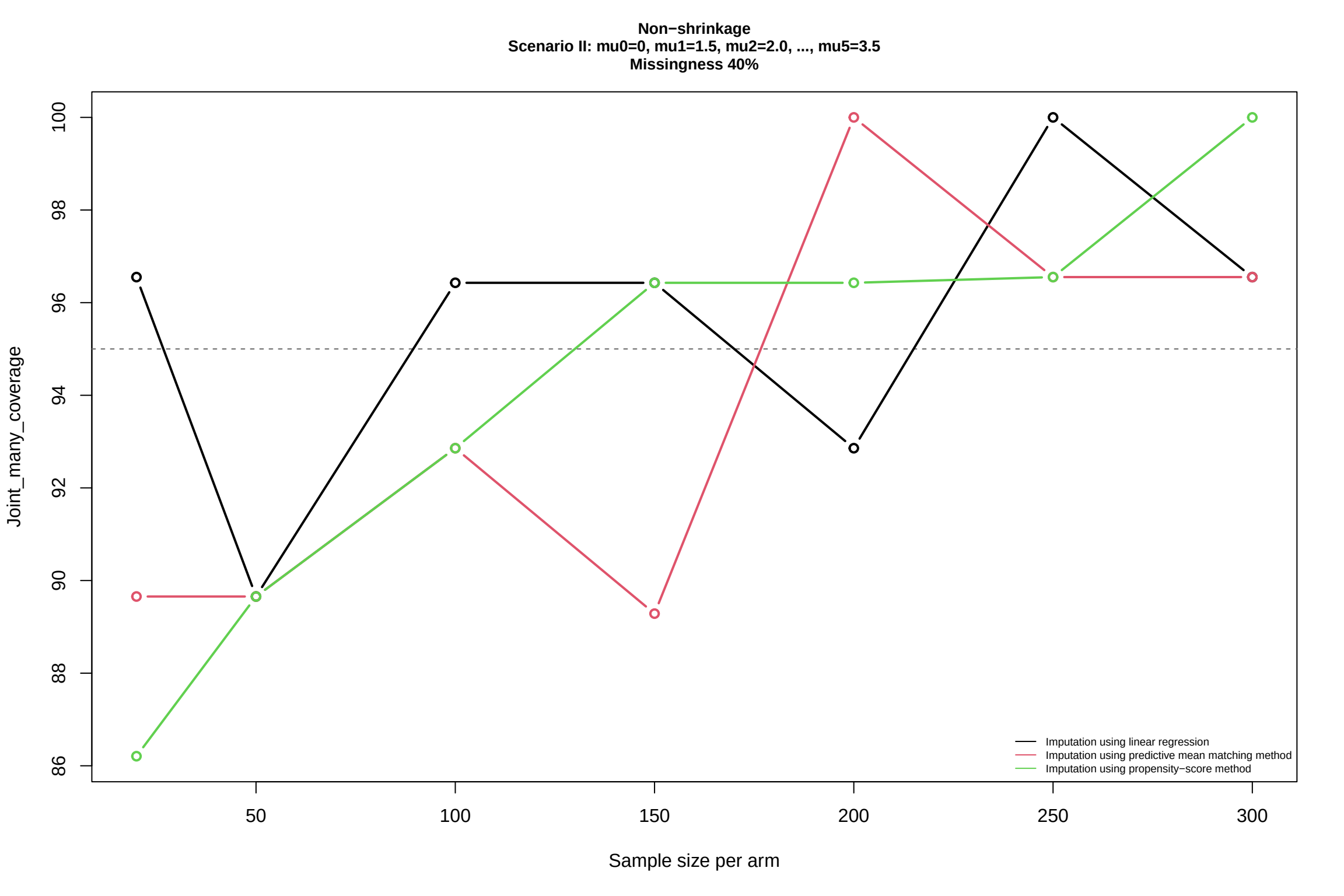

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

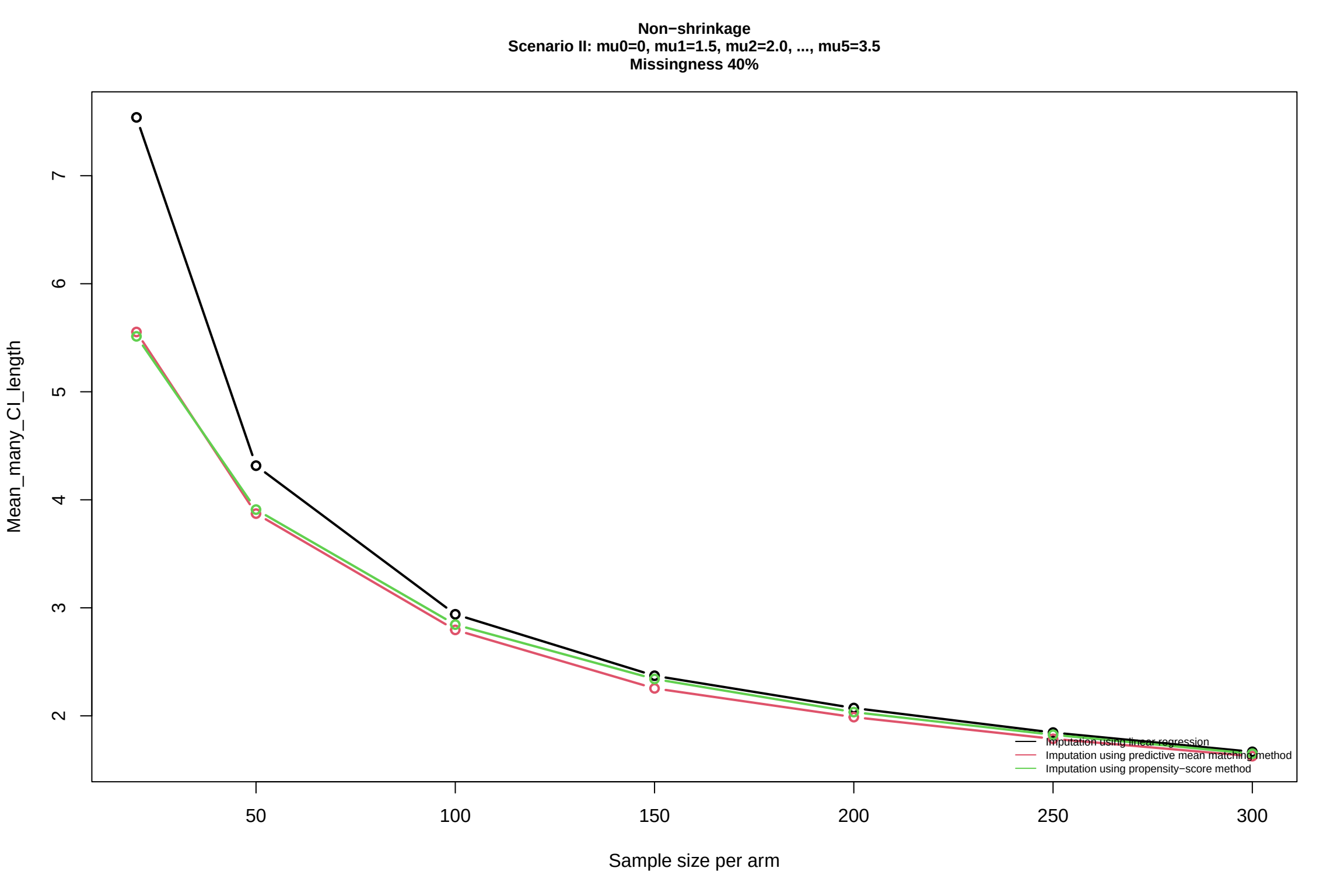

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Joint_many_coverage
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

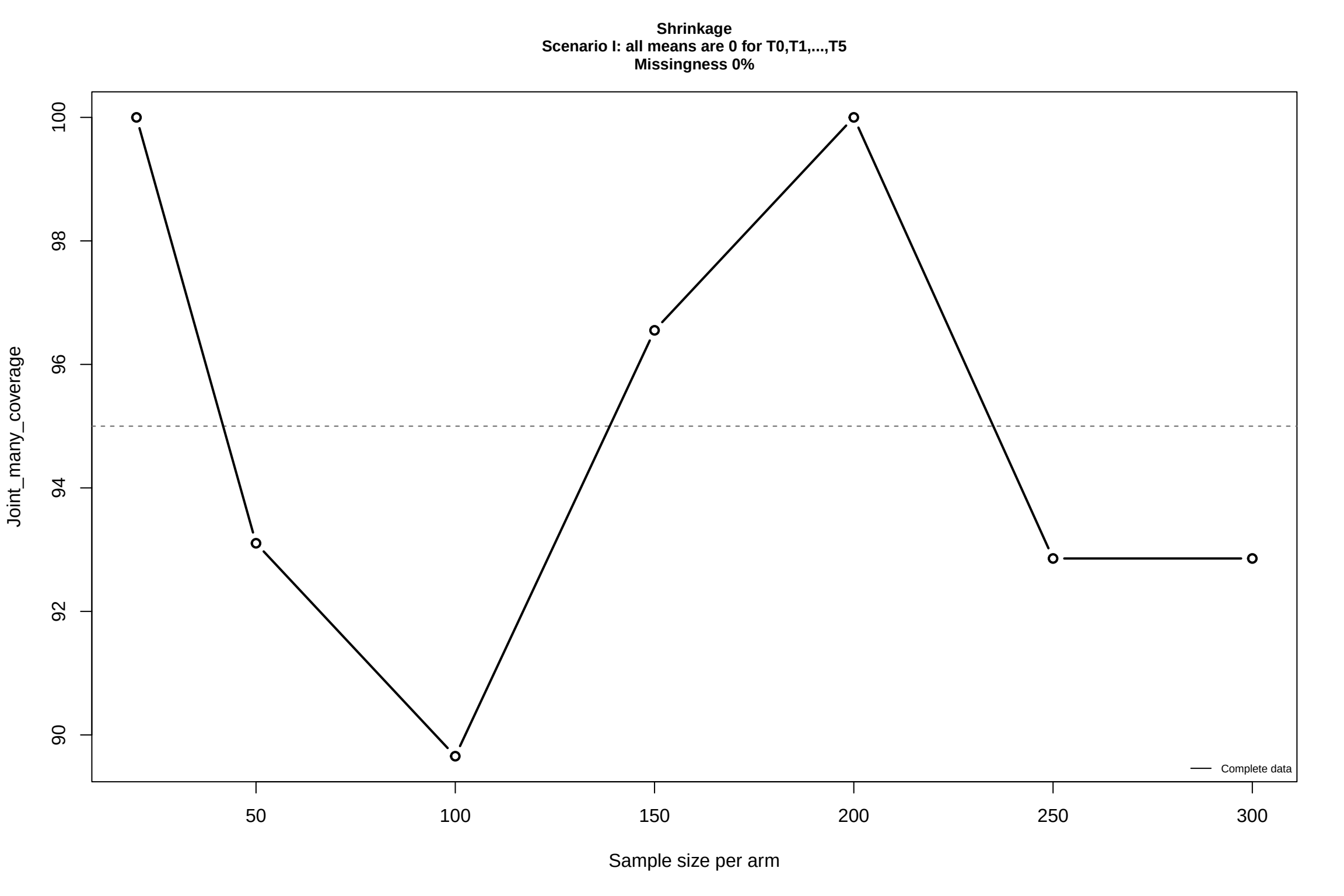

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

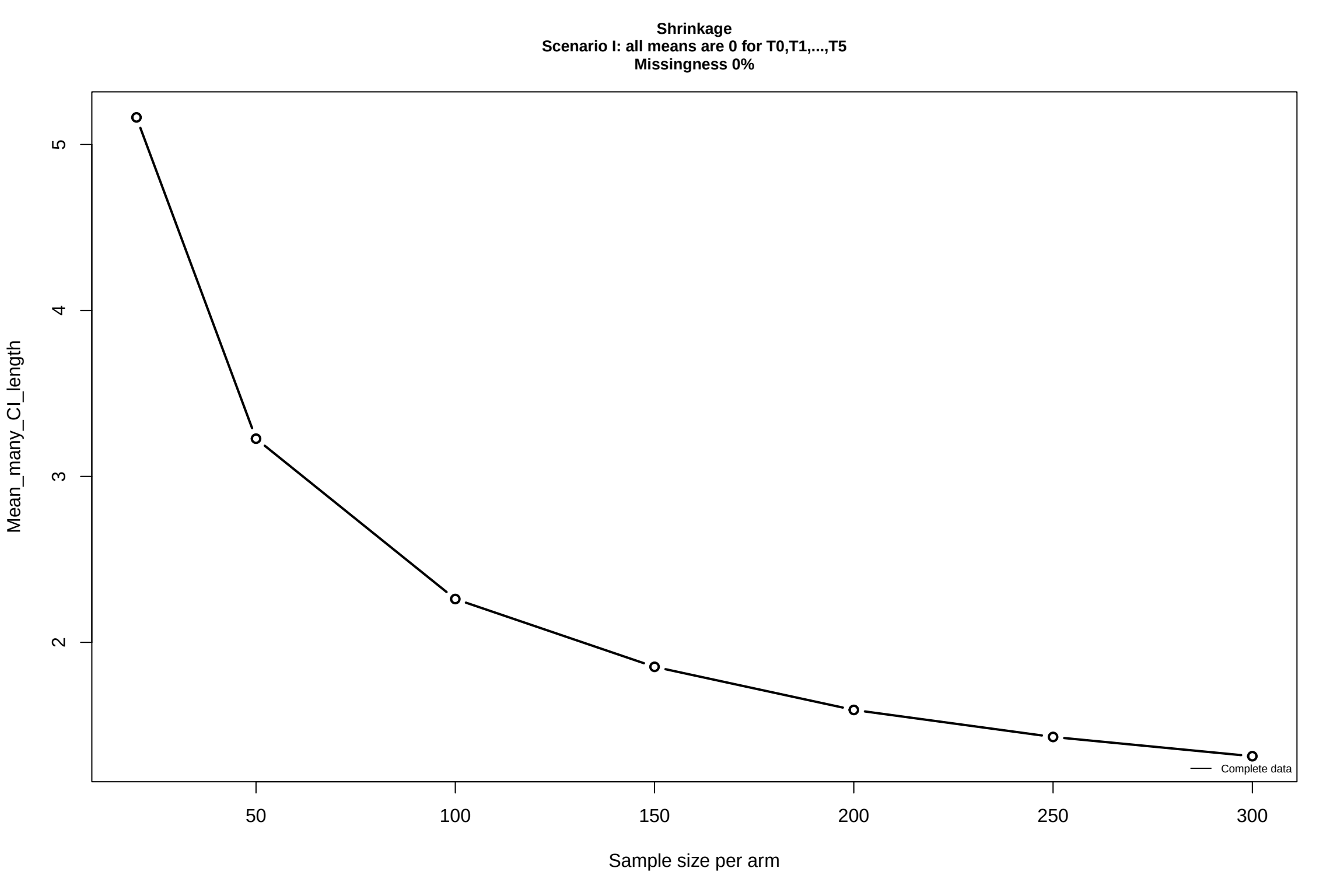

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

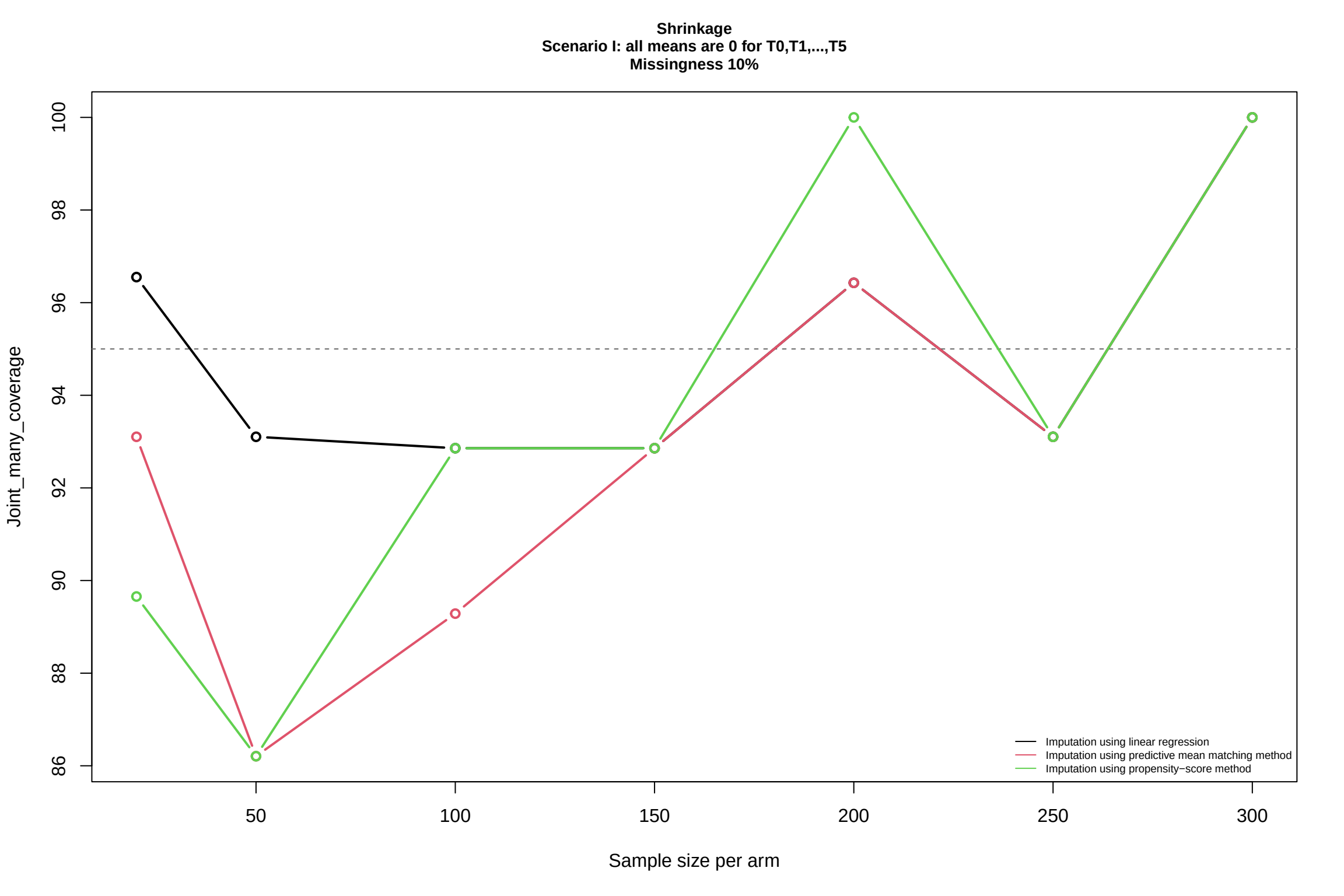

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

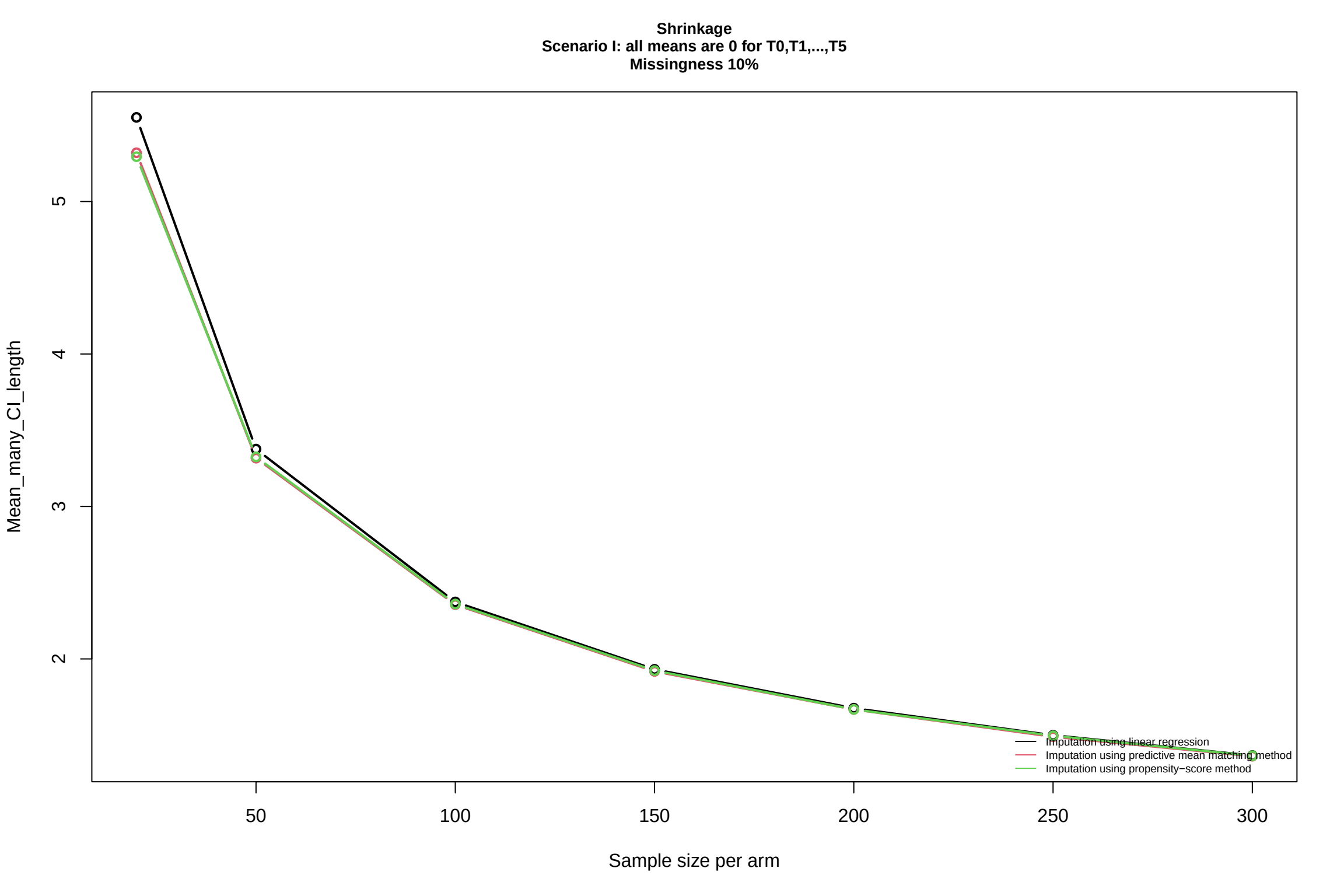

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

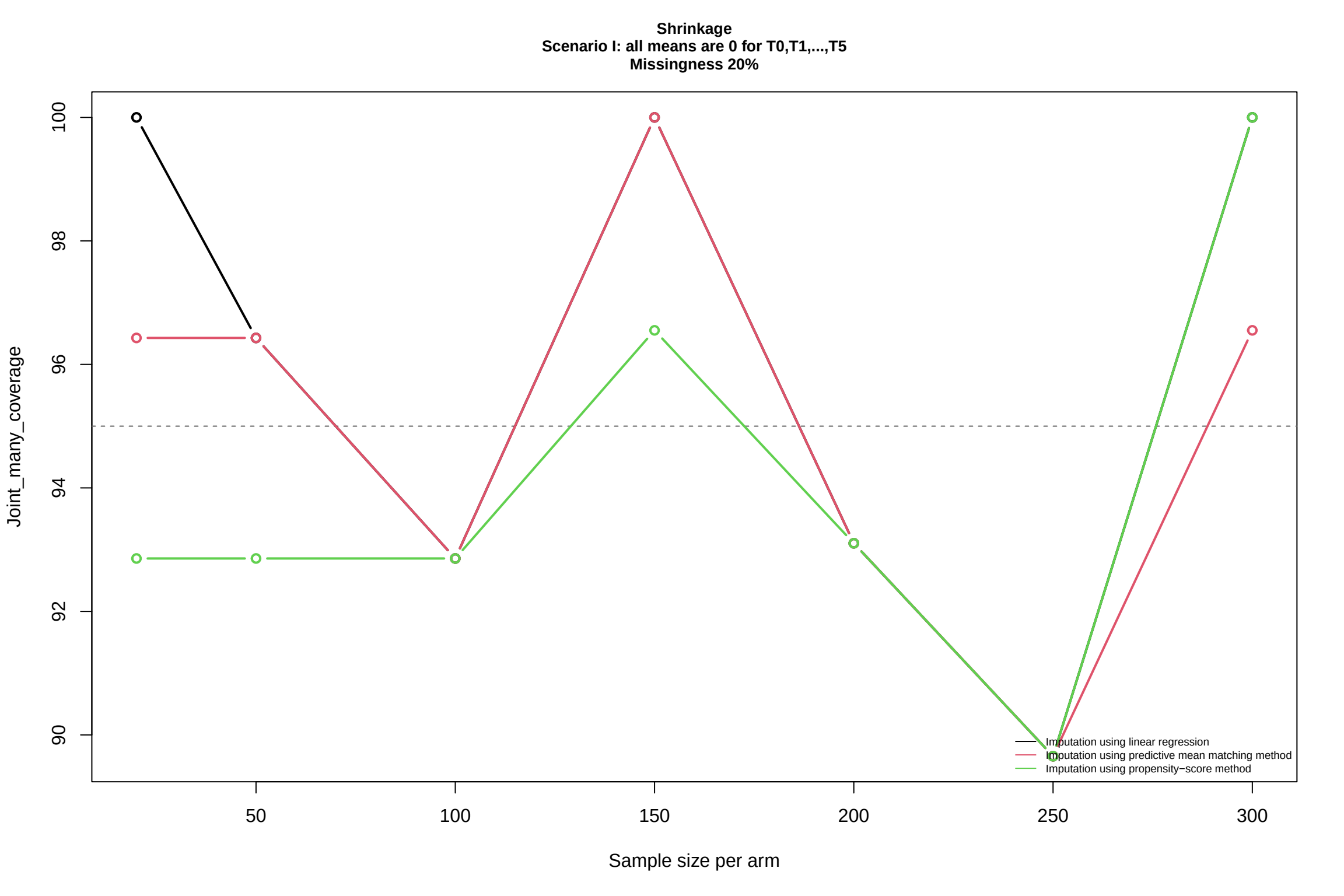

Non−shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

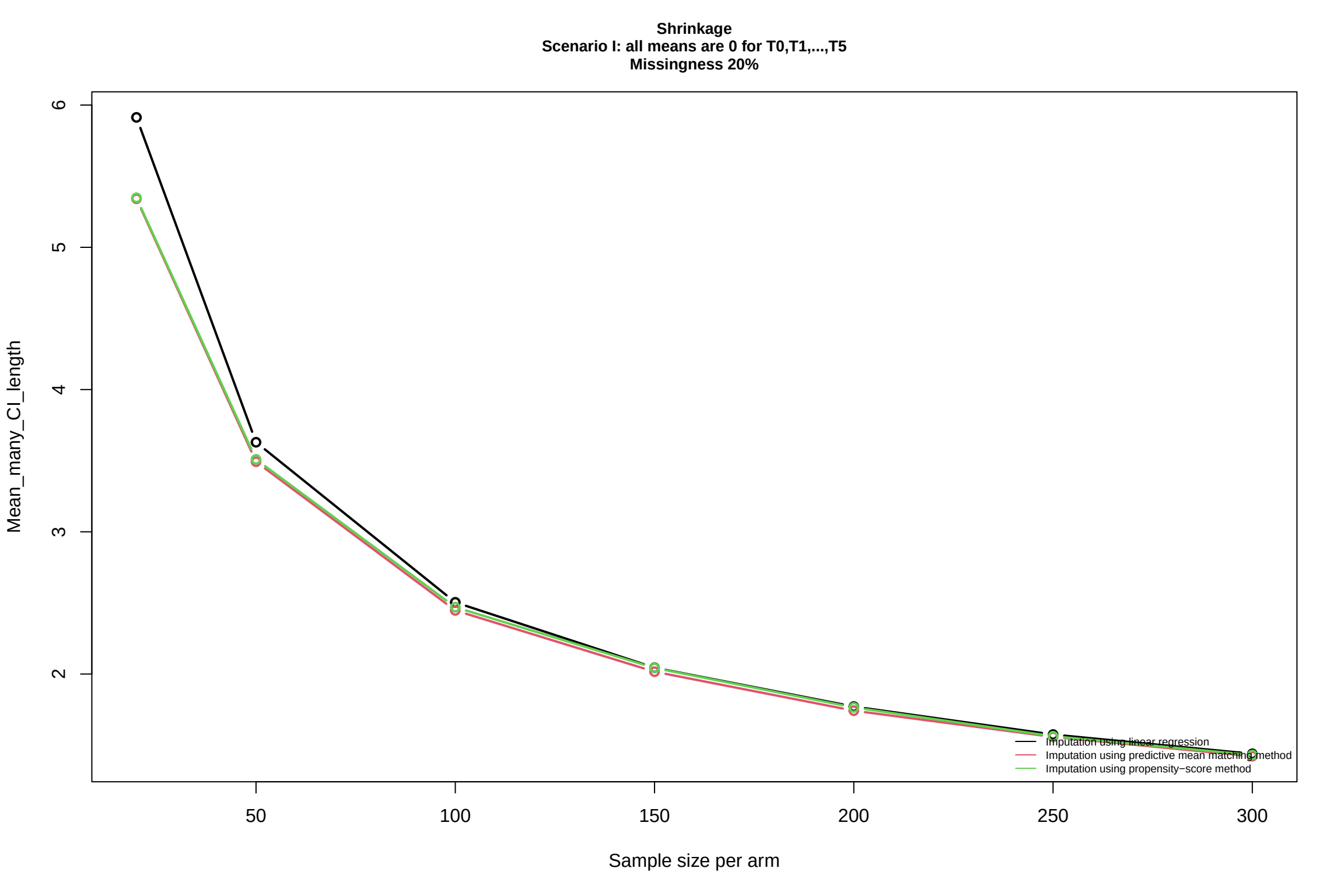

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Complete data

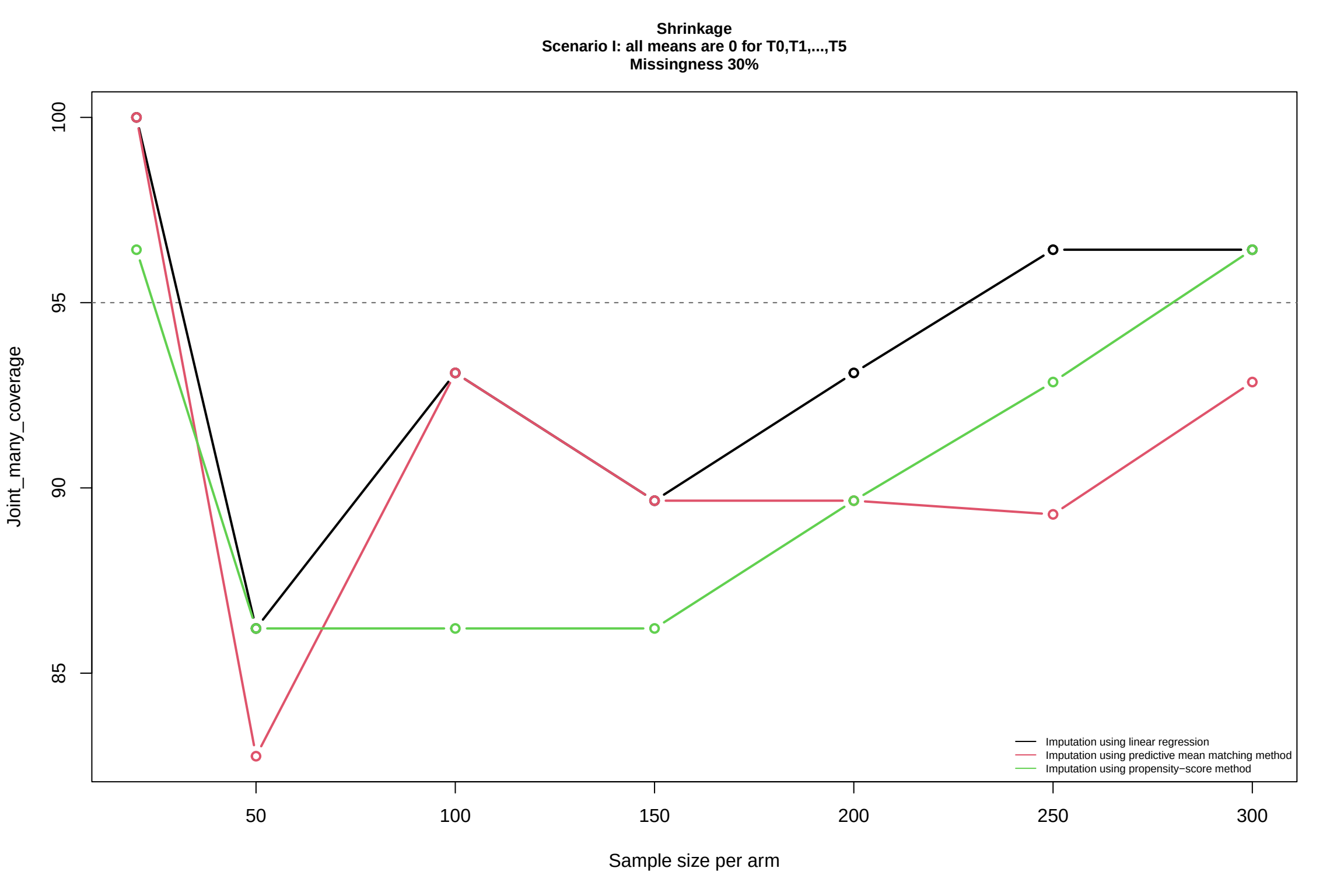

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

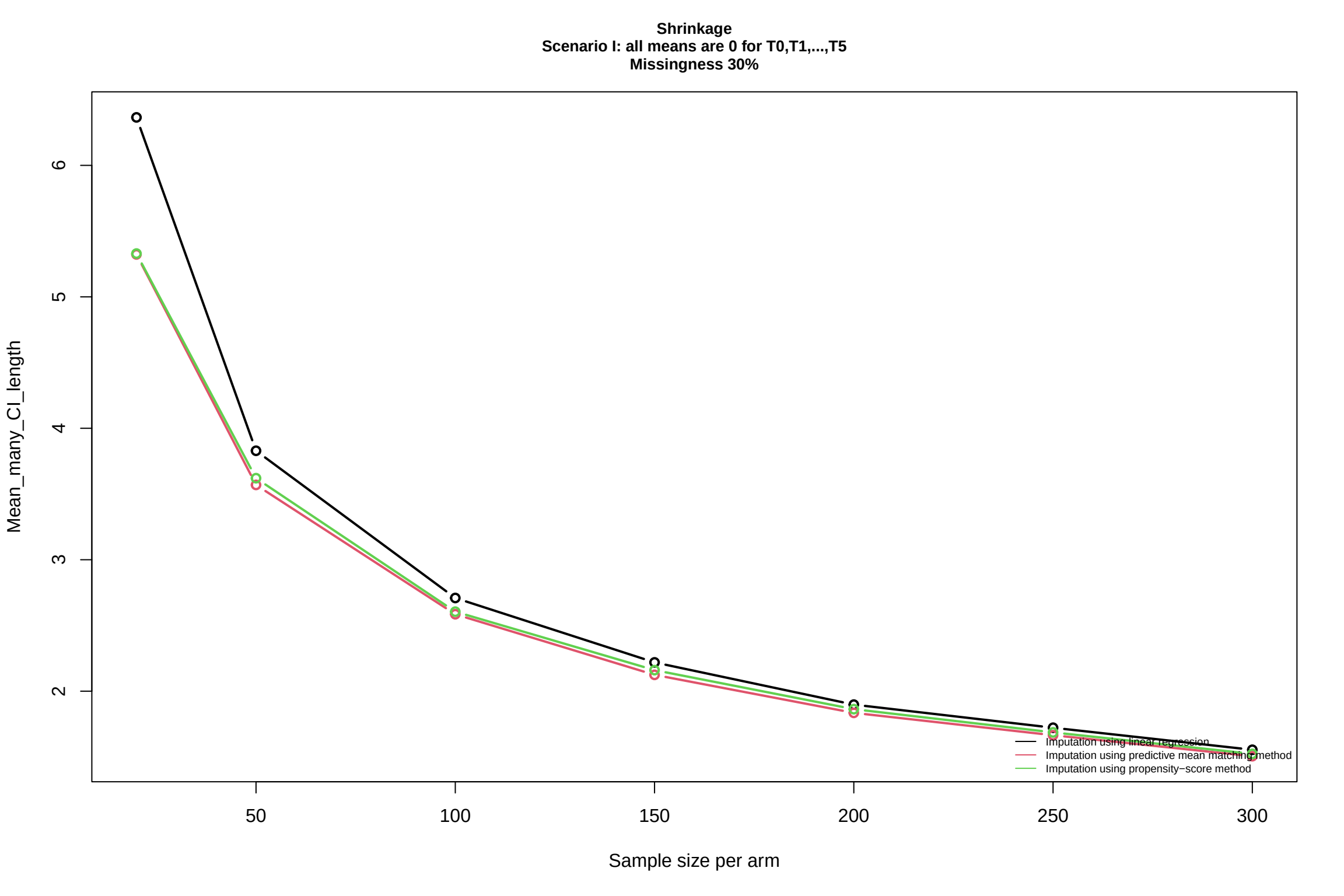

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

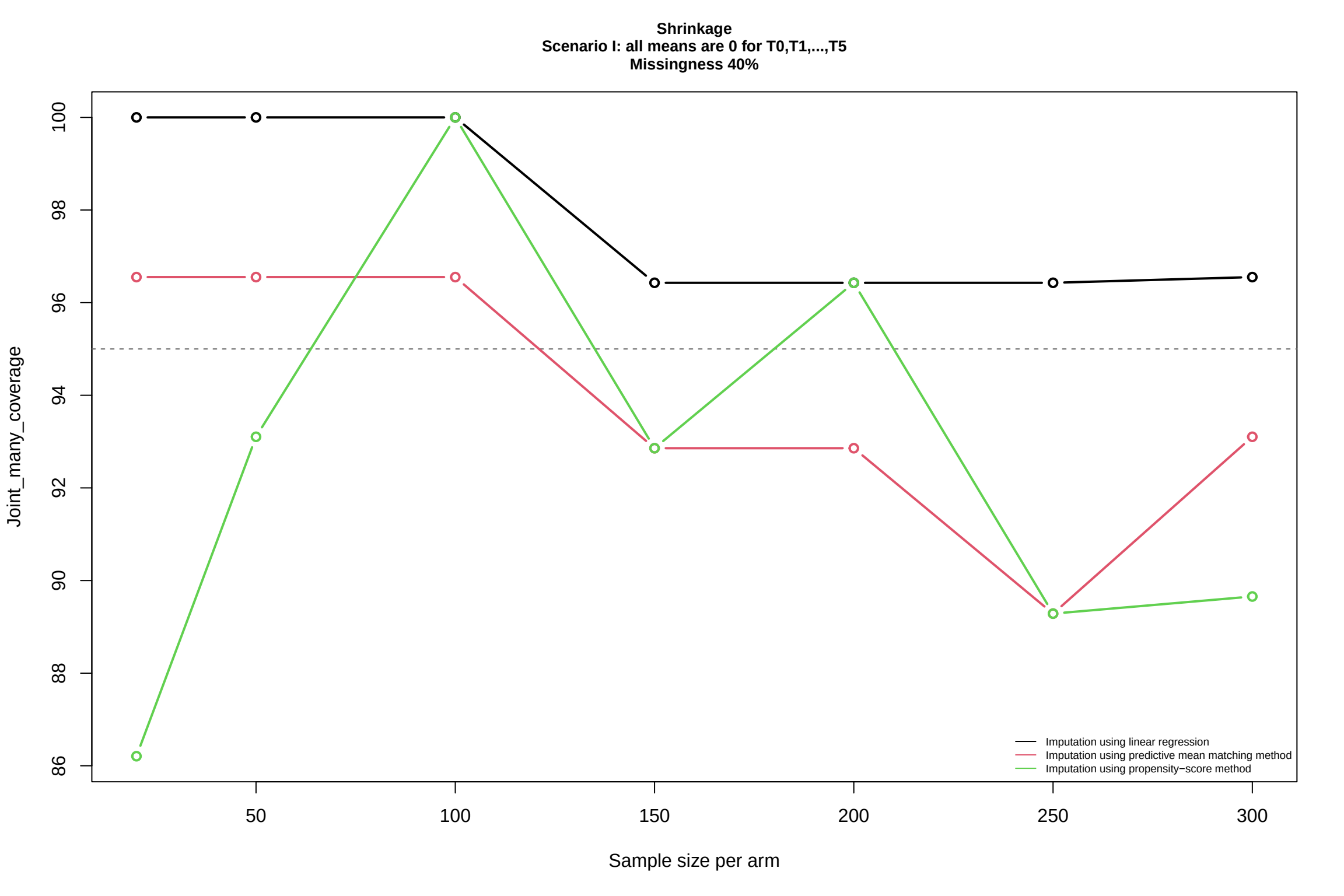

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 10%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

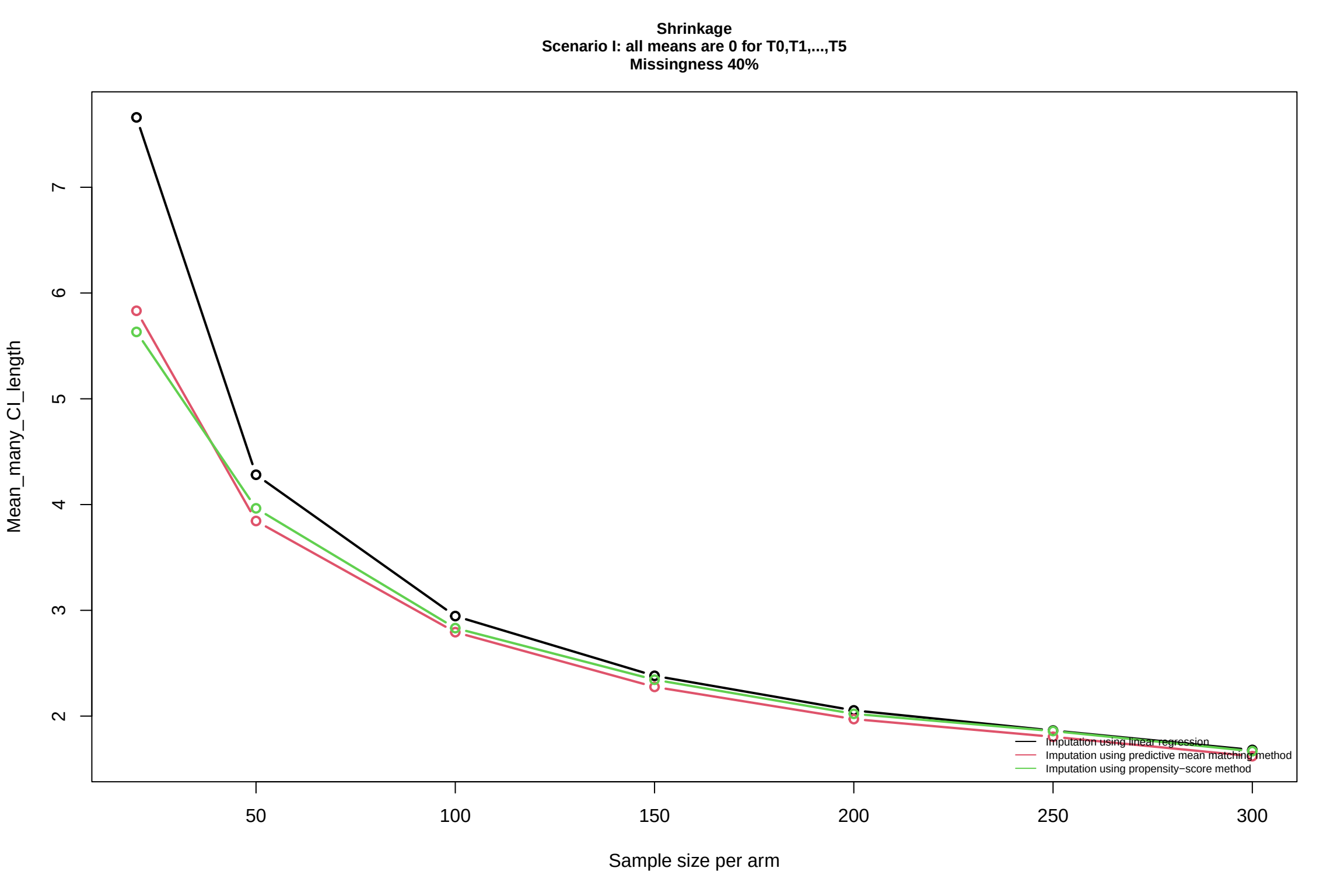

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Joint_many_coverage
100
98
96
94
92
90
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

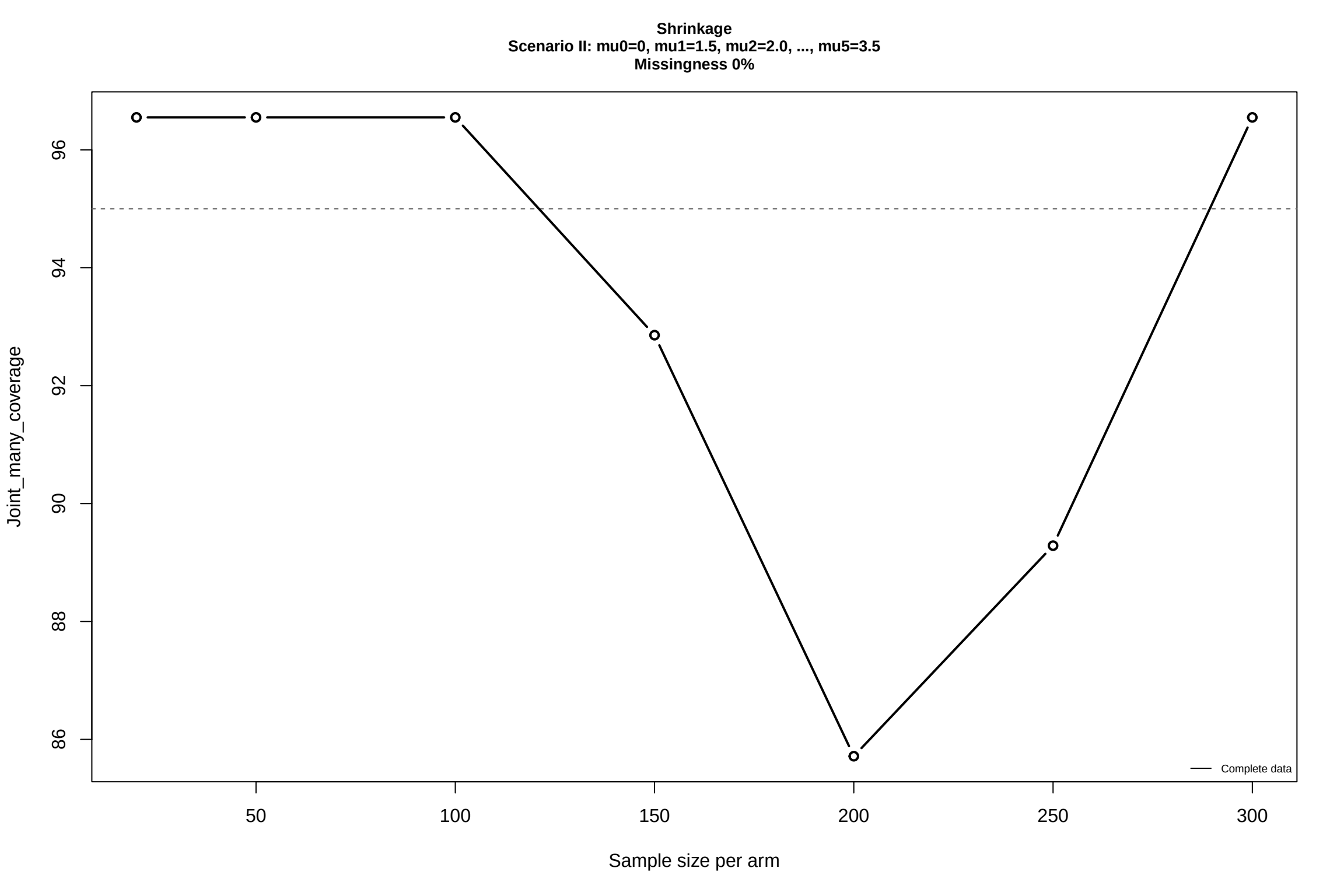

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 20%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

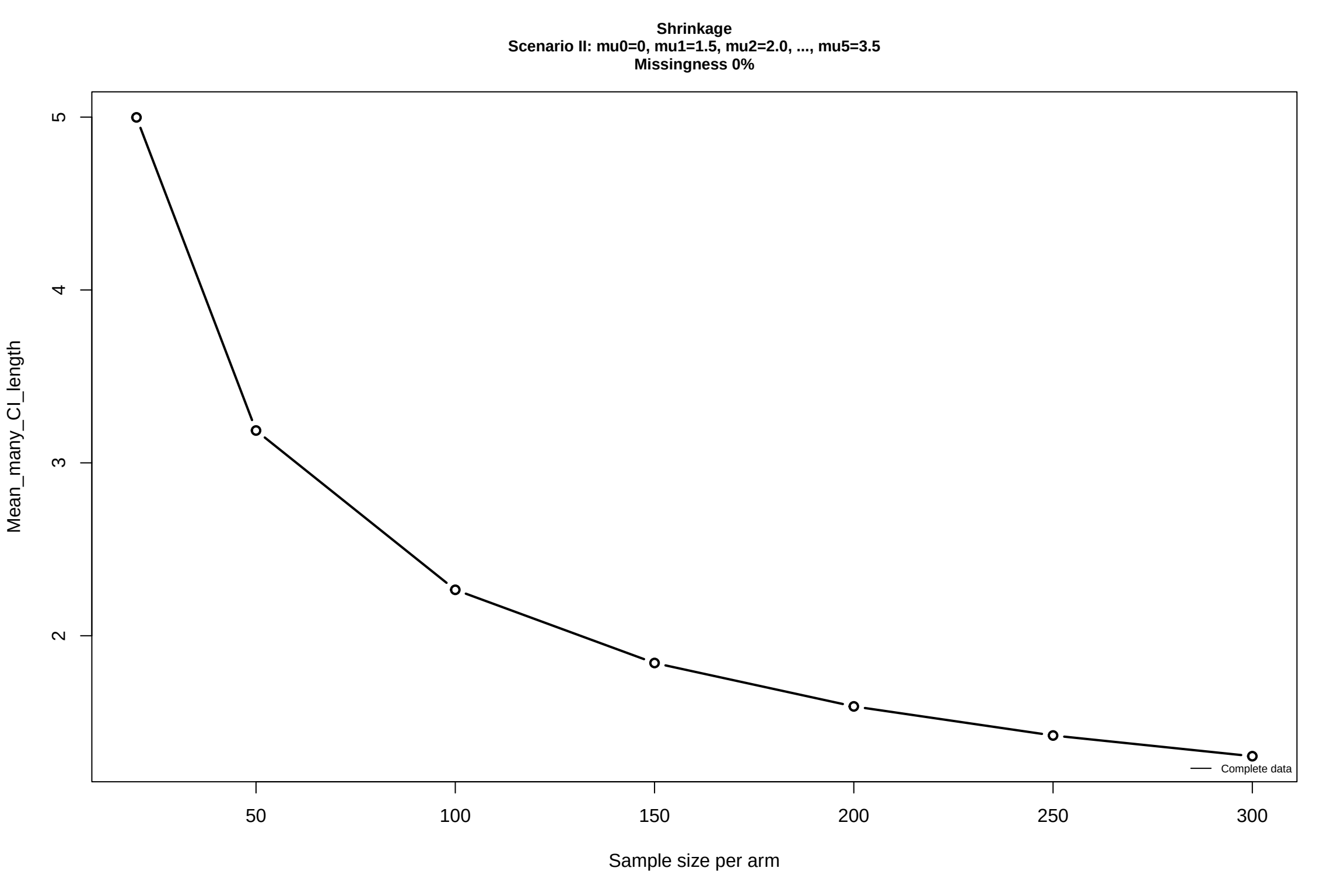

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Joint_many_coverage
100
95
90
85
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

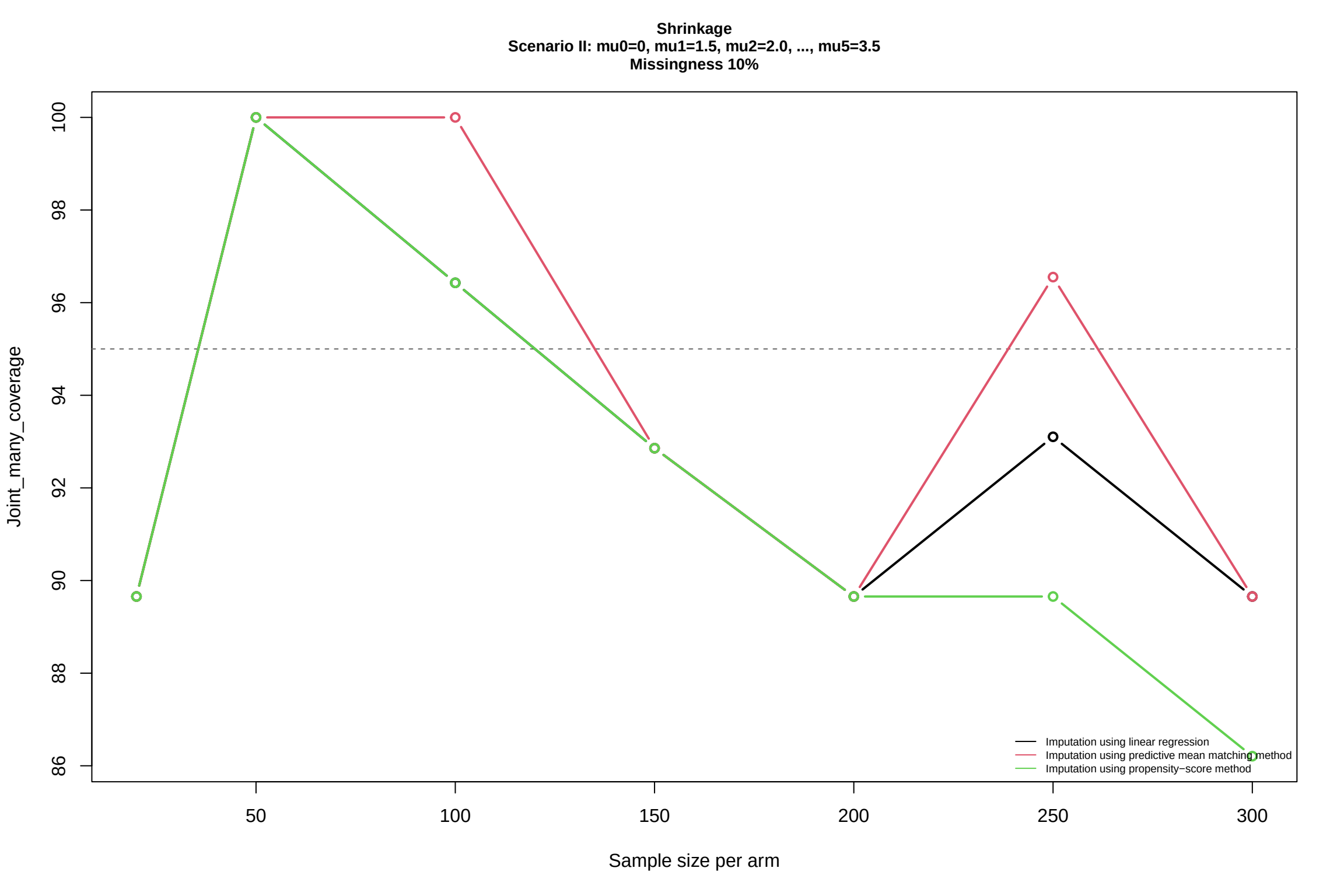

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

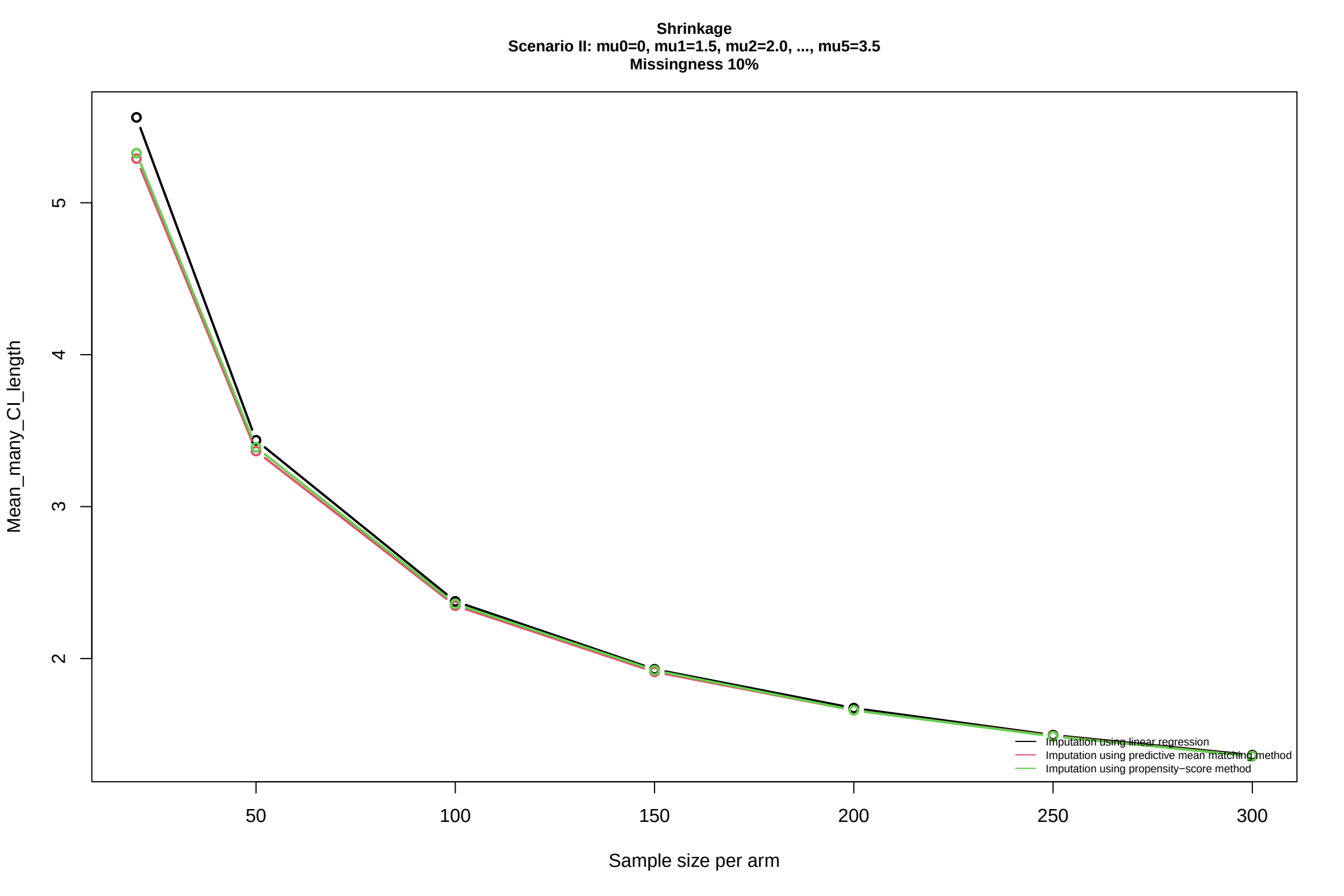

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

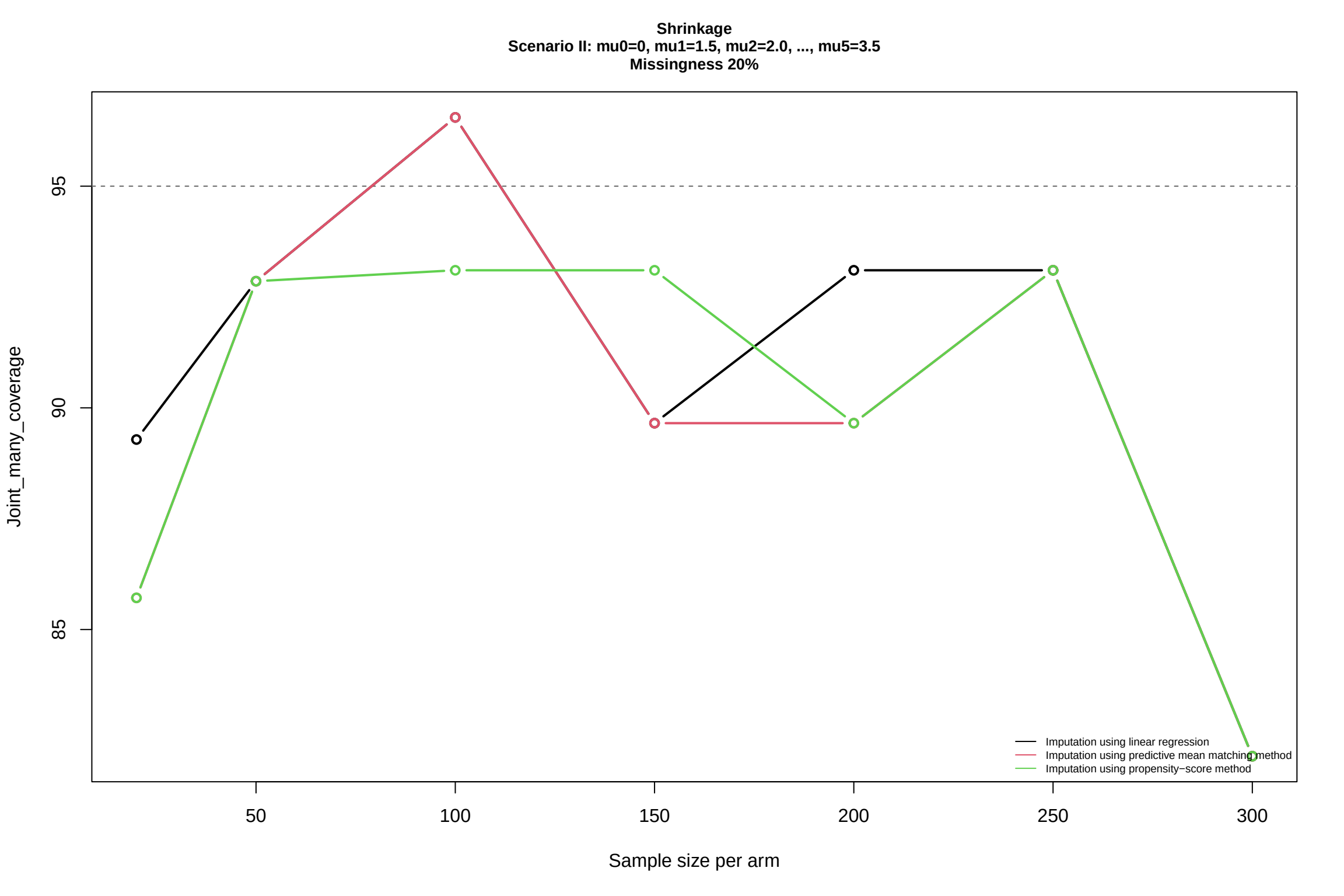

Shrinkage
Scenario I: all means are 0 for T0,T1,...,T5
Missingness 40%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

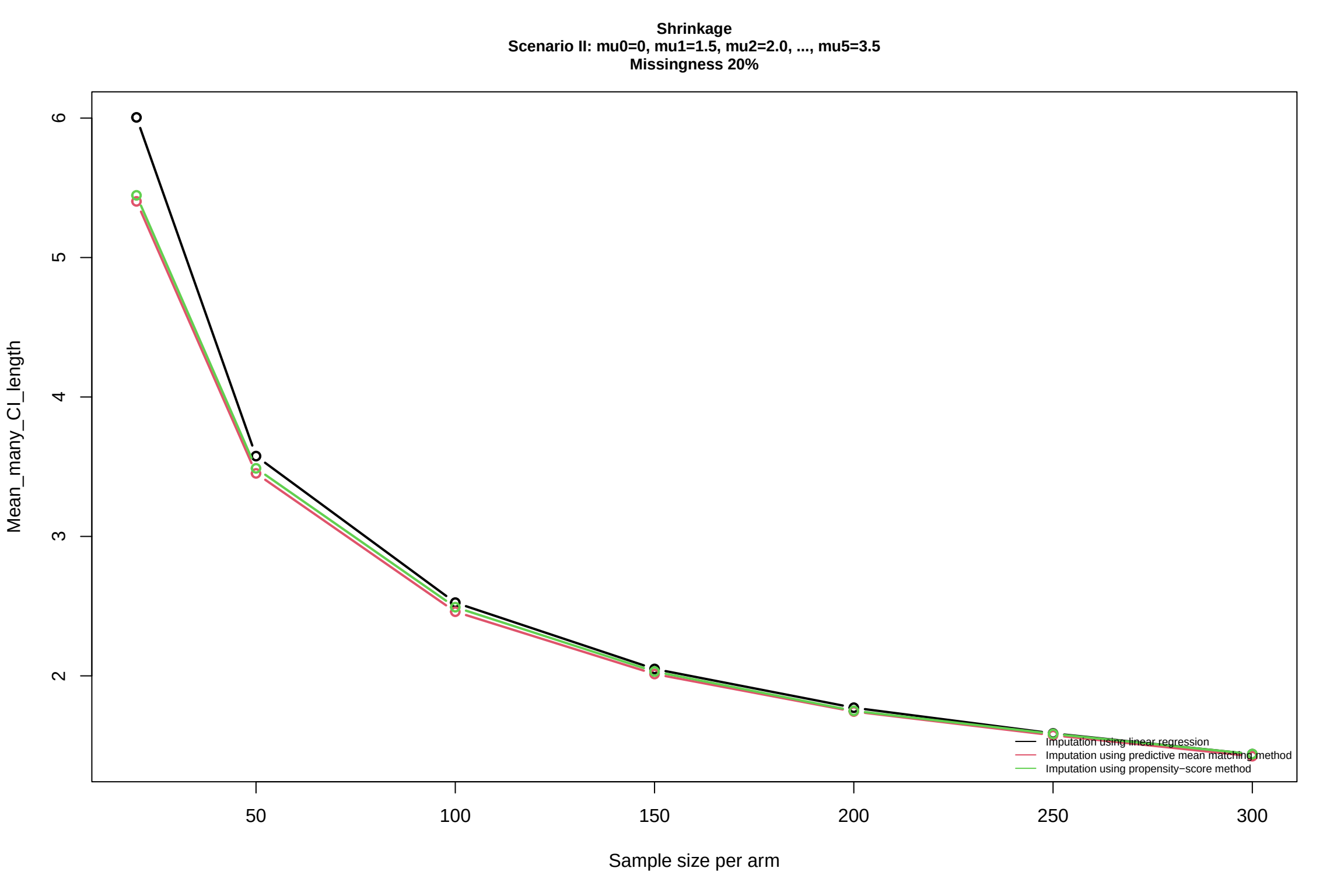

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Joint_many_coverage
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Complete data

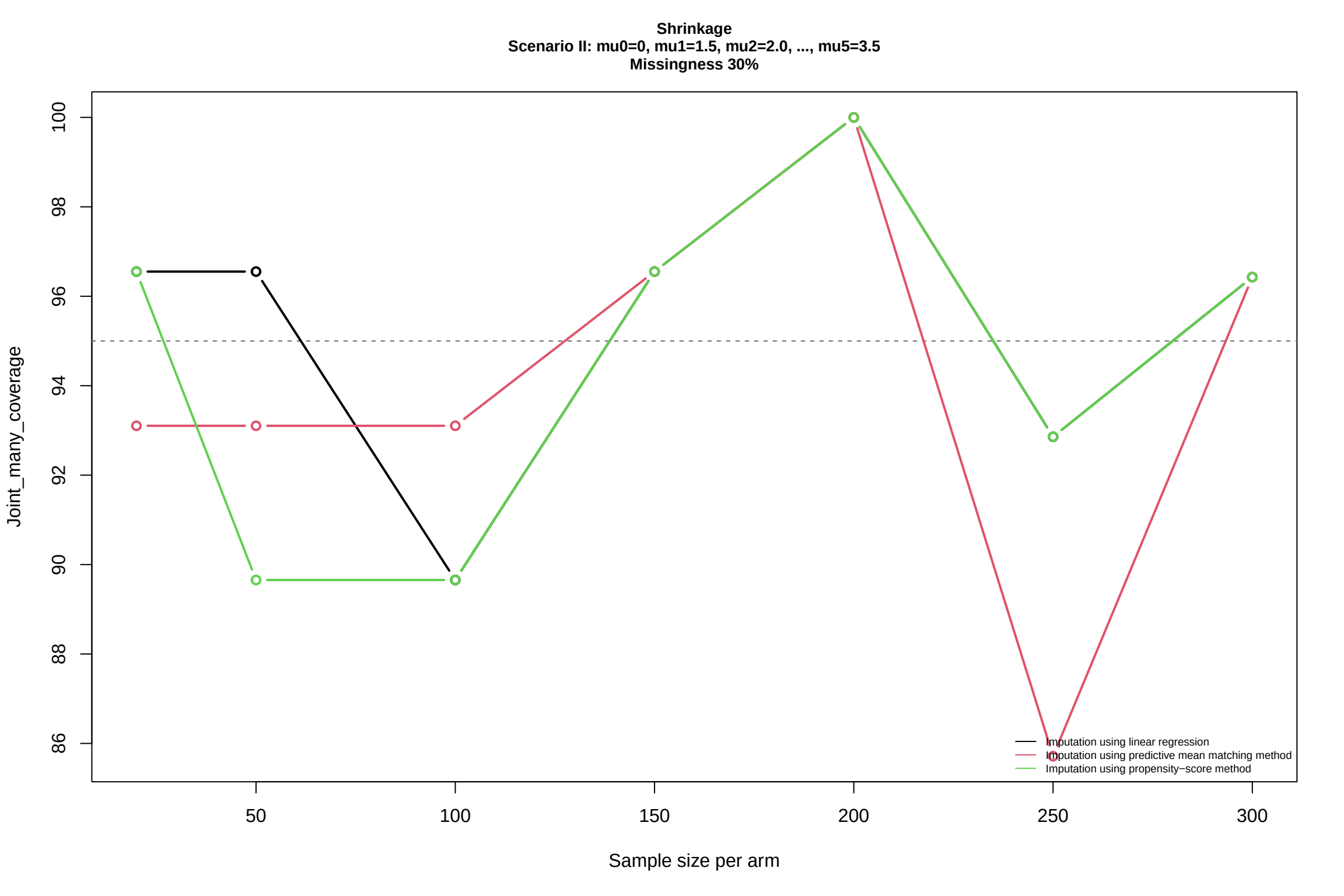

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 0%
Mean_many_CI_length
2
3
4
5
50
100
150
200
250
300
Sample size per arm
Complete data

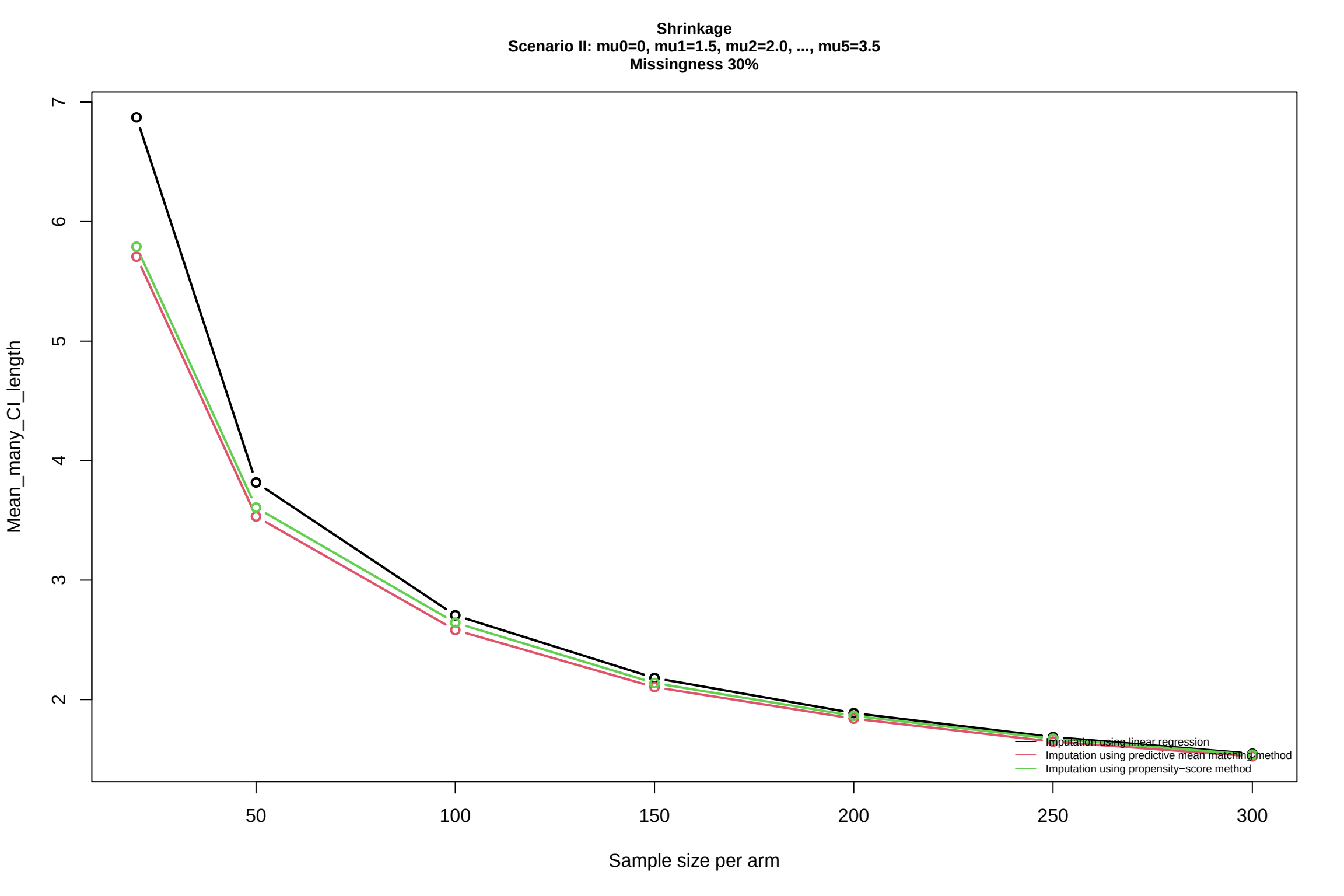

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Joint_many_coverage
100
98
96
94
92
90
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86
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100
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200
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Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method

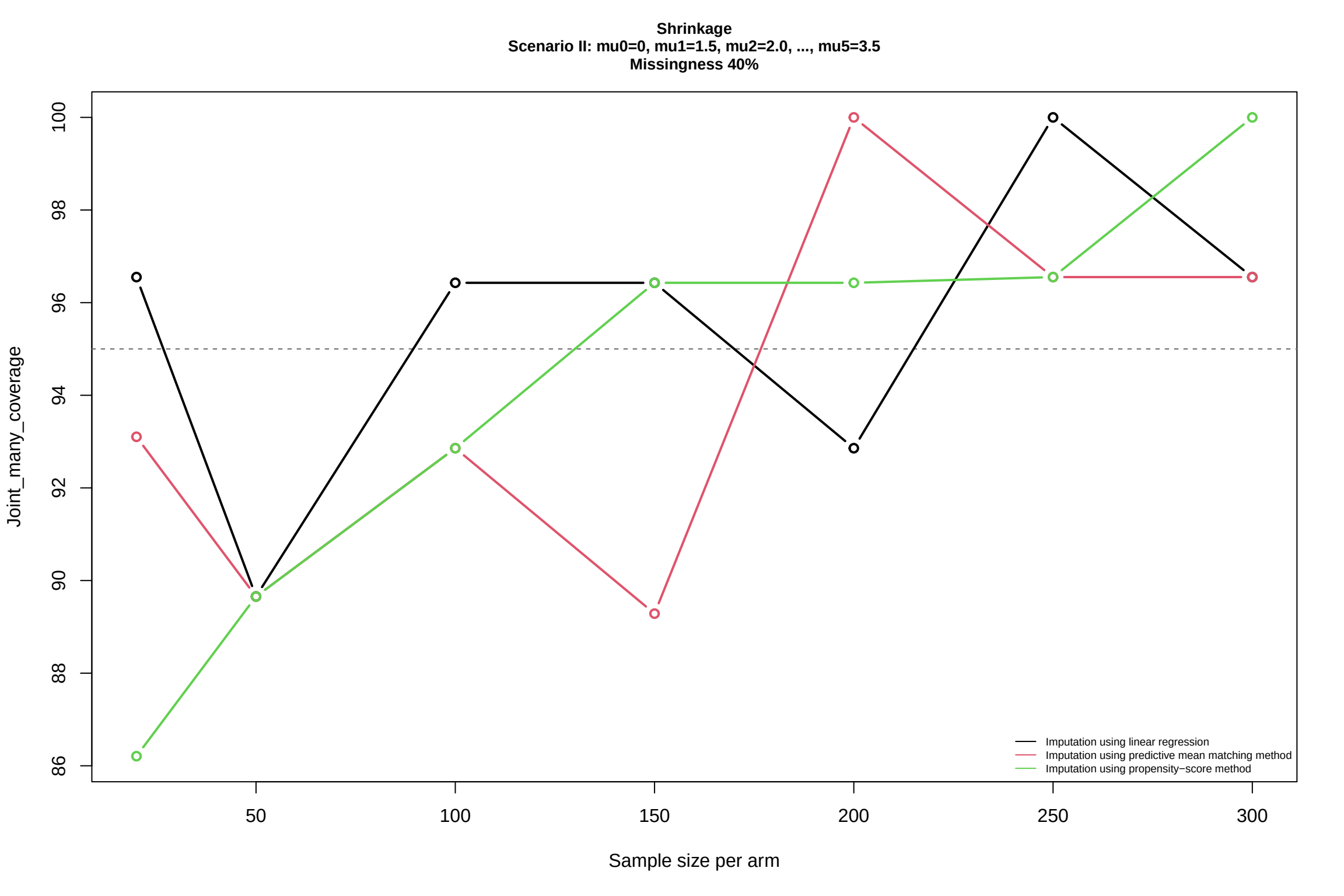

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 10%
Mean_many_CI_length
2
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Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
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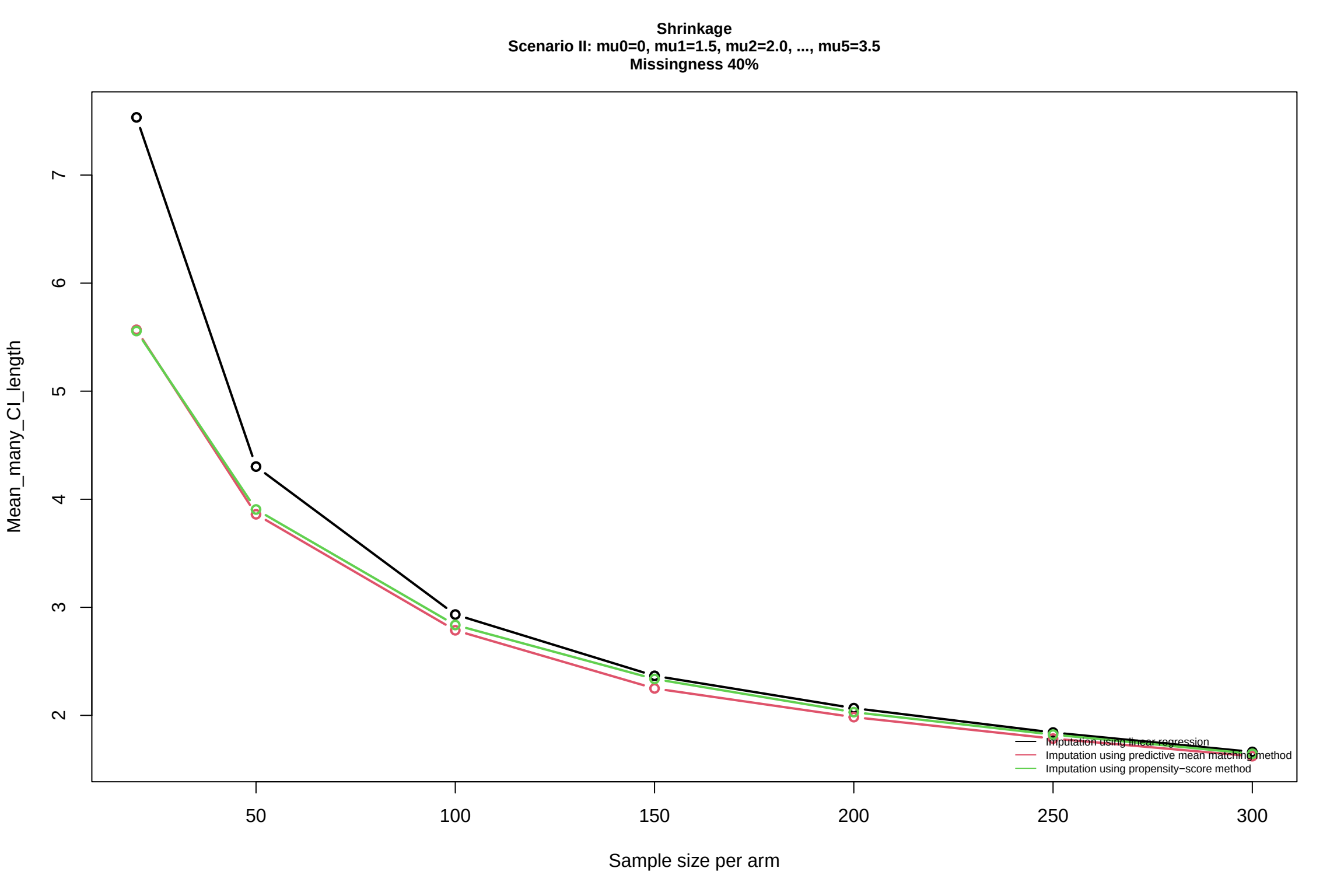

Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Joint_many_coverage
95
90
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Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 20%
Mean_many_CI_length
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3
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6
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Joint_many_coverage
100
98
96
94
92
90
88
86
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 30%
Mean_many_CI_length
2
3
4
5
6
7
50
100
150
200
250
300
Sample size per arm
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method


Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Joint_many_coverage
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Sample size per arm
Imputation using linear regression
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Shrinkage
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Missingness 40%
Mean_many_CI_length
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Sample size per arm
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Imputation using propensity−score method